# Optimal Dynamic Selection Under Costly Evaluation: Evidence from Graduate Admissions

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For the online appendix, see here

**Abstract**

This paper uses graduate admissions to study dynamic selection under costly evaluation, a problem faced in settings from undergraduate and professional admissions to hiring. Using application files from an economics PhD program, accessed through a privacy-protecting synthetic data protocol, I relate acceptance to success measures based on post-program job placements. I estimate structural models of an admissions committee's decision problem capturing first-round screening and matriculation, develop a test of whether the committee's decisions are optimal given a constraint on how many files it can read, and quantify gains from switching to the models' decision rules. The committee's decisions align with applicants' estimated success probabilities along most dimensions, but some characteristics, such as work experience and TOEFL/IELTS scores of non-native English speakers, appear underweighted. Reweighting these characteristics could have increased the admitted cohort's average success probability by 4 percentage points without reading more files, an estimate corrected for the winner's curse.

**Keywords:** Dynamic Optimization, Graduate Admissions, Hiring, Screening, Structural Estimation, Synthetic Data, Winner's Curse

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# 1 Introduction

Admissions and hiring committees must select a subset of applicants whose true quality is unobserved and costly to evaluate in terms of time and effort, and whose future success may be difficult to predict even using all available information. In such settings, committees generally rely on a coarse, low-cost screening of all the applicants, followed by a more intensive, high-cost evaluation of the subset of applicants who survive the screening, which may include reading files in full detail and interviewing the most promising applicants. Moreover, because applicants may have outside options, committees may also have to account for whether the applicants they accept will in turn take the offer. Other examples of selection with costly evaluation include conducting multi-stage clinical trials for medicines, messaging virtually on dating apps before meeting in real life, and simulating product prototypes on a computer before physically building them.

A fundamental question in such settings is how decision-makers should interpret the available information at each evaluation stage to select either the best subset of options or all options above a quality threshold. A related question is whether observed decision rules reflect optimal dynamic selection given evaluation costs and other constraints, or whether those rules deviate from optimal behavior. One could imagine a hiring committee undervaluing a skill or discriminating against a demographic. For example, Hoffman, Kahn, and Li (2018) find that managers who overrode job testing scores made worse hires. Moreover, if there are deviations, which rules should decision-makers implement instead? As Rambachan (2024) shows, comparing decision-makers' choices against a predictive model can only identify mistakes under assumptions about the decision-makers' preferences and information. Answering these questions therefore requires moving beyond identifying the determinants of success and explicitly modeling the decision problem.

Graduate admissions is a meaningful setting to study these questions, as shown by economics papers from Krueger and Wu (2000) to Bai, Esche, MacLeod, and Shi (2022). Graduate programs, especially PhD programs, invest at least a few years of time, along with tens of thousands of dollars of funding, into students' development. In addition, high-ranking academic placements boost a program's reputation, and so increase its ranking from sources such as U.S. News. Better students also lead to better coauthoring opportunities for faculty, and since people with graduate degrees often find themselves in influential positions, such as professorships, studying how best to admit them is important for programs and society. In particular, if schools are undervaluing applicants with certain characteristics (e.g. from a particular demographic), future applicants with such characteristics may receive opportunities they would not have otherwise. For example, while Chetty, Deming, and Friedman (2026) focus on undergraduate admissions, they find that applicants from high-income families have higher admissions rates at Ivy-Plus colleges even if their standardized test scores are no higher than those of less fortunate applicants.

Moreover, graduate admissions is a setting with costly evaluation. Admissions committees observe for all applicants a limited set of objective characteristics, such as standardized test scores, but typically lack the time to read every file in detail. As such, applicants are initially

screened using coarse signals, with more intensive evaluation reserved for the subset of applicants with promising objective characteristics. Final decisions may also involve waitlists or other forms of deferred acceptance, as uncertainty about applicant quality and matriculation status resolves over the period between the application and admissions deadlines. These features make graduate admissions well-suited for modeling dynamic selection under costly evaluation. Doing so for one economics PhD program, I find that while the committee's decisions are much better than random chance and align with many applicants' estimated success probabilities, the committee may misweight some applicant characteristics, and correctly weighting them could increase the admitted cohort's average success probability without reading any more files.

This paper addresses two specific research questions. Both require first determining which *ex-ante* factors in application files affect admission and success in graduate school. Following the literature, success may be defined in multiple ways, including program completion and job placement outcomes. I study these factors as comprehensively as possible by extracting all available data from application files, including recommendation letters, application essays, and CVs. My data consist of all applicants to a research university's economics PhD program from 2015 to 2019, for which I observe both the application files and the outcomes. In Online Appendix A, I supplement these data with the objective characteristics of all applicants to that university's economics, government, history, linguistics, and psychology PhD programs from 2020 to 2023 to compare admissions across programs. Because these data are confidential, I implement a privacy-protecting research protocol that uses synthetic data for model development and validation (see Section 2). This protocol also serves as a template for researchers who wish to work with sensitive data in general. It also has a side benefit, namely that I had to commit to my specifications on the synthetic data and could adjust them only sparingly after seeing results from the actual data.

The first research question asks whether admissions committees optimize or make mistakes when admitting applicants. Specifically, if a factor predicts admission but not success, the committee may be overvaluing it, whereas if a factor predicts success only, the committee may be undervaluing it. For instance, the committee may be putting too much or too little weight on elite undergraduate institutions, standardized test scores, or other applicant characteristics. To determine whether the committee is optimizing, it is necessary to model the committee's optimization problem, estimate the parameters, and develop a test of the hypothesis of optimality, which, to my knowledge, no other paper has done for a multi-stage selection problem. To make such models tractable, I extend a result in Bai et al. (2022) to reduce the committee's portfolio choice problem (i.e. from $N$ applicants, accept the optimal subset of $N_A$ applicants) to a cutoff rule problem (rank the $N$ applicants by their expected success probabilities and select the $N_A$ above the cutoff). I estimate multiple structural models in increasing order of complexity, modeling such features as first-round screening and matriculation. That said, evidence suggests that PhD program quality also affects success through "school value-added" (Angrist, Diederichs, & Ellison 2026), so it is important to account for that quality lest the model encourage the committee to put more weight on a variable that only predicts success by way of predicting admission into a highly-ranked

program. I discuss the tradeoffs of controlling for program quality in Section 3.

The second research question is whether committees can make better decisions without having to expend additional effort, and if so, how much better of a cohort they can admit. Perhaps as Simon (1956) suggests, if committees are not optimizing, they may instead be satisficing (i.e. aiming for a satisfactory result), namely because reading every application file would require too much effort.[1] To answer the second question, I use the structural models to calculate counterfactual deviation gains from adopting the optimal admissions strategy, while respecting the committee's time constraint of not being able to read every application file in detail. If the committee rejected likely matriculants who went on to succeed, that would be evidence of suboptimality.

I perturb the success parameter estimates about their asymptotic distribution when comparing the average success probability of the cohort admitted by the model's decision rule vs. the committee's. I then show that if the model's cohort is exogenously chosen, for example through the use of a holdout set to evaluate that cohort out of sample, then the fraction of perturbations in which the model's cohort does not improve on the committee's converges to the p-value of a $z$ test of the null hypothesis that the committee's decision rule is optimal. However, in economics PhD admissions, applicant pools are not large enough to reserve a holdout set, making the deviation gains susceptible to what I. Andrews, Kitagawa, and McCloskey (2024) deem the "winner's curse." Essentially, even if the estimated success probabilities are unbiased, the average success probability of the set selected using those estimates will be biased upward. Therefore, I use the perturbations to develop and implement a parametric bootstrap correction to separate the true deviation gain from the winner's curse. Finally, there is the possibility that committees have other priorities than maximizing success probabilities, such as admitting a diverse class across demographics. As such, I show how the models can be extended to factor in those priorities.

The structural estimates show that along most dimensions, the success probabilities implied by the committee's decisions align with the success probabilities estimated from outcomes. However, the committee may have misweighted some applicant characteristics. Holding fixed the number of files the committee read in each round and the number of applicants it admitted in each year, so that the counterfactual committee only re-ranks the applicants, the bootstrap-corrected average success probability of the model's decision rule's acceptance set is greater than the average success probability of the committee's decision rule's acceptance set by about 4 percentage points. When the models account for matriculation, so that the relevant set is the matriculation set, the corrected average success probability also increases by about 4 percentage points, though I cannot establish whether either gain is statistically significant. Because the committee's own subjective score for each application enters the models as a proxy for information in the files that the variables in my dataset miss, these gains are unlikely to reflect private information. Consistent with this hypothesis, the committee's actual matriculants succeeded at a rate no higher

[1] Satisficing is distinct from rational inattention (see Section 1.1), which assumes that decision-makers are optimizing and so embeds the choice of how much effort to allocate into an optimization framework.

than their estimated success probabilities, so if the committee has information that the model lacks, the committee does not appear to be using it. Conversely, the gains are measured relative to the information in my variables, so if the files contain success-relevant information that neither the committee nor my variables have, larger gains may be possible. In Monte Carlo simulated datasets, the same procedure finds larger gains when the simulated committee misweights applicant characteristics but not when it behaves nearly optimally (see Online Appendix B).

To determine which variables the committee may have misweighted, I reweight one variable at a time in the committee's estimated decision rule, holding the remaining weights fixed, and calculate the gain from the best reweighting. No individual variable yields a gain of more than than 2.5 percentage points, so the committee did not make any large mistakes. The committee appeared to underweight age, work experience, and English proficiency as measured by TOEFL/IELTS scores. In addition, in application essays, positive language may have been underweighted, whereas economics-specific language may have been overweighted. The committee also penalized applicants with more coding languages in their CVs although coding did not predict failure. Meanwhile, the committee may have overweighted elite U.S. undergraduate universities, though only for the acceptance set, as accepted applicants from such universities matriculated at lower rates. However, for most of these variables, the gain from reweighting turns negative for a minority of perturbations. Also, some misweightings might not hold going forward, such as essay-specific misweightings if future applicants use AI to help write their essays.

Moreover, I do not find evidence that the committee's constraint on how many application files it reads meaningfully affects the average success probability of its acceptance or matriculation set. To look for such evidence, I first graph the average success probability of the model's acceptance and matriculation sets as a function of the number of files read. Reading every application file rather than the roughly 40 percent the committee actually read barely raises the model's admitted cohort's average success probability. The committee's own decision rule, meanwhile, yields nearly the same average success probability regardless of how many files it reads.

Finally, given that the university in this study is not necessarily representative of all graduate programs, some of this paper's results might not generalize to other settings. However, the methods are generalizable to admissions, hiring, and any other setting where decision-makers must decide which options to select based on *ex-ante* information. If the sample sizes are high enough, such as in admissions at large law schools or hiring at large companies, then holdout sets can be used to avoid the winner's curse entirely and test optimality directly. But even in smaller settings, such as economics PhD admissions, the structural models all still apply, and methods such as the parametric bootstrap correction can account for bias resulting from the winner's curse.

## 1.1 Related Literature

Outside economics, Dawes (1971) illustrates, using small samples of matriculants, that a linear model predicting a psychology admissions committee's ratings of applicants from GRE scores,

grades, and undergraduate institutions can in turn predict post-matriculation faculty evaluations better than the ratings themselves. The model could also screen out applicants whom the committee would probably reject anyway, saving faculty time. In economics, Ehrenberg and Mavros (1995) study predictors of success found in application files. They analyze Cornell PhD students, including economics, and find that verbal but not quantitative GRE scores are predictive of program completion. Subsequent research implies the reason is likely that their sample is restricted to only the applicants who attended Cornell, nearly all of whose quantitative GRE scores were concentrated in the top percentiles. Two years later, Attiyeh and Attiyeh (1997) use a sample across 48 universities and multiple programs from 1990 to 1991 to study determinants of admission. They find that all else equal, committees prioritized underrepresented minorities.

Krueger and Wu (2000) and Grove and Wu (2007) then study admission and success at a single top economics department in 1989, including all applicants in their success regressions and tracking placement outcomes for non-matriculants. They find that admissions committee scores, GRE scores, and the prominence of recommendation letter writers predict good placements, but there is a lot of uncertainty. Finally, Athey, Katz, Krueger, Levitt, and Poterba (2007) analyze the determinants of success for Chicago, Harvard, MIT, Princeton, and Stanford economics PhD students from 1990 to 1999, adding graduate course grades into their models. While GRE scores and other *ex-ante* factors predict good placements, they do not when controlling for grades.

Stock, Finegan, and Siegfried (2006) begin a series of papers by compiling a 586-student dataset of Fall 2002 economics PhD matriculants. They start by looking at attrition rates in the first two years and find that low GRE scores predict attrition, but RAships and shared department office space help programs retain students. Siegfried and Stock (2007) compile another, larger dataset of everyone who earned a U.S. economics PhD from 1966 to 2003 and observe that top liberal arts colleges feed economics PhD programs at disproportionally high rates.

Stock, Finegan, and Siegfried (2009a) and Stock, Finegan, and Siegfried (2009b) follow up on the progress of the students in Stock et al. (2006) and find similar results but emphasize that there is a lot of uncertainty in predicting completion after five years. Stock, Siegfried, and Finegan (2011) and Stock and Siegfried (2014) track the students after eight years and summarize their multi-paper study, concluding that while some factors like high GRE scores, a liberal arts undergraduate institution, and financial support predict completion, uncertainty remains very high. Finally, Stock and Siegfried (2015) add more recent years to Siegfried and Stock (2007)'s larger dataset, noting that foreign undergraduate institutions have become more common over time.

Meanwhile, Grove, Dutkowsky, and Grodner (2007) study Syracuse economics PhD students, who have different characteristics than students attending top-ranked programs. They also incorporate factor analysis into their regressions, finding that different attributes predict success at different stages. For example, GRE scores predict passing comprehensive exams, whereas research motivation (a factor from their factor analysis) predicts program completion. Schlauch and Startz (2018) study economics PhD candidates from top 50 programs posting their CVs for the 2016–2017 job market. They find that while RA experience predicts attending a highly-ranked

PhD program, a master's degree does not, which aligns with my result that *Work Experience* but not *Graduate Degree* predicts success. Meanwhile, Jones, Schuhmann, Soques, and Witman (2020) survey individuals involved in admissions at 132 economics PhD programs. They find that higher-ranked programs care about different factors than lower-ranked programs. For example, they care more about undergraduate rankings. Also, while recommendation letters matter for admission, programming skills usually do not, though they may matter for success.

Bai et al. (2022) represent the frontier of economics graduate admissions research. They study economics PhD applicants at a single top program from 2013 to 2019 and like Krueger and Wu (2000) collect data on admissions and placement outcomes for all applicants. They start by proving that with no complementarities across applicants, a committee's portfolio choice problem simplifies to a cutoff rule problem, which is more tractable. I build on this result by extending it to two rounds with matriculation, directly modeling the resulting cutoff rule problem, and estimating the parameters, which is necessary to make claims as to whether committees are optimizing. In addition, Bai et al. perform textual analysis of recommendation letters and find that subjective factors, such as letters, matter for both admission and success more than objective factors, such as GREs. As such, the models in this paper incorporate subjective as well as objective factors.

This paper also relates to three methodological literatures: optimal dynamic selection with costly evaluation, evaluating whether decision-makers behave optimally, and synthetic data and privacy protection in empirical economics. In the first literature, earlier papers develop models of sequential search where agents face uncertain option quality and must decide when to stop searching given search costs. McCall (1970) and Mortensen (1970) model job search as a dynamic stopping problem in which workers optimally accept offers that exceed a reservation wage. Weitzman (1979) extends this logic to settings in which decision-makers can acquire information about multiple alternatives at a cost, which he denotes as "Pandora's Problem." Together, these papers establish that when information is costly, optimal decision rules are often threshold-based.

A more recent group of papers studies costly evaluation outside of search problems. In models of rational inattention, agents are fully optimizing but face constraints on their ability to process information. Sims (2003) introduces rational inattention as a framework in which information acquisition is costly, leading agents to optimally trade off decision accuracy with information costs. Matějka and McKay (2015) show that rational inattention can generate probabilistic discrete choice rules in line with multinomial logit models, while Caplin and Dean (2015) provide a revealed-preference characterization of optimal information acquisition that separates rational inattention from actual mistaken decisions. These models differ from Simon (1956)'s concept of satisficing, which proposes heuristic, "good enough" decision rules. In contrast, rational inattention models retain the mathematical framework of rationality but endogenize information acquisition. Notably, unlike this paper, none of these papers estimate their models on data.

This paper differs from both rational inattention and satisficing. I take the information environment faced by decision-makers as given, since admissions committees, at least in this framework, do not choose how much information is available at each stage of evaluation. Instead, they

interpret the information they observe at each stage and decide how to act on it. The contributions of this paper are to model this dynamic selection problem, estimate parameters governing the resulting decision rules, and assess whether observed behavior is optimal conditional on respecting the committee's constraints. By doing so, this paper makes optimality testable in dynamic selection problems and allows for the computation of counterfactual rules and deviation gains.

In the second literature, some papers use structural models to determine whether decision-makers behave optimally in contexts other than dynamic selection. For example, Cho and Rust (2010) find that a large car rental company could have increased profits by holding onto their rental cars for longer and charging lower prices to consumers who rent the older cars. In addition, Anderson, Rosen, Rust, and Wong (2025) find that some of the world's best professional tennis players could have increased their service game win probabilities by choosing different serve direction strategies than the ones they were using. In contrast, Chan, Gentzkow, and Yu (2022) estimate a structural model in which radiologists' pneumonia diagnostic thresholds are optimal given their skill and preferences. They find that variation in skill can explain 39 percent of the variation in diagnostic decisions. This paper provides a framework to determine whether decision-makers, such as admissions committees, optimally solve dynamic selection problems.

Other papers in this literature evaluate decision-makers against predictive models, which rank candidates by predicted performance and select those above a cutoff. Autor and Scarborough (2008) find that standardized job testing produced more productive hires than informal interviews without reducing minority hiring. Hoffman et al. (2018) show that managers' discretion in overriding job testing scores reflected bias or mistakes more than it reflected private information. Kleinberg, Lakkaraju, Leskovec, Ludwig, and Mullainathan (2018) study judges' bail decisions and find that reranking defendants by predicted risk could reduce crime without increasing jailing rates. They avoid the winner's curse by splitting their sample into training and test sets, as do Mullainathan and Obermeyer (2022) when studying physicians' testing decisions for patients at risk of heart attacks. Baron, Dobbie, Lombardo, Rambachan, and Ryan (2026) extend such comparisons to a two-stage setting in which child protective services screeners decide which allegations to investigate, and investigators decide which investigated allegations warrant opening a case. They find that evaluating each stage in isolation can overstate the effects of deploying an algorithm across the system as a whole. I also evaluate decisions by predicted outcomes, but I model the admissions committee's problem structurally, including evaluation costs and matriculation. With thousands rather than hundreds of thousands of observations, I cannot split my sample, so I correct for the winner's curse with a parametric bootstrap. Moreover, I observe placement outcomes for all applicants, avoiding Kleinberg et al. (2018)'s problem of observing crime outcomes only for released defendants. That said, the outcomes of applicants who attend other PhD programs partially reflect the characteristics of those programs (see Section 3).

Rambachan (2024) shows that comparisons against a predictive model, like Kleinberg et al.'s, identify a decision-maker's mistakes only under assumptions about its preferences and private information, and he finds that at least 20 percent of New York City judges make such mistakes.

His framework places weak restrictions on preferences and private information, but it takes as given that the decision-maker makes a separate choice for each individual with no capacity constraint, costly evaluation stage, or possibility of declined offers. In contrast, this paper specifies the committee's objective and proxies for its private information but also generalizes to capacity constraints, costly evaluation, and matriculation. My extension of Bai et al. (2022)'s cutoff rule result, which holds for large applicant pools, allows the committee's decisions to be analyzed one applicant at a time. Rambachan's and my approaches are therefore complements.

The third methodological literature addresses synthetic data and privacy protection. Rubin (1993) introduces synthetic data as a tool for protecting confidentiality by replacing sensitive microdata with simulated observations drawn from an estimated data-generating process. Before synthetic data, privacy protection often involved masking, or adding random noise to the data. Masking, however, induces a trade-off between privacy and information loss (Little 1993), and the rows in a masked dataset still correspond to real subjects. More recent work, such as Abowd and Schmutte (2015), emphasizes the challenges posed by restricted access to data. Meanwhile, Koenecke and Varian (2020) propose that researchers publicly release synthetic data, such as those generated by the Synthetic Data Vault (Patki, Wedge, & Veeramachaneni 2016), when they cannot release the actual data. Their proposed protocol differs from this paper's protocol, as this protocol is designed for a setting where the *researcher* cannot access the actual data.

The rest of the paper proceeds as follows. Section 2 describes the data and synthetic data protocol. Section 3 presents regression-based evidence of the determinants of acceptance, matriculation, and success, and shows why comparing the determinants of acceptance with those of success is insufficient to test optimality. Section 4 develops and estimates structural models of admissions that incorporate first-stage screening and matriculation probabilities. Section 5 computes deviation gains using the parametric bootstrap correction. Section 6 concludes.

# 2 Data

This paper uses two datasets: one consisting of all applicants to a research university's economics PhD program from 2015 to 2019, and another of all applicants to that university's economics, government, history, linguistics, and psychology programs from 2020 to 2023. Unlike the graduate school dataset, the economics department dataset contains subjective variables (i.e. those derived from textual documents such as CVs and recommendation letters) and program outcomes data (e.g. post-program job placements). Therefore, I estimate this paper's structural models on the economics department dataset only. However, the graduate school dataset, which still contains objective variables and each applicant's admission status, is useful for making regression-based comparisons across programs. All results from the graduate school dataset are in Online Appendix A. In this section, I describe the variables in the economics department dataset and the synthetic data protocol, which serves as a template for conducting research on confidential data.

## 2.1 Variables

There are three types of variables: dependent variables, objective independent variables, and subjective independent variables. Table 7 of Appendix A contains a full list of the variables. The dependent variables consist of measures of both admission and success. For admission, such measures include binary variables of whether the applicant makes it past the first-round screening stage, whether they are accepted into the program, whether they are waitlisted, whether they attend the university's open house, whether they matriculate at the university, and whether they attend (and were therefore accepted at) a different university's PhD program. I also have the U.S. News American and global rankings of the program they attended. If they attended a U.S. program, I use their American ranking, whereas if they attended a non-U.S. program, I use their global ranking times 3/8. The reason is that the maximum American ranking is 147, the maximum global ranking is 400, and I assign 150 to anyone who did not attend a PhD program. This procedure ensures that all U.S. and non-U.S. programs are ranked on the same scale.

For success in economics, I have a binary variable of whether they complete the program, along with variables for the type and ranking of their eventual job placement. Placement types include academic, government, consulting, and technology. For academic placements, I use the department's IDEAS/RePEc American and global rankings with a plus five penalty for business schools (Krueger & Wu 2000). For postdocs, I use a 2/3 to 1/3 weighted average of the university's ranking and the maximum ranking, which is 138 for American and 340 for global. For non-tenure-track positions, I use a 1/3 to 2/3 weighted average. For unranked tenure-track positions, I assign the maximum ranking plus one; for unranked postdocs and non-tenure-track positions, I assign 150 for American and 350 for global. Ultimately, for U.S. placements, I use the American ranking, whereas for non-U.S. placements, I use the global ranking times 3/7.

Meanwhile, for non-academic placements, I construct academia-equivalent rankings shown in Appendix A. Following Krueger and Wu (2000), the most prestigious non-academic placements (World Bank, IMF, Federal Reserve Board of Governors, European Central Bank, Amazon Science, Google Research, and Microsoft Research) are given a ranking of 40, with lower rankings down to 135 for less prestigious placements.[2] Per Eberhardt, Facchini, and Rueda (2026), economics PhD graduates, even those with Nobel laureate advisors, are increasingly taking non-academic jobs, especially in technology. As in Krueger and Wu (2000) and Bai et al. (2022), to avoid selection bias, I use publicly available LinkedIn profiles, graduate program websites, personal websites, PandaInUniv (2026), and García Guzmán (2026)'s *The Econ PhD placements dataset* to obtain outcomes data for all applicants, not just matriculants. The minority of applicants whose outcomes data are not available receive the lowest possible ranking of 150.

Regarding independent variables from the application files, I distinguish objective variables from subjective variables. Objective variables are those variables that the application software

[2] I use an average of the rankings provided by ChatGPT 5.5, Claude Opus 4.7, and Gemini 3 in temporary chat mode. The prompt and a link to screenshots of the outputs are available in Appendix A.

compiles into a spreadsheet for the committee's convenience, whereas subjective variables are those that the applicant uploads as pdf files that cannot be easily converted into a spreadsheet.[3] Because the admissions committee does not have time to read everyone's file, it uses the objective variables to screen out a predetermined percentage of the applicants before reading any files. For example, it screens out applicants with low quantitative GRE scores. For this reason, the structural models in this paper treat objective variables separately from subjective ones.

Objective variables include demographics, such as application year, age, gender, and a region of the world categorical variable, in which the categories are U.S., International Asian, and International Other. There are also performance variables, such as verbal, quantitative, and analytical GRE percentiles; undergraduate GPA; TOEFL/IELTS scores for non-native English speakers; two binary variables for having a graduate degree and work experience respectively; and a categorical variable for the type of undergraduate institution (elite U.S. university, elite U.S. liberal arts college, other U.S. college/university, elite foreign university, and other foreign university), where elite is always defined as top 50. Moreover, recommender variables, namely the number of professor (Krueger & Wu 2000) and prolific recommenders the applicant has, measure how well-connected the applicant is. A prolific recommender is one who writes a letter for at least one other applicant in the sample, which Bai et al. (2022) find predictive of acceptance and success.

Meanwhile, subjective variables in economics come from recommendation letters, application essays, CVs, and supplemental forms.[4] I use textual processing algorithms on the letters to measure the overall word count and rates of positive, negative, standout, ability, research, grindstone, teaching, communal, and agentic words (Bai et al. 2022). From application essays, the algorithms extract the word count; rates of positive, negative, economics, and research words; and whether the applicant mentioned a professor's name from this university. The last three are my analogues of Grove et al. (2007)'s *Specific Topic*, *Mention Paper*, and *Specific Member*.[5] From CVs, there are the word count, a binary variable for whether the applicant has a publication or working paper, and the number of programming languages the applicant listed. Jones et al. (2020) find that economics committee members, especially at high-ranking programs, do not put much weight on coding skills. But given the increasing prevalence of empirical research that uses techniques like webscraping and machine learning, along with the increasing use of AI tools in economics, such skills have become more commonplace. There is also a supplemental form, in which the applicant lists the number of math courses they took, as well as the names of advanced

[3] Bai et al. (2022) define objective variables as "verifiable performance measures" of applicant quality and subjective variables as committee member assessments. For most variables, our definitions coincide. While they do not categorize demographics, such as gender and nationality, I categorize demographics as objective variables.

[4] The department saved 430 of the applicants' textual documents, 102 of which were not transitioned to Round 2. For those without textual documents, I perform mean imputation and set *Missing Application Pdf* $= 1$.

[5] Grove et al. (2007)'s *Specific Topic* is a binary variable that equals 1 if the essay contains at least one economics topic. *Mention Paper* is a binary variable that equals 1 if the essay mentions any paper (including papers like undergraduate term papers that are not publications or working papers), and is designed to indicate a "demonstrated interest in doing economics research." Finally, *Specific Member* is the same as my binary specific professor variable.

math courses such as real analysis. In addition, I have a categorical variable of the applicant's indicated field of interest (e.g. microeconomics, macroeconomics, econometrics, etc.).

Finally, for applicants whose files the committee read, I have the subjective score from 0 (worst) to 10 (best) that their assigned faculty reader gave them relative to the scores that he or she gave the other applicants assigned to him or her.[6] This score is a proxy for information that the committee can observe but I cannot. Note that the committee can choose how heavily to weight the score in its decision-making process because the scores come from individual readers who do not have the sole authority to make accept vs. reject decisions. The committee score is available for only the applicants whose files were read, and the other subjective variables from the textual documents are available for about one-fifth of applicants. Per the 2015–2019 committee chair, the availability of these documents is mostly random, though the fraction of Round 1 survivors is higher for the applicants with available documents than for the full application pool.

## 2.2 Synthetic Data Protocol

In this section, I describe the synthetic data protocol used to protect the applicants' privacy, which I employ in conjunction with a university administrative office that is responsible for stewarding the data. While a handful of recent papers, such as Stanley and Totty (2024) and Carr, Wiemers, and Moffitt (2025), use synthetic data, the synthetic data in those papers had already been made available to researchers by the U.S. Census Bureau. In contrast, this paper's protocol covers all the steps from creation of the synthetic data to obtaining the final results on the actual data.

The steps of the protocol are as follows. First, the administrative office receives the data from the graduate admissions office and individual departments, such as economics. They remove all direct identifiers, such as names and email addresses, and run algorithms I provide to extract variables from textual documents, such as recommendation letters. Second, they use Patki et al. (2016)'s publicly-available Synthetic Data Vault (SDV) to generate the synthetic data. The SDV estimates a Gaussian copula to capture correlations between variables and then simulates a new, synthetic dataset from the copula using a random number generator. Crucially, the rows in the synthetic dataset do not correspond to real subjects, which prevents identity disclosure (Chapelle & Falissard 2023). The SDV works best when the final versions of the variables are already created (e.g. turning a list of country names into a categorical variable), so without seeing any actual data, I work with the office to create the variables before implementing the SDV.

However, there remains the possibility of attribute disclosure. For example, if an applicant from an unusual country has an unusually low GRE percentile, information about that applicant may leak into the synthetic dataset. As such, if any cell in the synthetic data has fewer than five other cells from that variable with the same value, the variable's observations are binned and in

[6] Some readers ranked their assigned applicants instead of scoring them from 0 to 10. For those readers, the committee chair converted their rankings into scores to ensure all applicants whose files were read received scores.

each bin, replaced with the bin's sample mean. Only at this point do I receive the synthetic data. I use the data to finalize my estimation code, which I then give to the administrative department. The department takes that code, runs it on the actual data, and returns the parameter estimates.

# 3 Regression Analysis

In this section, I present regression results for the economics department dataset with dependent variables for acceptance, matriculation, and success. For success, I have two dependent variables. The first variable is a binary variable for a top 100 or top 100 equivalent placement. *Top 100 Placement* $= 1$ for applicants who attained a top 100 U.S. placement, an international placement that maps to a top 100 U.S. placement, or a non-academic placement with a top 100 academia-equivalent ranking. Per Table 8 in Appendix B, *Top 100 Placement* $= 1$ for 19.71% of applicants vs. 34.18% of matriculants, indicating that the committee's matriculation set is much better than random chance would predict. The second variable is *Placement Rank*, which ranges from 1 to 150. For applicants whose placements I could not find, I set *Top 100 Placement* to 0, *Placement Rank* to 150, and *Missing Placement Rank* to 1. Likewise, for applicants whose PhD program I could not find, I set *Program Rank* to 150 and *Missing Program Rank* to 1.

Meanwhile, for the subjective variables in recommendation letters, I use principal component analysis to reduce the dimensionality of the letters part of the state space from 9 to 2. The loadings are in Table 1, and they show that PC1 can be interpreted as the degree to which the letters contain words from the dictionaries in general, with an emphasis on positive and ability words. Per Table 9 in Appendix B, the correlation between PC1 and the rate of positive words is 0.8114, and that between PC1 and the rate of ability words is 0.7486. Meanwhile, PC2 can be interpreted as the degree to which the letters are factual and unflattering. Its loadings are positive for negative, research (largest positive loading), grindstone, and teaching words and negative for positive, standout (largest negative loading), ability, communal, and agentic words.

Table 2 presents selected AMEs from the regressions,[7] and the full regression results are in Tables 10a and 10b in Appendix B.[8] I find that different independent variables best predict each dependent variable. For *Accept*, *Committee Score* is the best predictor; a one point increase in *Committee Score* makes an applicant 7.88 percentage points more likely to be accepted, and including it increases the Pseudo-$R^2$ from 0.2957 to 0.8108. However, other significant predictors of acceptance include demographics, *GRE Quantitative Percentile*, *Work Experience*, and *CV Coding*. Interestingly, *CV Coding*'s average marginal effect (AME) is negative, meaning that for each additional coding language an applicant indicates on their CV, they are 0.44 percentage

[7] The columns of Table 2 are the second, third, fifth, and sixth columns of Table 10b in Appendix B. The rows are the independent variables, excluding year dummies, with at least one AME significant at the 10% level.

[8] In Tables 11a and 11b in Appendix B, I present the regression results on synthetic data. For some variables, the synthetic estimates are similar to the actual estimates, but the synthetic estimates are generally less significant.

points less likely to be accepted. Despite the reduced sample size for *CV Coding*, it is still significant at the 10% level. Perhaps having more coding languages on a CV is associated with being from more of an industry background, which the committee might find less appealing.

Table 1: Principal Component Loadings

| | PC1 | PC2 |
|---|---|---|
| Positive | 0.5044 | -0.1769 |
| Negative | 0.1479 | 0.3759 |
| Standout | 0.2115 | -0.5082 |
| Ability | 0.4667 | -0.0177 |
| Research | 0.1214 | 0.6418 |
| Grindstone | 0.3879 | 0.3075 |
| Teaching | 0.3708 | 0.0743 |
| Communal | 0.1284 | -0.1983 |
| Agentic | 0.3768 | -0.1324 |
| Proportion of Variance | 0.2763 | 0.1456 |

Meanwhile, women are 1.63 percentage points more likely to be accepted, and applicants from East or South Asia are 2.14 percentage points less likely to be accepted. While *Female* and *International Asian* do not significantly predict success, I will show that the committee's decision rule with respect to these two demographics does not negatively affect the average success probability of the acceptance or matriculation set. Therefore, if this decision rule is part of a diversity objective, it is nonetheless still also consistent with a success-maximizing objective.

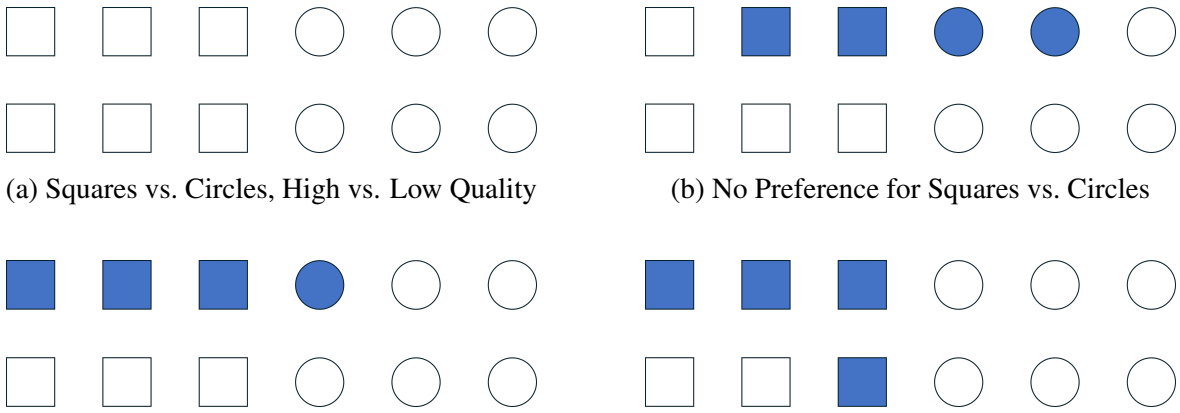
Figure 1 illustrates three different cases of a committee pursuing a diversity objective, where the demographics are squares and circles, and the applicants are high or low quality. Squares are the historically underrepresented demographic, so a diversity-inclined committee prefers them. In the first case, the committee is not diversity-inclined, so it accepts two high-quality squares and two high-quality circles. In the second case, it has a small preference for squares, so it accepts three high-quality squares and one high-quality circle. Because all four acceptances are high quality, the committee's decision rule is still consistent with maximizing the quality of its acceptance set. However, in the third case, the committee has a large preference for squares, forcing it to accept a low-quality square over a high-quality circle. In this case, the committee's decision rule is inconsistent with quality maximization. I find no evidence that the committee featured in this paper lands in the third case with respect to any observed demographic.

Meanwhile, predictors of acceptance often negatively predict matriculation for applicants who are accepted. For example, *International Asian*'s AME is significantly negative for *Accept* but significantly positive for *Matriculate*. In contrast, *GRE Quantitative Percentile*'s AME is significantly positive for *Accept* but significantly negative for *Matriculate*. Presumably, applicants with high GRE quantitative scores have more outside acceptances and so are less likely to matriculate at this university. However, *Work Experience* is significantly positive for both *Accept* and *Matriculate*, whereas *CV Coding* is significantly negative for both. That said, the best predictor

Table 2: Selected Logistic Regression AMEs

| Dependent Variable | Accept | Matriculate | T100 Placement | T100 Placement |
|---|---|---|---|---|
| Age | -0.0001 | -0.0223** | 0.0037 | 0.0053* |
| | (0.0011) | (0.0108) | (0.0031) | (0.0030) |
| Female | 0.0163** | -0.0490 | -0.0157 | -0.0032 |
| | (0.0068) | (0.0534) | (0.0172) | (0.0158) |
| U.S. | -0.0013 | 0.1472* | 0.0213 | 0.0192 |
| | (0.0084) | (0.0864) | (0.0291) | (0.0282) |
| International Asian | -0.0214*** | 0.1644** | 0.0298 | -0.0127 |
| | (0.0073) | (0.0691) | (0.0213) | (0.0195) |
| GRE Quantitative Percentile | 0.0011** | -0.0147** | 0.0028** | 0.0006 |
| | (0.0005) | (0.0063) | (0.0011) | (0.0011) |
| Undergraduate GPA | 0.0195 | -0.1026 | -0.0721 | -0.1096** |
| | (0.0235) | (0.1820) | (0.0587) | (0.0558) |
| TOEFL/IELTS | -0.0022 | -0.0631 | 0.0219 | 0.0409* |
| | (0.0111) | (0.0776) | (0.0238) | (0.0221) |
| Work Experience | 0.0140** | 0.1591** | 0.0271 | 0.0358** |
| | (0.0064) | (0.0700) | (0.0194) | (0.0178) |
| #Prolific Recommenders | 0.0039 | -0.0426 | 0.0137* | 0.0076 |
| | (0.0031) | (0.0285) | (0.0081) | (0.0074) |
| Committee Score | 0.0788*** | -0.0371 | 0.0144** | 0.0007 |
| | (0.0045) | (0.0322) | (0.0069) | (0.0062) |
| Advanced Math | 0.0246 | -0.1950** | -0.0317 | -0.0287 |
| | (0.0168) | (0.0793) | (0.0402) | (0.0372) |
| Letters Terms (%) PC1 | 0.0055 | 0.0671** | 0.0038 | 0.0159 |
| | (0.0054) | (0.0283) | (0.0114) | (0.0115) |
| CV Coding | -0.0044* | -0.0414** | 0.0108 | 0.0125 |
| | (0.0026) | (0.0193) | (0.0081) | (0.0082) |
| Essay Economics Terms (%) | 0.0001 | 0.0366** | -0.0175** | -0.0104 |
| | (0.0016) | (0.0162) | (0.0081) | (0.0078) |
| Essay Positive Terms (%) | -0.0013 | -0.0072 | 0.0118* | 0.0123* |
| | (0.0019) | (0.0189) | (0.0070) | (0.0068) |
| Missing Committee Score | -0.0096 | | -0.0528** | -0.0277 |
| | (0.0120) | | (0.0212) | (0.0191) |
| Open House | | 0.2615*** | | |
| | | (0.0512) | | |
| Program Rank | | | | -0.0011*** |
| | | | | (0.0003) |
| Missing Program Rank | | | | -0.3965*** |
| | | | | (0.0512) |
| Sample | All | Accept = 1 | All | All |
| Observations | 2329 | 257 | 2329 | 2329 |
| Pseudo-$R^2$ | 0.8108 | 0.2877 | 0.0353 | 0.2430 |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.



(a) Squares vs. Circles, High vs. Low Quality

(b) No Preference for Squares vs. Circles

(c) Small Preference for Squares

(d) Large Preference for Squares

Figure 1: Diversity Objective

of matriculation is whether the accepted applicant attended the department's open house. Those who attended were 26.15 percentage points more likely to matriculate than those who did not. Attending the open house is a positive signal that the applicant is interested in this university.

Finally, the best predictor of success is *Program Rank*. Applicant characteristics equal, attending a PhD program ranked 10 places better (i.e. 10 places closer to rank 1) increases the probability of a top 100 placement by 1.1 percentage points and improves the expected placement rank (i.e. toward rank 1) by 0.8 places. However, other significant predictors include *Age*, *TOEFL/IELTS*, *Work Experience*, and *Essay Positive Terms (%)*.[9] Interestingly, *Age*, *TOEFL/IELTS*, and *Work Experience* are significant only when controlling for *Program Rank*. This finding aligns with a scenario in which other universities underweighted those variables, resulting in such applicants attending lower-ranked programs, which may in turn have decreased their chances of success. Such a scenario would be in accordance with evidence that PhD program quality causally affects success via school value-added (Angrist et al. 2026).[10] Consistent with this scenario, *Work Experience* significantly predicts matriculation among accepted applicants since many of them may have lacked better outside offers.[11] In turn, omitting *Program Rank* could bias those AMEs downward and conceal the true effect of those variables on success. Conversely, if *Program Rank* also reflects ability I do not observe, then controlling for *Program Rank* could bias those AMEs upward, by the same logic I apply to GPA below.

Specifically, a high *Undergraduate GPA* significantly predicts failure when controlling for

[9]The age distribution of applicants drops off sharply past their early 30s to the point that a quadratic *Age* term is unidentified. To avoid extrapolating *Age*'s positive AME to the small fraction of much older applicants, I winsorize *Age* at its 95th percentile, which is 33 years old. Also, I use percentiles to convert TOEFL's 0–120 scores to IELTS's 1–9 scores, ensuring that both tests are on the same scale before combining them into one variable.

[10]Angrist et al. (2026) find program quality matters even after controlling for their version of *Committee Score*.

[11]In contrast, age, which is positively correlated with work experience, negatively predicts matriculation, which, holding work experience fixed, could reflect the higher opportunity cost of leaving an established career.

*Program Rank*. If GPA does not provide much signal for success, which is plausible since GPAs are not standardized across institutions, but admissions committees in general select positively on them, then applicants with equal observed characteristics who were selected into the same program tier may exhibit a negative correlation between GPA and success due to unequal unobserved characteristics. However, this paper's structural models are meant for pre-selection decision-making, so I follow Bai et al. (2022) and remove GPA after this section. Meanwhile, *GRE Quantitative Percentile*, *#Prolific Recommenders*, and *Committee Score* significantly predict success only when not controlling for *Program Rank*. If other universities select on similar criteria as this university, then these variables may predict success partly by predicting acceptance into higher-ranked programs, in which case controlling for *Program Rank* would be necessary to avoid omitted variable bias. But conversely, by the same logic I apply to GPA, if *Program Rank* also reflects ability I do not observe, controlling for it could bias their AMEs downward.

In Table 12 of Appendix B, I present lasso logistic regression results with 5-fold cross-validation for variable selection. For acceptance and matriculation, lasso selects most of the variables, but it selects fewer variables for success. In the table's sixth column, with *Top 100 Placement* as the dependent variable and controlling for *Committee Score* and *Program Rank*/*Missing Program Rank*, lasso selects the following ten variables: *Age*, *Undergraduate GPA*, *TOEFL/IELTS*, *Work Experience*, *Letters Terms (%) PC1*, *Essay Economics Terms (%)*, *Essay Positive Terms (%)*, *Missing Committee Score*, *Program Rank*, and *Missing Program Rank*.

Readers familiar with Table 18 of Bai et al. (2022) may notice that their success regressions without controlling for PhD program rank fit substantially better than this paper's success regressions without *Program Rank* and *Missing Program Rank*. It is only after including *Program Rank* and *Missing Program Rank* that the Pseudo-$R^2$s in Table 2 become comparable to those in Bai et al. (2022), though it is worth noting that adding PhD program rank controls does not further increase their Pseudo-$R^2$s by much. This pattern occurs because they define success as attaining a top 10 or top 20 assistant professorship. It is very hard to attain such a professorship without attending a top-ranked PhD program. Per their Table 17, 83% of top 10 assistant professor placements in their data came from top 10 economics PhD programs, an additional 8% came from non-economics PhD programs which may have also been top 10 in their disciplines, and only one applicant attained a top 10 placement from a PhD program ranked lower than this paper's university. It is also the case that variables in application files, such as GRE scores, predict acceptance at top PhD programs. Therefore, their application file variables predict placement indirectly by predicting program rank, and adding program indicators contributes little because the file has already revealed most of that information. In contrast, while attending a PhD program is necessary for a top 100 or top 100-equivalent placement, attending a top-ranked program is not. Graduates of programs across many tiers attain it, in part because non-academic research institutions (e.g. IMF, Federal Reserve, etc.) account for many of this paper's top 100 placements. As such, application file variables do not indirectly predict success as they do in Bai et al. (2022), making it necessary to control for program rank to increase the Pseudo-$R^2$ in Table 2.

I briefly address inference vs. prediction, as well as sample selection. Both are important when running regressions with success measures as dependent variables. Regarding inference vs. prediction, omitted independent variables make it challenging to obtain consistent estimates in a success regression. For example, I do not observe the effort that students put in after matriculating. Even effort, let alone any other variables about the applicants that I do not observe, can bias success parameter estimates upward if effort is correlated with the variables corresponding to those parameters and correlated with success irrespective of those variables.

However, there are reasons for reassurance. First, I incorporate *Committee Score* as an independent variable, which proxies variables that the committee observes and I do not. I also follow Bai et al. (2022) and include as a proxy of effort the number of "grindstone terms," such as depend* and diligen*, in recommendation letters. In addition, in this paper, prediction matters as much as inference. Mullainathan and Spiess (2017) distinguish $\hat{\beta}$ (e.g. the effect on educational outcomes of hiring an additional teacher) from $\hat{y}$ (e.g. which teacher to hire) problems. For graduate admissions, if *Work Experience* predicts success, then the committee should consider it regardless of whether the committee knows the mechanism through which it affects success.

Moreover, the determinants of success conditional on applying (see Krueger and Wu 2000, and Grove and Wu 2007) are different from those conditional on matriculating (see Athey et al. 2007). Bai et al. (2022) prove that if $P(\text{Success}|\text{Skill})$ is an S-curve, then $\hat{\beta}$ will be smaller for more selected samples, a theoretical finding consistent with the significance of the quantitative GRE being weaker in studies on matriculants only. This problem is akin to height not predicting success in a sample of professional basketball players. Like Bai et al. (2022), I collect placement outcomes for all applicants, not just those who attended this university. But there are other measures of success, such as course grades and passing comprehensive examinations, that are only available for matriculants, and that have been used in papers like Athey et al. (2007).

As such, for these measures, one can implement a Heckman (1979) correction. Notationally, let $y_n^* = x_n\beta + u_n$, $d_n = \mathbb{1}(z_n\gamma + v_n > 0)$, and $y_n = d_n y_n^*$. $y$ is a measure of success, $x$ are variables in applications, and $d$ is matriculation. The first step is to run a probit regression of $d$ on $z$ and compute the inverse Mills ratio $\hat{\lambda}_n = \frac{\phi(z_n\hat{\gamma})}{\Phi(z_n\hat{\gamma})}$. The second step is to run OLS or heckprobit of $y$ on $(x, \hat{\lambda})$ to obtain $\hat{\beta}$. For a Heckman instrument, one needs a variable that affects $d$ but not $y$. An example is a variable measuring the applicant's enthusiasm about the program, such as whether they included a specific professor's name in their application essay. In Section 4, I discuss the implementation of Heckman corrections in a subset of this paper's structural models.

In addition, in Online Appendix A, I implement a Wald test of whether the determinants of acceptance are the same across this university's graduate programs, and I find that they are not the same. Could I similarly implement a Wald test of whether the determinants of acceptance are the same as those of success within a program? For example, does the quantitative GRE have the same AME for acceptance as for success? Intuitively, provided the committee wants to select applicants based on a "success" model, predictors of admission only may be overvalued, and vice versa. Such a test would not work, however. First, there may be unwanted rejections of the

null. For example, the committee may value diversity if it does not hurt success probabilities too much, in which case it may favor demographics not strictly more likely to succeed. There may also be unwanted non-rejections. If there is discrimination in the job market, the committee may rationalize discriminating in the same way, which would make the test not reject the null.

But more importantly, this test would not account for the mechanics of the admissions process. As a simple example, suppose the following conditions hold. First, *Work Experience* has a large positive effect on success, and *Graduate Degree* has a smaller positive effect. Second, 25% of applicants have both characteristics and therefore the highest success probability, 25% have *Work Experience* only (second-highest), 25% have *Graduate Degree* only (third-highest), and 25% have neither (lowest). Third, this university accepts half the applicants, and the other half attend a lower-ranked program, which ensures that everyone's outcomes are observable. Fourth, unlike applicant quality, program quality does not affect outcomes, at least in this example.

Under these suppositions, an optimal committee will accept the applicants with *Work Experience* and ignore *Graduate Degree*. As such, an acceptance regression will have a degenerate *Work Experience* AME of 1 and a *Graduate Degree* AME of 0, while a success regression will have AMEs as follows: $1 >$ *Work Experience* $>$ *Graduate Degree* $> 0$. Also, for the acceptance AMEs to equal the success AMEs, the committee would suboptimally have to accept applicants without work experience. Therefore, unlike with, for instance, wages equaling marginal revenue product in a competitive labor market which allows for cross-regression comparisons, the acceptance and success AMEs need not be equal for the committee to be optimal. In turn, to draw conclusions about optimality, the relationship between the AMEs must be interpreted through models of the committee's selection problem, which are the basis for Section 4.

# 4 Structural Analysis

There are two main reasons to estimate structural models of the admissions committee's decision problem. First, in order to make any claims about whether committees are optimizing, it is necessary to define what optimization is by explicitly modeling their problem. Second, structural models can uncover feasible counterfactual deviation decision rules to help committees admit stronger applicants, and the models can quantify any deviation gains from such rules.

The question then is how to set up the models. Bai et al. (2022) prove that if all the variables in an application file can be reduced to a single number (i.e. an expected success probability),[12] then accepting the optimal subset of applicants is the same as using a cutoff rule. A cutoff rule problem is far more tractable than a portfolio choice problem because I do not have to consider all the subsets of the hundreds of applicants in each year who can be accepted; I only have to rank the applicants and choose the ones above the cutoff. Because committees may be more selective

[12]The assumption that committees select based on expected success may have become more realistic in light of the 2023 Supreme Court ruling against race-based affirmative action in *Students for Fair Admissions v. Harvard*.

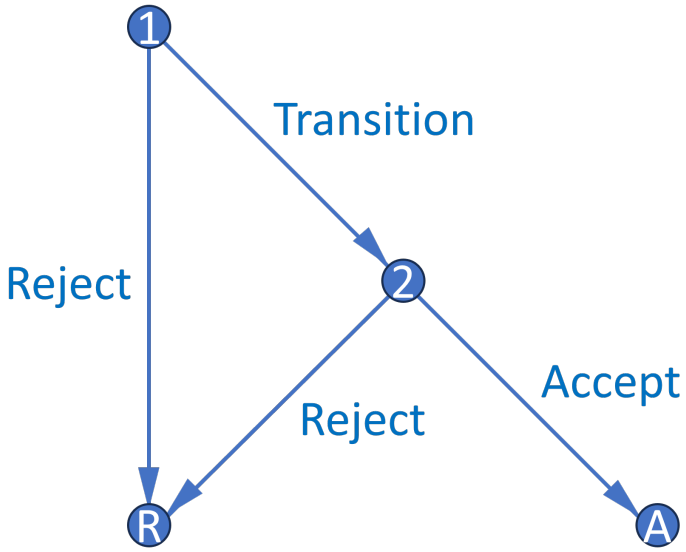



Figure 2: Decision Graph

in some years than others, I allow the cutoff probability to vary across years.

After generalizing into a two-round dynamic framework Bai et al. (2022)'s proof that the committee's portfolio choice problem reduces to a cutoff rule problem, I start by modeling the committee's problem as follows. Per Figure 2, the committee observes in Round 1 only each applicant's objective variables, such as GRE scores. Based on those variables, it can transition the applicant to Round 2, or it can issue an immediate rejection. If the applicant survives Round 1, the committee then takes the time to read the application file and access all variables, both objective and subjective (e.g. recommendation letters). Finally, it accepts or rejects the applicant.

Rather than setting up a two-round model, why do I not rank the applicants based on their success probabilities conditional on all data and compare this one-round model's ranking to the committee's? The reason is that a one-round model is unfair to the committee because the committee does not have time to read everyone's files (though in the future, perhaps committees can use textual processing algorithms or local LLMs to "read" files quickly and generate additional objective variables without too much effort). Moreover, two-round models are useful not just for graduate admissions but for any other setting with first-stage screening, such as hiring, etc.

I start with two different two-round models: dynamic and static. In Round 2 of the dynamic model, the committee accepts the applicants who survived Round 1 and have success probabilities greater than the Round 2 cutoff probability. In Round 1, it forms a conditional expectation about those who survive to Round 2, meaning that the Round 1 cutoff is the marginal transitioned applicant's expected Round 2 value. As such, the Round 1 cutoff can take a value greater than one. In contrast, in the static model, the committee ranks the applicants based on their objective variables, reads the files of those whose expected success probabilities are above the first cutoff

probability, re-ranks the surviving applicants based on all their data, and accepts those above the second cutoff probability. I call this model static because each round is a separate static problem. The static model is useful for its simplicity and interpretable cutoff probabilities in both rounds.

The dynamic model appears more sophisticated because in Round 1, it favors applicants whose objective variables predict good subjective variables. However, the static model also correctly favors those applicants. While the static Round 1 success probability parameter estimates are inconsistent because the subjective variables are omitted, the fitted values estimate the success probability averaged over the subjective variables given the objective ones. Therefore, the component of the subjective variables that is predictable from the objective variables is absorbed into the estimates. Only the myopic model, which is identical in setup to the dynamic model except that the parameters for the objective variables in the distribution of the subjective variables are restricted to zero, does not favor such applicants in Round 1. The static model also better describes what real-world committees do, as they do not solve Round 2 conditional expectations.

Later in this section, when I present maximum likelihood estimation results of these models, I include the Akaike Information Criterion (AIC) $= 2[K - \log(\hat{\mathcal{L}}^R)]$, which can be used to compare each restricted model with its unrestricted counterpart. $K$ is the number of parameters, and $\hat{\mathcal{L}}^R$ is the restricted likelihood's value; lower AIC values are preferred. AIC values are comparable only between the restricted and unrestricted specifications of the same model. The reason is that the static objective is a composite likelihood over decisions and placement outcomes, with every outcome entering once in each of the Rounds 1 and 2 success blocks, whereas the dynamic likelihoods also include the densities of the subjective variables conditional on the objective variables.

Still later in this section, I present dynamic and static models in which not every accepted applicant matriculates, and which are also derived from two-round portfolio choice models, albeit models in which the objective is to maximize the sum of the success probabilities of the matriculants rather than the acceptances. In Round 2, the committee still accepts the applicants who survived Round 1 and have success probabilities greater than the Round 2 cutoff probability. No weight is placed on matriculation probabilities in Round 2 because while likely matriculants are more likely to produce a positive payoff, they are also more likely to take up cohort spots and so are more "expensive." However, weight is placed on matriculation probabilities in Round 1 because if the committee chooses to read the files of only the applicants who are most likely to succeed, it could find itself in a situation in Round 2 in which the only accepted applicants who matriculate are those with strong objective variables and weak subjective variables. This situation becomes increasingly likely the lower-ranked the committee's university is.

## 4.1 Portfolio Choice to Cutoff Rule

Bai et al. (2022) prove that selecting the best $N_A$ applicants out of $N$ is equivalent to selecting

the $N_A$ whose success probability exceeds a cutoff.[13] Their proof covers a single round in which every accepted applicant matriculates, and in this section, I extend the result to two rounds, matriculation, and preference shocks. Let $x$ be a vector of objective variables (e.g. GRE scores), and $z$ be a vector of subjective variables (e.g. variables from recommendation letters). Also, let $d_n$ denote the accept or reject decision for the $n$th applicant, and $y_n$ denote that applicant's outcome, where $T =$ transition, $A =$ accept, $R =$ reject, $S =$ success, and $F =$ failure. The committee solves a constrained optimization problem, in which it maximizes the sum of the success probabilities of the accepted applicants under the constraint that it cannot accept more than $N_A$ applicants. For simplicity, suppose there is one round in one year, and all accepted applicants matriculate:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} P(y_n = S|x_n,z_n)\mathbb{1}(d_n = A), \text{ where } \sum_{n=1}^{N} \mathbb{1}(d_n = A) \leq N_A. \tag{1}$$

Because success probabilities are never negative, the unconstrained objective increases every time $d_n = A$, which means that the constraint binds. The Lagrangian with multiplier $\lambda$ is:

$$\begin{aligned} \mathcal{L}(d_1,\ldots,d_N;\lambda) &= \sum_{n=1}^{N} P(y_n = S|x_n,z_n)\mathbb{1}(d_n = A) + \lambda\left\{N_A - \left[N - \sum_{n=1}^{N} \mathbb{1}(d_n = R)\right]\right\} \\ &= \sum_{n=1}^{N} \left[P(y_n = S|x_n,z_n)\mathbb{1}(d_n = A) + \lambda\mathbb{1}(d_n = R)\right] + \lambda(N_A - N). \end{aligned} \tag{2}$$

Suppose without loss of generality that the applicants are sorted in decreasing order such that $P(y_1 = S|x_1,z_1) \geq \cdots \geq P(y_N = S|x_N,z_N)$. To ensure that the committee accepts exactly $N_A$ applicants, it must be the case that $\lambda^* \in \left[P(y_{N_A+1} = S|x_{N_A+1},z_{N_A+1}), P(y_{N_A} = S|x_{N_A},z_{N_A})\right]$. That is, in this setup, $\lambda^*$ is set identified.[14] Then the solution is to accept all applicants whose success probability exceeds $\lambda^*$ and reject the rest, which means that $\lambda^*$ is a cutoff probability:

$$\forall n \in \{1,\ldots,N\},\ d_n(x_n,z_n|\lambda^*) = \begin{cases} A & \text{if } P(y_n = S|x_n,z_n) > \lambda^* \\ \{A,R\} & \text{if } P(y_n = S|x_n,z_n) = \lambda^* \\ R & \text{if } P(y_n = S|x_n,z_n) < \lambda^*. \end{cases} \tag{3}$$

If $P(y_n = S|x_n,z_n) = \lambda^*$, then the committee is indifferent between acceptance and rejection, which means that in the event of a tie at the margin, the committee will randomly accept enough applicants to reach $N_A$ acceptances. Also, while $\lambda^*$ is set identified in this setup, I show in the proof of Theorem 2 in the next subsection how to point identify it with admissions data by modeling a committee that compares each applicant's success probability to a single-valued cutoff,

[13] To relax the assumption that the committee only cares about success, its objective can be generalized to maximizing the sum of a convex combination of success probabilities and other factors, such as diversity.

[14] If there were a continuum of applicants as in Bai et al. (2022)'s proof, the set's length would be 0, and so $\lambda^*$ would be point identified. But all the models in this paper have $N$ applicants, as there are in the data.

and then by exploiting the observed committee's exact selectivity. Finally, given a large number of applicants, $\lambda^*$ can be interpreted as the marginal accepted applicant's success probability. Therefore, I denote it going forward as $P(y^* = S)$, or the success probability of that applicant.

This version of the portfolio choice problem contains just one round. But the committee's actual problem has two rounds: first-round screening on $x$ only, along with second-round final selections on $x$ and $z$. In Appendix D.1, I generalize, for both the dynamic and static setups, the one-round portfolio choice problem to a two-round portfolio choice problem, and show that it also reduces to a two-round cutoff rule problem. The Round 2 cutoff rule problem, in which the committee chooses among the $N_T$ applicants who survive Round 1, is:

$$\max_{(d_{2,1},\ldots,d_{2,N_T})\in\{A,R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S|x_n,z_n)\mathbb{1}(d_{2,n} = A), \text{ where: } \sum_{n=1}^{N_T} \mathbb{1}(d_{2,n} = A) \leq N_A. \tag{4}$$

Its solution with $\lambda_2^* = P(y_2^* = S) \in \left[P(y_{N_A+1} = S|x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S|x_{N_A}, z_{N_A})\right]$ is:

$$\forall n \in \{1,\ldots,N_T\},\ d_{2,n}(x_n,z_n|\lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S|x_n,z_n) > \lambda_2^* \\ \{A,R\} & \text{if } P(y_n = S|x_n,z_n) = \lambda_2^* \\ R & \text{if } P(y_n = S|x_n,z_n) < \lambda_2^*. \end{cases} \tag{5}$$

I show in Appendix D.1 that under an independence assumption, the set-identified interval for $\lambda_2^*$ collapses to a point, denoted $\overline{\lambda_2^*}$. Let $EV_{2,n}(x_n) = \mathbb{E}[\max\{P(y_n = S|x_n,z_n), \overline{\lambda_2^*}\}|x_n] = \int_{\widetilde{z}_n} \max\{P(y_n = S|x_n,\widetilde{z}_n), \overline{\lambda_2^*}\} dF_{z|x}(\widetilde{z}_n|x_n)$. Then the Round 1 portfolio choice problem is:

$$\max_{(d_{1,1},\ldots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} EV_{2,n}(x_n)\mathbb{1}(d_{1,n} = T), \text{ where: } \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = T) \leq N_T. \tag{6}$$

Its solution with $\lambda_1^* = L(y_1^* = S) \in \left[EV_{2,N_T+1}(x_{N_T+1}), EV_{2,N_T}(x_{N_T})\right]$ is:

$$\forall n \in \{1,\ldots,N\},\ d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T,R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^*. \end{cases} \tag{7}$$

That said, it is possible to add the preference shocks directly to the portfolio choice problem and carry them through the derivation, in which case the result is exactly the cutoff rule problems in this section. For simplicity, I do not do so throughout this paper, but I prove in Appendix E.1 that for the one-round portfolio choice problem without matriculation (i.e. the portfolio choice problem I first presented), the preference shocks can be carried through the derivation:

**Theorem 1.** *For large N, the committee's portfolio choice problem with preference shocks has a cutoff rule solution with logistic conditional choice probabilities.*

**Proof.** See Appendix E.1.

### 4.1.1 Dynamic Model

The dynamic model's notation is as follows. $r \in \{1,2\}$ is the round, $x$ is a vector of objective variables, $z$ is a vector of subjective variables, $t$ is the year, $\bar{k}$ is this university's U.S. News program rank ($k$ is the applicant's observed future program rank), and $\varepsilon_r$ is a type 1 extreme value (T1EV) shock with scale $\sigma$. The shocks let the model fit the data, as a real-world committee will not always accept the applicants with the highest success probabilities.

But what do the shocks represent? Mechanically, $\varepsilon_r$ is the part of the committee's decision that the observables $(x,z,t,\bar{k})$ do not explain. It is convenient to call $\varepsilon_r$ a preference shock, but the label should not be read as pure idiosyncratic taste. The shock reflects my ignorance of what the committee conditioned on, not randomness in the committee's behavior, and is therefore a residual defined relative to the observables. As such, anything the committee responded to that is absent from the observables is folded into $\varepsilon_r$, which means that what $\varepsilon_r$ contains is an empirical matter. The more of the committee's information I observe, the closer $\varepsilon_r$ comes to genuine taste, and the more I do not observe, the more $\varepsilon_r$ stands in for private information that the committee acted on. This gap matters most when such variables also predict success, since the model then omits determinants of success. I discuss private information and its implications in Section 5.

Starting in Round 2, the committee chooses $d_2 \in \{\text{Accept}, \text{Reject}\}$. If it accepts an applicant, it gets the applicant's success probability $P(y = S|x,z,t,\bar{k})$; the probability of succeeding at this university is a function of applicant characteristics, the year, and this university's quality. Whereas if the committee rejects the applicant, it gets the Round 2 cutoff probability $P(y_2^* = S|t)$, which is the success probability of $t$'s marginal accepted applicant, whose success probability is conditional on $(x,z,t,\bar{k})$. There are various measures of success, from completing the program to obtaining a job placement above a threshold ranking. The Round 2 value function is:

$$v_2(x,z,t,\bar{k},\varepsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x,z,t,\bar{k},d_2) + \varepsilon_2(d_2), \text{ where:}$$
$$u_2(x,z,t,\bar{k},d_2) = P(y = S|x,z,t,\bar{k})\mathbb{1}(d_2 = A) + P(y_2^* = S|t)\mathbb{1}(d_2 = R). \tag{8}$$

Meanwhile in Round 1, the committee chooses $d_1 \in \{\text{Transition}, \text{Reject}\}$. If it transitions an applicant to Round 2, it gets $\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}]$, which is the expected value of Round 2, where $z$ is integrated out conditional on $(x,t)$. However, if it rejects the applicant, it gets $L(y_1^* = S|t)$, or the expected Round 2 value of $t$'s marginal transitioned applicant. Because this

value can exceed 1, it cannot be interpreted as a probability. The Round 1 value function is:

$$v_1(x,t,\bar{k},\varepsilon_1) = \max_{d_1 \in \{T,R\}} u_1(x,t,\bar{k},d_1) + \varepsilon_1(d_1), \text{ where:}$$

$$u_1(x,t,\bar{k},d_1) = \mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}]\mathbb{1}(d_1 = T) + L(y_1^* = S|t)\mathbb{1}(d_1 = R), \text{ where:} \tag{9}$$

$$\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}] = \int_{\widetilde{z}} \sigma \log\left( \sum_{\widetilde{d}_2 \in \{A,R\}} e^{u_2(x,\widetilde{z},t,\bar{k},\widetilde{d}_2)/\sigma} \right) dF(\widetilde{z}|x,t).$$

Each round's conditional choice probabilities (CCPs) then have the closed-form expression:

$$P(d_r|x,(z),t,\bar{k}) = \frac{\exp(u_r(x,(z),t,\bar{k},d_r)/\sigma)}{\sum_{\widetilde{d}_r \in D_r} \exp(u_r(x,(z),t,\bar{k},\widetilde{d}_r)/\sigma)}. \tag{10}$$

To estimate this model, I first set up the unrestricted likelihood function. I parameterize the unrestricted CCPs ($P_{d_1}$,$P_{d_2}$), success probability ($P_y$), and cutoffs ($L_{y_1^*}$,$P_{y_2^*}$) with logistic and exponential specifications containing parameters $\theta$, where $\Lambda(u) = e^u/(1+e^u)$ is the logistic function:

$$P_{d_1} = P\big(d_1 = T\big|x,t;\theta_T\big) = \Lambda\big(f_T(x,t)\theta_T\big), \quad P_{d_2} = P\big(d_2 = A\big|x,z,t;\theta_A\big) = \Lambda\big(f_A(x,z,t)\theta_A\big) \tag{11}$$

$$P_y = P\big(y = S\big|x,z,t,\bar{k};\theta_S\big) = \Lambda\big(f_S(x,z,t,\bar{k})\theta_S\big) \tag{12}$$

$$L_{y_1^*} = L\big(y_1^* = S\big|t;\theta_{S_1^*}\big) = \exp\big(f_{S_1^*}(t)\theta_{S_1^*}\big), \quad P_{y_2^*} = P\big(y_2^* = S\big|t;\theta_{S_2^*}\big) = \Lambda\big(f_{S_2^*}(t)\theta_{S_2^*}\big). \tag{13}$$

Each of $f_T$, $f_A$, $f_S$, $f_{S_1^*}$, and $f_{S_2^*}$ is a row vector of an intercept and the variables in its arguments, with $t$ entering as year dummies. As such, each specification is $\Lambda(\cdot)$ or $\exp(\cdot)$ of a linear combination of those variables (Table 7 of Appendix A lists the variables in $x$ and $z$). In particular, $f_{S_1^*}(t)$ and $f_{S_2^*}(t)$ contain only an intercept and year dummies, so each year has its own Round 1 and Round 2 cutoffs, and the exponential and logistic forms serve to keep $L_{y_1^*}$ positive and $P_{y_2^*}$ between 0 and 1. The cutoffs enter only the restricted likelihood, which I describe below. Letting $F_j$ be the conditional CDF for each $z_j$, I parameterize the integral in Equation (9) as:

$$\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}] = \int_{\widetilde{z}_1} \cdots \int_{\widetilde{z}_J} \Big[ \cdot \Big] dF_1(\widetilde{z}_1|\widetilde{z}_2,\ldots\widetilde{z}_J,x,t) \cdots dF_J(\widetilde{z}_J|x,t). \tag{14}$$

The conditional density $f_{z|x,t}$ is the product of one block per $z_j$. Each binary $z_j$ is modeled by a logistic regression on $(x,t)$, so the binary $z$'s are independent given $(x,t)$, and the continuous $z$'s are jointly normal given $(x,t)$ and the binary $z$'s. $\theta_F$ contains the parameters of these blocks. Because $P_y$ depends on the continuous $z$'s only through $f_S(x,z,t,\bar{k})\theta_S$, which is therefore normal as well, I compute the integral in Equation (9) by summing over every configuration of the binary $z$'s and, within each configuration, applying 15-node Gauss–Hermite quadrature to $f_S(x,z,t,\bar{k})\theta_S$.

Let $N_T$ be the number of $N$ total applicants who survive to Round 2. Also, $z^c$ = *Committee Score*, whereas $z^{\neg c}$ = all the other subjective variables. The unrestricted likelihood is then:

$$\begin{aligned}\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} f\big(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_F\big) P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)} \\ \prod_{n=N_T+1}^{N} \big(1 - P_{d_{1,n}}(\theta_T)\big) \prod_{n=1}^{N} f\big(z_n^{\neg c} | x_n, t_n; \theta_F\big) P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}.\end{aligned} \tag{15}$$

There is a complication, namely that *Committee Score* is unobservable for the $N - N_T$ of $N$ applicants who were rejected in Round 1.[15] The missing data pose a problem when estimating $f(z_n | x_n, t_n; \theta_F)$ in the likelihood because *Committee Score* is part of $z$. Because I cannot obtain *Committee Score* for everyone, I tried estimating the following Heckman-corrected likelihood:

$$\begin{aligned}\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \Phi(x_n, t_n; \theta_H) f\Big(z_n | x_n, t_n, \frac{\phi(x_n, t_n; \theta_H)}{\Phi(x_n, t_n; \theta_H)}; \theta_F\Big) P_{d_{1,n}}(\theta_T) \\ P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)} \prod_{n=N_T+1}^{N} \big(1 - \Phi(x_n, t_n; \theta_H)\big)\big(1 - P_{d_{1,n}}(\theta_T)\big) \\ f\Big(z_n^{\neg c} | x_n, t_n, \frac{-\phi(x_n, t_n; \theta_H)}{1 - \Phi(x_n, t_n; \theta_H)}; \theta_F\Big) \prod_{n=1}^{N} P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}.\end{aligned} \tag{16}$$

The probit selection block is $\Phi(x_n, t_n; \theta_H)$, and the inverse Mills ratio is $\frac{\phi(x_n, t_n; \theta_H)}{\Phi(x_n, t_n; \theta_H)}$. However, there are no available Heckman instruments that affect selection but not the distribution of $z$ (year dummies affect $z$ as well as selection). Therefore, identification must come from the nonlinearity of the inverse Mills ratio. This identification is weak, and the corrected results are nearly identical to the uncorrected ones, so the uncorrected likelihood is my preferred specification.

After estimating the unrestricted likelihood, I estimate the restricted likelihood $\mathcal{L}^R$ by substituting in the model's CCPs, which are functions of the success probability $(\theta_S)$ and structural $(\sigma, \theta_{S_1^*}, \theta_{S_2^*})$ parameters. To save computation time, I use $(\theta_F^U, \theta_S^U)$ as an initial guess of $(\theta_F^R, \theta_S^R)$. Because I can observe whether the committee read each applicant's file and whether the committee accepted each applicant, I identify the cutoff parameters $(\theta_{S_1^*}, \theta_{S_2^*})$ for each year by how selective the committee is that year. Without loss of generality, every time $d_2 = A$, the term $\frac{\exp(P(y=S|x,z,t,\bar{k})/\sigma)}{\exp(P(y=S|x,z,t,\bar{k})/\sigma) + \exp(P(y_2^*=S|t)/\sigma)}$ enters $\mathcal{L}^R$. This term decreases in $P(y_2^* = S|t)$, so every time an applicant is accepted, it puts downward pressure on $P(y_2^* = S|t)$, considering the objective is to maximize the likelihood. Intuitively, the committee's selectivity identifies the cutoff probability. Also, if the committee engages in sorting, or selecting the applicants with the highest success probabilities $P(y = S|x, z, t, \bar{k})$, the restricted likelihood improves. Formally:

[15] For those applicants, I perform mean imputation for *Committee Score*. Also, in the restricted Round 2 CCPs, I set their *Missing Committee Score* = 0 so as not to penalize them for not being in the committee's transition set.

Table 3: Restricted Dynamic Cutoffs

| | R1 Cutoff Level | R2 Cutoff Probability | Sigma |
|---|---|---|---|
| 2015 | 0.3731 | 36.58% | |
| | (0.3067, 0.4539) | (29.81%, 43.92%) | |
| 2016 | 0.3795 | 37.21% | |
| | (0.3025, 0.4761) | (29.20%, 45.99%) | |
| 2017 | 0.4126 | 40.27% | |
| | (0.3563, 0.4778) | (34.52%, 46.30%) | |
| 2018 | 0.3836 | 37.68% | |
| | (0.3271, 0.4497) | (31.79%, 43.97%) | |
| 2019 | 0.3887 | 38.14% | |
| | (0.3268, 0.4624) | (31.89%, 44.81%) | |
| Sigma | | | 0.0020 |
| | | | (0.0013) |
| Observations = 2329 | Parameters = 529 | LL = -14131.0 | AIC = 29319.9 |

**Theorem 2.** *For simplicity, suppose there is one round. The maximum likelihood estimate* $\hat{P}_{y^*} = \hat{P}(y^* = S|t)$ *is such that the number of acceptances* $N_A$ *equals the expected* $N_A$*. Also, for any* $\hat{P}_{y^*}$ *and* $N_A$*,* $\mathcal{L}^R$ *is highest when the committee accepts the* $N_A$ *highest* $P_{y_n} = P(y_n = S|x_n, z_n, t_n, \overline{k}_n)$*.*

**Proof.** See Appendix E.1.

When estimating this model, it is necessary to use public outcomes data to track success for all applicants, including those who did not matriculate at this university. Bai et al. (2022) prove that success parameter estimates will be biased toward zero if the sample is restricted to matriculants only. The reason is that all matriculants must have been accepted, and accepted applicants tend to have high values of the determinants of success. For example, conditional on matriculating, the relationship between the quantitative GRE score and success is weakened because nearly all matriculants have very high scores (analogous to height not predicting success in the NBA).

Along with tracking success for all applicants, it is also necessary to track which, if any, graduate program each applicant attended in order to avoid assuming that success depends only on applicant characteristics. For instance, a rejected applicant who succeeded at some other university might not have succeeded at this one, and so the committee may not have made a mistake by rejecting them. If a variable predicts success only through its effect on matriculation at a higher-ranked university, then this university should not use that variable in admissions. In estimation, I use each applicant's observed future program rank $k$ in the success probability block and this university's fixed program rank in the restricted conditional choice probabilities.

In Table 3, I present the dynamic model's estimated cutoffs with 95% confidence intervals, as well as the estimated T1EV scale parameter $\sigma$ and the AIC. While the Round 1 cutoffs are not probabilities, they and the Round 2 cutoffs show that the committee was similarly selective across years. $\hat{\sigma}$ is also small, meaning that the committee's decisions are relatively deterministic with respect to the variables in the application files. A larger set of the unrestricted and restricted parameter estimates is in Tables 13 and 14 of Appendix C. I also present results from simulated data in Tables 7 and 8 of Online Appendix B, which illustrate features I embed in the simulated

data-generating process, namely that the simulated committee's weights are more aligned with the success probability parameters in the optimal simulated dataset than the suboptimal one.

### 4.1.2 Static Model

I now present the static model. Relative to the dynamic model, this model omits $z$ in the Round 1 success probability specification, meaning that $z$'s parameter estimates are inconsistent (though crucially, the fitted values are still correct). In addition, for any multiyear model, such as those in Online Appendix D, the Round 1 value function must be dynamically linked to the Round 2 value function, meaning that the static framework cannot work across years. However, this model has two advantages. First, it more accurately describes how real-world committees behave. Second, the Round 1 cutoff can be interpreted as a probability and compared to the Round 2 cutoff probability. The static model's Round 2 value function is the same as the dynamic's:

$$\begin{gathered} v_2(x,z,t,\bar{k},\varepsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x,z,t,\bar{k},d_2) + \varepsilon_2(d_2), \text{ where:} \\ u_2(x,z,t,\bar{k},d_2) = P(y=S|x,z,t,\bar{k})\mathbb{1}(d_2=A) + P(y_2^*=S|t)\mathbb{1}(d_2=R). \end{gathered} \tag{17}$$

But unlike in the dynamic model, if the committee transitions an applicant in Round 1, it gets $P(y=S|x,t,\bar{k})$, or the applicant's success probability unconditional on $z$, which approximates the expected Round 2 value in the dynamic model (see Appendix D.1) and may differ from $P(y=S|x,z,t,\bar{k})$. Meanwhile, if the committee rejects the applicant, it gets $P(y_1^*=S|t)$, or the success probability of $t$'s marginal transitioned applicant, whose success probability is conditional on $x$ only. $P(y_1^*=S|t)$ can be less or greater than $P(y_2^*=S|t)$ depending on how much the committee values its time vs. admitting the best possible applicants. The Round 1 value function is:

$$\begin{gathered} v_1(x,t,\bar{k},\varepsilon_1) = \max_{d_1 \in \{T,R\}} u_1(x,t,\bar{k},d_1) + \varepsilon_1(d_1), \text{ where:} \\ u_1(x,t,\bar{k},d_1) = P(y=S|x,t,\bar{k})\mathbb{1}(d_1=T) + P(y_1^*=S|t)\mathbb{1}(d_1=R). \end{gathered} \tag{18}$$

The restricted CCPs are the same as in the dynamic model:

$$P(d_r|x,(z),t,\bar{k}) = \frac{\exp(u_r(x,(z),t,\bar{k},d_r)/\sigma)}{\sum_{\widetilde{d}_r \in D_r} \exp(u_r(x,(z),t,\bar{k},\widetilde{d}_r)/\sigma)}. \tag{19}$$

Rather than write out the specifications for the unrestricted CCPs, success probabilities, and cutoff probabilities, I reference Equations (11)–(13). That said, there are two differences from the dynamic framework. First, there are two estimated success probabilities $(P_{y_1}, P_{y_2})$, and second, the Round 1 cutoff is now a probability rather than a level. The unrestricted likelihood is then:

$$\begin{aligned} \mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)} \prod_{n=N_T+1}^{N} \big(1 - P_{d_{1,n}}(\theta_T)\big) \\ \prod_{n=1}^{N} \big[P_{y_n}(\theta_{S_1}) P_{y_n}(\theta_{S_2})\big]^{\mathbb{1}(y_n=S)} \big[(1 - P_{y_n}(\theta_{S_1}))(1 - P_{y_n}(\theta_{S_2}))\big]^{\mathbb{1}(y_n=F)}. \end{aligned} \tag{20}$$

Table 4: Restricted Static Cutoffs

| | R1 Cutoff Probability | R2 Cutoff Probability | Sigma |
|---|---|---|---|
| 2015 | 29.36% | 39.20% | |
| | (23.20%, 36.38%) | (32.75%, 46.05%) | |
| 2016 | 29.17% | 40.53% | |
| | (22.21%, 37.27%) | (33.55%, 47.92%) | |
| 2017 | 37.06% | 41.69% | |
| | (32.10%, 42.29%) | (34.83%, 48.90%) | |
| 2018 | 38.12% | 39.30% | |
| | (33.98%, 42.44%) | (31.80%, 47.33%) | |
| 2019 | 34.04% | 39.62% | |
| | (28.67%, 39.85%) | (32.54%, 47.16%) | |
| Sigma | | | 0.0064 |
| | | | (0.0035) |
| Observations = 2329 | Parameters = 80 | LL = -2992.0 | AIC = 6144.1 |

The static model's estimated cutoff probabilities, $\hat{\sigma}$, and AIC are in Table 4, and the unrestricted and restricted estimates are in Tables 15 and 16 of Appendix C (with simulated estimates in Tables 9 and 10 of Online Appendix B). Both rounds' cutoffs are probabilities,[16] and in all five years, the Round 2 cutoff is greater than the Round 1 cutoff. This ordering is expected because both forecasts are scaled to the same outcomes over the same applicant pool, Round 2 accepts about a tenth of that pool, and Round 1 transitions between one third and three fifths of it. The committee read 59% of the files in 2018 vs. 35–43% in the other years, but 2018's Round 1 cutoff is not lower than the other years' cutoffs because the committee's decisions alone identify only which applicants fell above and below the cutoff. The success probabilities are necessary to identify the cutoff's level, but they do so relatively weakly, as the confidence intervals show.

The myopic model is a special case of the dynamic model in which the parameters for $x$ in every block of $f_{z|x,t}$ are restricted to equal zero, while the intercepts, parameters for $t$, dependence parameters among the components of $z$, and variance parameters remain unrestricted. In this model, if an objective variable is associated with $z$ in addition to being associated with success in its own right, the committee puts no extra weight on that variable in Round 1. The myopic estimates on actual data are in Tables 17 and 18 of Appendix C, and those on simulated data are in Tables 11 and 12 of Online Appendix B. The myopic model is the one model where the restricted AIC is worse than the unrestricted AIC, which is evidence against myopia.

## 4.2 Portfolio Choice with Matriculation

In the portfolio choice problem in Section 4.1, as well as in Bai et al. (2022), there is an assumption that every accepted applicant will matriculate. But in practice, many accepted applicants do not matriculate and instead attend another university. For committees at the very top universities,

[16]The cutoffs are with respect to restricted success probabilities. The average restricted success probability of the acceptance set is greater than its average unrestricted counterpart because the restricted model assumes the accepted applicants are likeliest to succeed. As such, the restricted success probability parameter estimates shift accordingly.

matriculation probabilities may be less of a concern, as accepted applicants who do not matriculate likely do so for idiosyncratic reasons (e.g. choosing Harvard over MIT). But at the majority of universities, the accepted applicants with the highest success probabilities will often have relatively low matriculation probabilities, as the lower-ranked universities were likely safety options from the outset. The question then is whether the committee should select applicants based on matriculation probabilities as well as success probabilities, or whether it should still accept the applicants with the highest success probabilities irrespective of matriculation.

As such, I generalize the portfolio choice problem to incorporate matriculation probabilities. The committee now accepts the subset of applicants with the highest sum of success times matriculation probabilities subject to a capacity constraint on the expected number of matriculants. For each applicant $n$, the committee makes an acceptance or rejection decision given:

$$\begin{aligned} &\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} P(y_n = S|x_n,z_n)P(m_n = M|x_n,z_n)\mathbb{1}(d_n = A), \\ &\text{where: } \sum_{n=1}^{N} P(m_n = M|x_n,z_n)\mathbb{1}(d_n = A) \leq N_M, \end{aligned} \tag{21}$$

where each applicant's matriculation decision $m$ is to either matriculate ($M$) or decline ($D$), $P(m = M|x,z)$ is each applicant's matriculation probability, and $N_M$ is the maximum number of matriculants. Due to the linearity of expectations, the objective function is equivalent to the expected sum of the success probabilities of the accepted applicants who matriculate. Because success and matriculation probabilities are never negative, the unconstrained objective increases every time $d_n = A$, which means that the constraint binds. The Lagrangian with multiplier $\lambda$ is:

$$\begin{aligned} \mathcal{L}(d_1,\ldots,d_N;\lambda) &= \sum_{n=1}^{N} P(y_n = S|x_n,z_n)P(m_n = M|x_n,z_n)\mathbb{1}(d_n = A) \\ &\quad + \lambda\left\{ N_M - \left[ \sum_{n=1}^{N} P(m_n = M|x_n,z_n) - \sum_{n=1}^{N} P(m_n = M|x_n,z_n)\mathbb{1}(d_n = R) \right] \right\} \\ &= \sum_{n=1}^{N} \left[ P(y_n = S|x_n,z_n)\mathbb{1}(d_n = A) + \lambda\mathbb{1}(d_n = R) \right] P(m_n = M|x_n,z_n) \\ &\quad + \lambda\left[ N_M - \sum_{n=1}^{N} P(m_n = M|x_n,z_n) \right]. \end{aligned} \tag{22}$$

From the Lagrangian, it is clear that if there is a cutoff rule solution, the solution is on the success probabilities alone. While applicants with higher matriculation probabilities are more valuable in terms of the objective, they are also more expensive in terms of the capacity constraint on the number of matriculants. Intuitively, if the committee were to prioritize applicants with higher matriculation probabilities, it would not be able to give as many acceptances.

Moreover, the only way to get a cutoff rule solution that incorporates matriculation probabilities would be to have the constraint be on the number of acceptances. That is, the constraint

would be $\sum_{n=1}^{N} \mathbb{1}(d_n = A) \leq N_A$, the Lagrangian would equal $\sum_{n=1}^{N} \big[P(y_n = S|x_n, z_n)P(m_n = M|x_n, z_n)\mathbb{1}(d_n = A) + \lambda\mathbb{1}(d_n = R)\big] + \lambda(N_A - N)$, the applicants would be ranked based on $P(y_n = S|x_n, z_n)P(m_n = M|x_n, z_n)$, and likely matriculants would receive more acceptances.

Crucially, however, because not every accepted applicant will matriculate, this exact problem does not have a cutoff rule solution. For example, suppose $N = 2$, $N_M = 1$, $P(y_1 = S|w_1) = 0.9$, $P(m_1 = M|w_1) = 0.5$, $P(y_2 = S|w_2) = 0.89$, and $P(m_2 = M|w_2) = 0.89$. Since $0.5 + 0.89 > 1$, the committee can accept only one applicant. The cutoff rule solution would be to accept Applicant 1, but Applicant 2 would yield a higher value of the objective since $0.89 \times 0.89 > 0.9 \times 0.5$. But fortunately, any exceptions to the cutoff rule solution disappear once $N$ becomes sufficiently large; that is, when there is a continuum of applicants as in Bai et al. (2022)'s setup.

**Theorem 3.** *For large N, the committee's portfolio choice problem with matriculation probabilities has a cutoff rule solution on success probabilities.*

**Proof.** See Appendix E.1.

Given this result, it appears that committees should not factor in matriculation probabilities when making admissions decisions. In a one-round model, that is the case. However, in a two-round portfolio choice model with matriculation, the cutoff rule is such that the committee should consider matriculation probabilities in Round 1. This rule holds in particular for lower-ranked programs. For example, if a lower-ranked program's committee chooses to read only the files of applicants with perfect GRE scores, then it may have trouble filling its cohort in Round 2, considering that the best applicants are unlikely to matriculate. And even if the committee can fill its cohort, that cohort is likely to be filled with applicants who have weak subjective variables. In contrast, if the committee had considered matriculation probabilities in Round 1, it may have transitioned applicants with lower GRE scores but strong subjective variables who will actually matriculate. I provide the full derivation in Appendix D.2, but I present the dynamic portfolio choice problem and cutoff rule solution here. The Round 2 problem is:

$$\begin{aligned} \max_{(d_{2,1},\ldots,d_{2,N_T}) \in \{A,R\}^{N_T}} & \sum_{n=1}^{N_T} P(y_n = S|w_n)P(m_n = M|w_n)\mathbb{1}(d_{2,n} = A), \\ \text{where: } & \sum_{n=1}^{N_T} P(m_n = M|w_n)\mathbb{1}(d_{2,n} = A) \leq N_M. \end{aligned} \tag{23}$$

Edge cases aside, its solution with $\lambda_2^* = P(y_2^* = S)$ is:

$$\forall n \in \{1,\ldots,N_T\},\ d_{2,n}(w_n|\lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S|w_n) > \lambda_2^* \\ \{A,R\} & \text{if } P(y_n = S|w_n) = \lambda_2^* \\ R & \text{if } P(y_n = S|w_n) < \lambda_2^*. \end{cases} \tag{24}$$

Let $EV_{2,n}(x_n) = \int_{\widetilde{z}_n} \max\{P(y_n = S|x_n, \widetilde{z}_n) - \overline{\lambda_2^*}, 0\}P(m_n = M|x_n, \widetilde{z}_n)dF_{z|x}(\widetilde{z}_n|x_n)$, where $\overline{\lambda_2^*}$ is

the collapsed interval for $\lambda_2^*$. Then the Round 1 portfolio choice problem is:

$$\max_{(d_{1,1},\ldots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} EV_{2,n}(x_n)\mathbb{1}(d_{1,n}=T), \text{ where: } \sum_{n=1}^{N}\mathbb{1}(d_{1,n}=T)\leq N_T. \tag{25}$$

Edge cases aside, its solution with $\lambda_1^* = L(y_1^*=S, m_1^*=M)$ is:

$$\forall n\in\{1,\ldots,N\},\ d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T,R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^*. \end{cases} \tag{26}$$

As I show in Appendix D.2, $EV_{2,n}(x_n)$ puts considerably more weight on success than matriculation. Therefore, the matriculation-dynamic and dynamic models have similar decision rules, and the best static approximation for the matriculation-dynamic model is the static model.

### 4.2.1 Matriculation Models

Because matriculation probabilities do not matter in Round 2, the matriculation-dynamic and static Round 2 value functions are the same as the previous section's Round 2 value functions:

$$\begin{gathered} v_2(x,z,t,\bar{k},\varepsilon_2) = \max_{d_2\in\{A,R\}} u_2(x,z,t,\bar{k},d_2)+\varepsilon_2(d_2), \text{ where:} \\ u_2(x,z,t,\bar{k},d_2) = P(y=S|x,z,t,\bar{k})\mathbb{1}(d_2=A)+P(y_2^*=S|t)\mathbb{1}(d_2=R). \end{gathered} \tag{27}$$

However, the Round 1 value function contains matriculation probabilities:[17]

$$v_1(x,t,\bar{k},\varepsilon_1) = \max_{d_1\in\{T,R\}} u_1(x,t,\bar{k},d_1)+\varepsilon_1(d_1), \text{ where:} \tag{28}$$

$$u_1(x,t,\bar{k},d_1) = \mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}]\mathbb{1}(d_1=T)+L(y_1^*=S,m_1^*=M|t)\mathbb{1}(d_1=R), \text{ where:}$$

$$\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}] = \int_{\widetilde{z}} \sigma\log\left(e^{[P(y=S|x,\widetilde{z},t,\bar{k})-P(y_2^*=S|t)]/\sigma}+1\right)P(m=M|x,\widetilde{z},t)dF(\widetilde{z}|x,t).$$

The derivation of the Round 1 value function is in Appendix D.2. The object I denote by $\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}]$ is the Round 1 transition payoff, as opposed to the literal conditional expectation of $v_2$. As such, the resulting social surplus function contains $e^{[P(y=S|x,\widetilde{z},t,\bar{k})-P(y_2^*=S|t)]/\sigma}+1$ instead of the usual $e^{P(y=S|x,\widetilde{z},t,\bar{k})/\sigma}+e^{P(y_2^*=S|t)/\sigma}$. Due to the matriculation term $P(m=M|x,\widetilde{z},t)$,

[17] I exclude *Open House* from $P(m=M|x,z,t)$ because the committee accepts some applicants before the open house occurs. But *Open House* could be used for late acceptances in a three-round waitlist model like those in Online Appendix C. Also, the matriculation regression results in Section 3 are robust to excluding *Open House*.

Table 5: Restricted Matriculation-Dynamic Cutoffs

| | R1 Cutoff Level | R2 Cutoff Probability | Sigma |
|---|---|---|---|
| 2015 | 0.0071 | 36.35% | |
| | (0.0026, 0.0194) | (29.47%, 43.84%) | |
| 2016 | 0.0071 | 38.25% | |
| | (0.0028, 0.0180) | (31.05%, 46.00%) | |
| 2017 | 0.0105 | 40.65% | |
| | (0.0041, 0.0267) | (35.15%, 46.39%) | |
| 2018 | 0.0095 | 36.59% | |
| | (0.0038, 0.0240) | (30.01%, 43.71%) | |
| 2019 | 0.0095 | 37.41% | |
| | (0.0035, 0.0255) | (31.04%, 44.26%) | |
| Sigma | | | 0.0020** |
| | | | (0.0010) |
| Observations = 2329 | Parameters = 568 | LL = -14260.9 | AIC = 29657.8 |

the incorrect latter expression does not result in the same ordering of applicants as the correct former expression. The restricted CCPs are the same as in the dynamic and static models:

$$P(d_r|x,(z),t,\bar{k}) = \frac{\exp(u_r(x,(z),t,\bar{k},d_r)/\sigma)}{\sum_{\widetilde{d}_r \in D_r} \exp(u_r(x,(z),t,\bar{k},\widetilde{d}_r)/\sigma)}. \tag{29}$$

Rather than write out the specifications for the unrestricted CCPs, success probability, and cutoff probabilities, I reference Equations (11)–(13). But the specification for the matriculation probability is: $P_m = P\big(m = M\big|x,z,t;\theta_M\big) = \Lambda\big(f_M(x,z,t)\theta_M\big)$. Because $P_m$ also depends on the continuous $z$'s, I compute the integral in Equation (28) as in the dynamic model, except over two jointly normal quantities, $f_S(x,z,t,\bar{k})\theta_S$ and $f_M(x,z,t)\theta_M$, instead of one. To do so, I use Gauss–Hermite quadrature with 15 nodes for each quantity. The unrestricted likelihood is then:

$$\begin{aligned}\mathcal{L}_N^U(\theta) = &\prod_{n=1}^{N_A} P_{m_n}(\theta_M)^{\mathbb{1}(m_n=M)}(1-P_{m_n}(\theta_M))^{\mathbb{1}(m_n=D)} \\ &\prod_{n=1}^{N_T} f\big(z_n^c|z_n^{\neg c},x_n,t_n;\theta_F\big)P_{d_{1,n}}(\theta_T)P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)}(1-P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)} \\ &\prod_{n=N_T+1}^{N}\big(1-P_{d_{1,n}}(\theta_T)\big)\prod_{n=1}^{N} f\big(z_n^{\neg c}|x_n,t_n;\theta_F\big)P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1-P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}.\end{aligned} \tag{30}$$

As with the restricted dynamic likelihood, I use $(\theta_F^U,\theta_S^U)$ as an initial guess of $(\theta_F^R,\theta_S^R)$ for the restricted matriculation-dynamic likelihood. Also, a Heckman correction can be implemented for $f_{z|x,t}$ because *Committee Score* is missing for the applicants rejected in Round 1, but due to the lack of an available Heckman instrument, I do not implement one in my preferred specification.

The matriculation-dynamic model's estimated cutoffs, $\hat{\sigma}$, and AIC are in Table 5, and the unrestricted and restricted estimates are in Tables 19 and 20 of Appendix C (with simulated estimates in Tables 13 and 14 of Online Appendix B). The Round 1 cutoffs are much lower in the

matriculation-dynamic model than in the dynamic model, but that is because of the expression for $\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}]$. Both rounds' cutoffs and $\sigma$ are well-identified in all five of the years.

Nevertheless, discussions with members of this university's committee indicate that the committee does not consider matriculation probabilities in Round 1 because this university is ranked highly enough that in general, it can fill its cohort with applicants who have strong objective and subjective variables. However, this university has at times considered matriculation probabilities when making admissions decisions; it would have ignored matriculation probabilities entirely only in a world of perfect information about each applicant's intention. Suppose, for example, that every time the committee accepts an applicant, it immediately hears back from that applicant whether the applicant will matriculate. Then in Round 2, the committee would start with the applicant with the highest success probability, immediately hear back, proceed to the applicant with the second-highest success probability, and continue until the cohort is filled.

But in the real world, committees do not immediately hear back from applicants. Therefore, if a committee does not consider matriculation probabilities when giving early non-waitlisted acceptances, by the time it hears back from the accepted applicants (most of whom will not matriculate since success and matriculation probabilities are negatively correlated), relatively good applicants who would have matriculated will have moved on to other universities. As a result, the committee may have to settle for weaker applicants than if it had considered matriculation probabilities. In addition, the committee can adjust the number of late acceptances it gives based on how many early matriculants it receives to help it attain the target number of matriculants. These dynamics motivate models of the waitlist process that I present in Online Appendix C.

# 5 Optimality and Gains

Suppose the committee were to have chosen the acceptance or matriculation set that was chosen by the dynamic, static, myopic or matriculation-dynamic model. Would that set have an average success probability greater than the set chosen by the committee's decision rule, which for brevity I refer to throughout this section as the committee's set, and if so, how much greater?[18] To answer these questions, I solve the model with $\sigma = 0$ in the conditional choice probabilities (CCPs), compute the average success probabilities of the model's acceptance or matriculation set, and compare those probabilities to the probabilities of the committee's acceptance or matriculation set. When I solve the model, I plug in the unrestricted success probability parameter estimates, not the restricted estimates. The reason is that the restricted estimates may be biased to rationalize suboptimal behavior (see Anderson et al. 2025). For example, if the committee puts more weight on the quantitative GRE score than the outcomes imply, then its restricted parameter estimate

[18]The average success probability of the acceptance (matriculation) set chosen by the committee's decision rule is 0.7 (0.5) percentage points greater than that of the committee's observed set, making the reported gains conservative. Using the committee's decision rule also enables counterfactuals like letting $N_T$ vary, which I do later in this section.

will be greater since the restricted CCPs assume that the committee is behaving optimally.

For the dynamic model, I rank the applicants in each year by the expected Round 2 social surplus function $\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}] = \int_{\widetilde{z}} \sigma \log\left(e^{P(y=S|x,\widetilde{z},t,\bar{k})/\sigma} + e^{P(y_2^*=S)/\sigma}\right) dF(\widetilde{z}|x,t)$,[19] transition the number of transitioned applicants in the data, rank the survivors by $P(y=S|x,z,t,\bar{k})$, and accept the number of accepted applicants in the data. Crucially, when ranking applicants, the model uses this university's fixed *Program Rank* $\bar{k}$ for all applicants and not each applicant's observed rank $k$. The reason is that the committee does not know each applicant's observed rank *ex-ante*, so the model cannot be allowed to know it either. Meanwhile, for the static model, I rank the applicants in Round 1 by $P(y=S|x,t,\bar{k})$. After that, the algorithm is the same as the dynamic model's. Once I have determined which applicants each model accepts, I compute their average success probability, pooled across years, and compare it to the average success probability of the applicants that the committee's decision rule selects.[20] Determining which applicants the committee's rule selects is straightforward. In Round 1, I rank the applicants by $P(d_1 = T|x,t)$, and in Round 2, I rank the survivors by $P(d_2 = A|x,z,t)$. The number of acceptances is unchanged.

For the matriculation models, the algorithm is more involved because the purpose of these models is to evaluate the committee's performance based on the applicants who matriculate at the university. Accepting strong applicants who do not matriculate due to getting offers from a higher-ranked university is not accomplishing anything. For the matriculation-dynamic model, I rank the applicants in each year by $\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}] = \int_{\widetilde{z}} \sigma \log\Big(e^{[P(y=S|x,\widetilde{z},t,\bar{k}) - P(y_2^*=S)]/\sigma} + 1\Big) P(m=M|x,\widetilde{z},t) dF(\widetilde{z}|x,t)$, transition the number of transitioned applicants in the data, rank the surviving applicants by $P(y=S|x,z,t,\bar{k})$, and accept them in descending order, each matriculating with probability $P(m=M|x,z,t)$ until the number of matriculants in the data matriculate.[21] I then compute the average success probability of the matriculation set across matriculants and across 500 simulations, and compare it to the average success probability of the committee's set. As I show in Section 4, the best static approximation of the matriculation-dynamic model is the static model, so I also run the matriculation set's algorithm for the static model's optimal decision rule.

Suppose I find deviation gains. Could committees have incorrect beliefs about which applicants are most likely to succeed? Also, if so, can they do a better job admitting applicants by changing the weights they put on some of the variables? If there are gains in the dynamic, static, or myopic models, the matriculation rate of the model's (stronger) acceptance set may be less than that of the committee's (weaker) acceptance set. In that case, the estimated gains would be an upper bound. However, the matriculation model accepts those likelier to matriculate to prevent losing good applicants with weaker observables by rejecting them before reading their files.

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[19] For the myopic model, the expected social surplus is $\int_{\widetilde{z}} \sigma \log\left(e^{P(y=S|x,\widetilde{z},t,\bar{k})/\sigma} + e^{P(y_2^*=S)/\sigma}\right) dF(\widetilde{z}|t)$.

[20] I evaluate the model and committee's sets with success probabilities and not realized outcomes because of *Program Rank*'s effect on success. With success probabilities, I am able to set the rank of all applicants in either set to $\bar{k}$ to assess how those applicants would have done at this university, which cannot be done with realized outcomes.

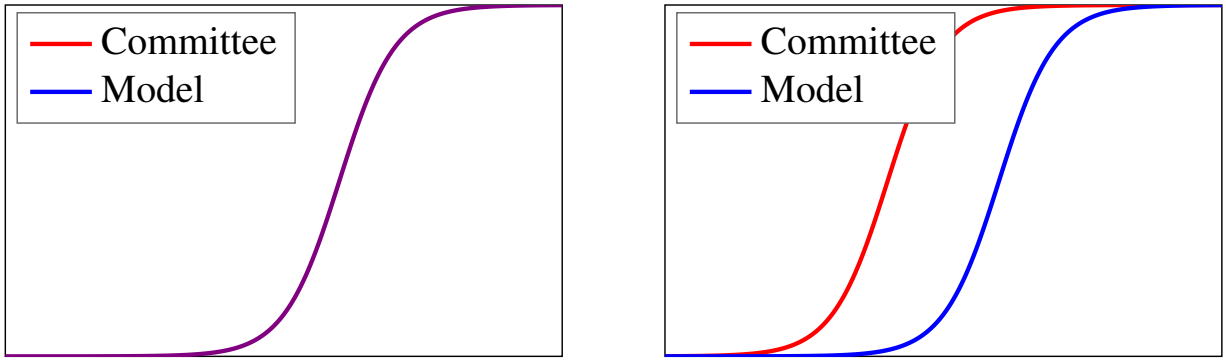
[21] This paper's university is ranked highly enough that it has never run out of applicants to accept in Round 2.

In addition, this procedure can credit the committee only for information I observe. For example, the committee may personally know a recommendation letter writer, or the letter may contain information that algorithms cannot detect, such as exactly how rigorous an applicant's master's program was. The procedure evaluates both the committee's and the model's acceptance or matriculation set with the same measure, the unrestricted success probability $P(y = S|x,z,t,\overline{k})$, which is estimated from the observables. If the committee acts on a success-predicting variable I do not observe, the measure cannot detect that variable, so any edge the committee's private information provides goes unmeasured. The end result would be an overstated deviation gain.

The protection against this limitation is the comprehensiveness of the data. The application files I observe are the same files the committee read, and from those files, I construct (in addition to the objective variables $x$) measures of mathematical preparation, the substance of the application essay, and the content and tone of the recommendation letters. Most importantly, because $z$ includes *Committee Score*, conditioning the success probability on $z$ absorbs much of the residual private information.[22] Because *Committee Score* also enters the model's decision rule, the model can weight it as heavily as the success estimates warrant, just as the committee chair and other committee members can also choose how heavily to weight it. In this way, I effectively merge human and data-driven approaches. While humans may have access to more information and can use their intuition to evaluate less quantifiable applicant characteristics, they may also fall prey to human biases that models do not. Nonetheless, no set of observables is guaranteed to span the committee's information set, so the deviation gains are best read as conditional on the observed variables closely reconstructing what the committee saw. That said, the realized success rate of the committee's observed matriculation set is no higher than the average success probability of the committee's decision rule's matriculation set, meaning that any private information the committee had did not measurably help the committee attain a better matriculation set.

To compare the model's and committee's acceptance or matriculation sets, I use the following notation. For brevity, write $P_{y_n}(\theta_S) = P(y_n = S|x_n,z_n,t_n,\overline{k}_n;\theta_S)$ as applicant $n$'s success probability at parameter $\theta_S$, with program rank fixed at this university's rank. Throughout this section, the superscript $U$ refers to the unrestricted likelihood, in which the committee's CCPs are estimated freely. In particular, $\hat{\theta}_S^U$ is the vector of unrestricted success probability parameter estimates, and $d^U$ is the committee's decision rule as described by the unrestricted CCPs. The superscript $R$ refers to the model's optimal decision rule $d^R$, which is the rule that the restricted likelihood in Section 4 assumes the committee follows. A feasible acceptance set $\mathcal{A}$ is a subset of the applicant pool of size $N_A$ that respects the per-cohort quotas and the constraint on the transition set, and $\mathcal{A}$'s average success probability at $\theta_S$ is $\overline{P}_{\mathcal{A}}(\theta_S) = \frac{1}{N_A}\sum_{n\in\mathcal{A}} P_{y_n}(\theta_S)$. The model's decision rule $d^R$

[22] In Table 12 of Appendix B, *Missing Committee Score* survives the lasso with a negative sign, and has AMEs of $-0.0528^{**}$ (*Program Rank* excluded) and $-0.0277$ (rank included) in Table 2 of Section 3. As such, application variables equal, applicants rejected in Round 1 might have been less likely to succeed. In Round 1, the committee could have observed each applicant's country of origin, undergraduate institution, and letter writers' names. Aside from that, it did not observe anything not in my dataset until Round 2, when *Committee Score* proxies such information.




Optimal Committee Suboptimal Committee

Figure 3: Deviation Gains CDFs

selects, at any $\theta_S$, the feasible set $\mathcal{A}^R$ with the highest average success probability:

$$\mathcal{A}^R(\theta_S) = \arg\max_{\mathcal{A}} \overline{P}_{\mathcal{A}}(\theta_S). \tag{31}$$

The committee's rule $d^U$ selects the set $\mathcal{A}^U$, which does not depend on $\theta_S$. The deviation gain evaluates these two sets, with the model's set fixed at the estimates, at any parameter value:

$$g(\theta_S) = \overline{P}_{\mathcal{A}^R(\hat{\theta}_S^U)}(\theta_S) - \overline{P}_{\mathcal{A}^U}(\theta_S). \tag{32}$$

As such, the observed deviation gain is $\hat{g} = g(\hat{\theta}_S^U)$ because it is evaluated at the estimates.

If I hold the Round 1 transition set fixed at the committee's, then the observed deviation gain can never be negative; that is, $\hat{g} \geq 0$. This result follows from how the model chooses its set, not from anything about the committee's behavior. The model accepts the $N_A$ transitioned applicants with the highest estimated success probabilities, so no other set of $N_A$ acceptances, including the committee's, can have a higher average estimated success probability. For the purposes of this claim, I hold the Round 1 transition set fixed because the model chooses its transition set before the files are read, based on what it expects the files to contain rather than what they turn out to contain. Without this assumption, it is therefore possible in rare cases for the committee's set to come out ahead if, for example, the committee favors high-upside applicants in Round 1, and it turns out that favoring such applicants ultimately helps it attain a better set of accepted applicants.

As such, the sign of the observed deviation gain is uninformative by itself. What is informative is whether the gain is large relative to the sampling error in $\hat{\theta}_S^U$, since the model's set is guaranteed to be the best set only at the estimates with which it was chosen. To gauge how much the deviation gain depends on this sampling error, I evaluate both sets at perturbations of the estimates. A perturbation $\tilde{\theta}_{S,b}^U$ is a random draw of the success probability parameters from the estimated asymptotic distribution of $\hat{\theta}_S^U$, namely $\mathcal{N}(\hat{\theta}_S^U, \hat{\Sigma})$, so each perturbation is a value of the parameters that is plausible given the data. At a perturbation, the sign of the gain can flip:

$$g(\tilde{\theta}_{S,b}^U) = \overline{P}_{\mathcal{A}^R(\hat{\theta}_S^U)}(\tilde{\theta}_{S,b}^U) - \overline{P}_{\mathcal{A}^U}(\tilde{\theta}_{S,b}^U) < 0. \tag{33}$$

The reason is that perturbing the parameters changes the applicants' success probabilities and therefore how they rank, so the model's set, which stays fixed at $\mathcal{A}^R(\hat{\theta}_S^U)$, need not be the best set at $\tilde{\theta}_{S,b}^U$. As a simple example, suppose that the two sets differ only in that the committee accepts Applicant 1, whose estimated success probability is 0.45, whereas the model accepts Applicant 2, whose estimated success probability is 0.47. If Applicant 1 has a higher quantitative GRE score than Applicant 2, then a perturbation that happens to increase the weight on the quantitative GRE score may raise Applicant 1's success probability above Applicant 2's, in which case the committee's set now has the higher average success probability. In general, the sign flips more often when the committee's and model's sets are similar in composition, and the applicants on which they differ have similar success probabilities. If the committee optimizes or comes close to it, the sign will flip for many perturbations, whereas if the committee is far from optimal, the model's set will be better by a wide enough margin that few perturbations flip the sign.

I can therefore use such perturbations of the success parameters to assess whether the deviation gains are robust to sampling error. Following Krinsky and Robb (1986), across $B = 500$ perturbations, I plot the CDFs of the average success probabilities of the acceptance and matriculation sets of the dynamic, static, myopic, and matriculation-dynamic models, as well as the CDFs of the sets implied by the committee's decision rule. While the acceptance and matriculation sets are selected with the actual parameter estimates, the average success probabilities are computed with the perturbed estimates. Each perturbation therefore yields one distinct value of each set's average success probability, and each CDF features the range of values that the set's average success probability can plausibly take given the sampling error in the estimates.

Figure 3 illustrates what such CDFs look like in general. When a committee switches from a suboptimal decision rule to the optimal decision rule, the actual decision rule's CDF shifts right to match the model's CDF. I provide a demonstration on the simulated data, where per Figures 1 and 2 in Online Appendix B, in the dataset where the committee is nearly optimal, the models' CDFs are very close to the committee's. However, in the dataset where the committee is not optimal, the models' CDFs strongly stochastically dominate the committee's CDF.

In addition, if neither set is chosen using $\hat{\theta}_S^U$, a condition to which I return below, the perturbations yield a hypothesis test of whether the deviation gain is statistically significant. Let $\theta_S^0$ denote the true success probability parameters, so that $g_0 = g(\theta_S^0)$ is the true deviation gain of the model's set over the committee's set. The null hypothesis is that there is no deviation gain, $H_0\colon\ g_0 = 0$, and the alternative hypothesis is that the model's set is better than the committee's, $H_1\colon\ g_0 > 0$. The test statistic is the observed deviation gain $\hat{g}$, which is approximately normal with mean $g_0$ and a standard deviation $\sigma_g$ that I estimate with the standard deviation of the perturbed gains $g(\tilde{\theta}_{S,b}^U)$. As in any one-sided test, the p-value is the probability, if $H_0$ is true, of a gain at least as large as the one I observe. Under $H_0$, $\hat{g}/\sigma_g$ is approximately standard normal, so this probability is $1 - \Phi(\hat{g}/\sigma_g)$, which is the p-value of a one-sided $z$ test that uses the standardized gain $\hat{g}/\sigma_g$ as the test statistic, since dividing by $\sigma_g$ is a monotone transformation.

With two fixed sets, the simplest way to conduct this test would be to compute $\sigma_g$ in closed

form with the delta method and convert the $z$ statistic $\hat{g}/\hat{\sigma}_g$ into a p-value.[23] Then the test would not require perturbations. However, as I explain below, the model's set is not fixed, since it is chosen using the same estimates with which its average success probability is computed. My correction for the resulting upward bias in $\hat{g}$, which I present later in this section, solves the model again at each perturbation. This step is something that the delta method cannot replicate because the model's set changes discontinuously with $\theta_S$. As such, the average success probability of the model's set is not differentiable in $\theta_S$, which prevents the delta method from working.

Therefore, I work with the perturbations going forward, and before I correct gains evaluated at perturbations, which like the observed gain are biased upward if the model's set is not fixed, I show that when the committee and model's sets are both fixed, the fraction of perturbations in which the model's set does no better than the committee's reproduces the $z$ test's p-value. This way, the corrected fraction in the next subsection is a modification of a statistic with a known interpretation, even though it is not itself a p-value. Given fixed sets, the uncorrected fraction reproduces the $z$ test's p-value because the perturbed gains are spread around $\hat{g}$ the same way the observed gain would be spread around zero under $H_0$. Since under $H_0$, zero is as far below the perturbed gains' mean of $\hat{g}$ as $\hat{g}$ is above the mean of zero, the symmetry of the normal distribution implies that a perturbed gain at or below zero is as likely as a gain at or above $\hat{g}$. Formally:

**Theorem 4.** *For two sets that are not chosen using $\hat{\theta}_S^U$, as $B \to \infty$, the fraction of perturbations in which the model's set does no better than the committee's, $g(\tilde{\theta}_{S,b}^U) \leq 0$, converges to the p-value of the test of $H_0\colon g_0 = 0$ vs. $H_1\colon g_0 > 0$ that uses the observed gain $\hat{g}$ as the test statistic.*

**Proof.** See Appendix E.2.

However, there is the aforementioned consideration that must be accounted for. For the magnitude of the deviation gains and either version of the $z$ test to be unbiased, both sets, in particular the model's, must be exogenously chosen, i.e. chosen without using the estimates with which their average success probabilities are computed. The committee's set meets this condition since it is determined by the committee's decisions, which were made before any outcomes were realized, but the model's does not. As a result, even if the success probability parameter estimates and therefore the individual success probabilities are unbiased, the average success probability of the model's set will be biased upward, which I. Andrews et al. (2024) call the "winner's curse."

The intuition is that the model's selection favors applicants whose success probabilities are overestimated. The estimated weight on each variable in the success probability contains sampling error, so some applicants' estimated success probabilities are too high by chance, whereas others' are too low by chance. An applicant whose success probability is overestimated is more likely to end up above the model's cutoff and be accepted, whereas an applicant whose success probability is underestimated is more likely to end up below it and be rejected. Therefore, the errors in the estimated success probabilities of the model's accepted applicants are positive on

[23] Since both sets are fixed, $\nabla g(\hat{\theta}_S^U) = \frac{1}{N_A}\left[\sum_{n \in \mathcal{A}^R(\hat{\theta}_S^U) \setminus \mathcal{A}^U} \nabla P_{y_n}(\hat{\theta}_S^U) - \sum_{n \in \mathcal{A}^U \setminus \mathcal{A}^R(\hat{\theta}_S^U)} \nabla P_{y_n}(\hat{\theta}_S^U)\right]$, where $\nabla P_{y_n}(\theta_S) = P_{y_n}(\theta_S)\big(1 - P_{y_n}(\theta_S)\big) f_S(x_n, z_n, t_n, \bar{k})'$. Only applicants where the two sets disagree enter $\hat{g}$ and its standard deviation.

average, even though the errors across all applicants are not, and scoring the model's set with the same estimates that chose it overstates its average success probability. The name comes from common-value auctions, in which the winning bidder tends to be the one who most overestimated the item's value. In contrast, the committee could not have selected on the errors in my estimates, so the errors for its accepted applicants average out. Simply put, the model's set appears best at the estimates partly because it was chosen with the same errors that are used to evaluate it.

If the number of observations is large relative to the number of parameters, the magnitude of the curse will be small, but with 2,329 observations to over 30 parameters in the admissions dataset, it is not negligible here. One way to obtain an exogenously-chosen set for the model would be to evaluate the performance of the model's decision rule on a holdout set. An example would be using 2015–2018 success probability parameter estimates to select the model's acceptance or matriculation set for 2019, computing the average success probability of that set with 2019 success probability parameter estimates, and repeating this procedure while holding out each of 2015, 2016, 2017, and 2018. However, there are too few observations in any single year to reliably estimate the success probability parameters on one year's worth of data. Even the sample sizes from splits with three years of training and two years of testing are too small. Therefore, I develop and implement different methods to correct for the winner's curse and accurately estimate the magnitude of the gains, starting as follows with the parametric bootstrap correction.

## 5.1 Parametric Bootstrap Correction

In this subsection, I start by showing that when the model's acceptance or matriculation set is selected using the same estimates that the observed deviation gain is computed from, that gain is biased upward. To show this, I consider the hypothesis that the committee is optimal, $H_0$: $\mathcal{A}^U = \mathcal{A}^R(\theta_S^0)$ up to ties at the cutoff, meaning that its set is the one that is optimal at the true parameters. Equivalently, the gain from switching to the truly optimal set, $\overline{P}_{\mathcal{A}^R(\theta_S^0)}(\theta_S^0) - \overline{P}_{\mathcal{A}^U}(\theta_S^0)$, is zero. This gain can never be negative, since no set beats the truly optimal set at the true parameters, so the committee is optimal exactly when the gain is at its lower bound of zero. This hypothesis differs from the null hypothesis in Theorem 4, which compares two fixed sets rather than the committee's set and the truly optimal set. For two parameter values $\theta$ and $\theta'$, define:

$$\Delta(\theta;\theta') = \overline{P}_{\mathcal{A}^R(\theta)}(\theta) - \overline{P}_{\mathcal{A}^R(\theta')}(\theta). \tag{34}$$

$\Delta(\theta;\theta')$ is the amount by which the set that is optimal at $\theta$ beats the set optimal at $\theta'$ when both sets are evaluated at $\theta$. Because $\mathcal{A}^R(\theta)$ maximizes $\overline{P}_{\mathcal{A}}(\theta)$ by construction, it is the case that $\Delta(\theta;\theta') \geq 0$. Moreover, under $H_0$, the expected observed deviation gain is $\Delta(\hat{\theta}_S^U;\theta_S^0)$ because:

$$\begin{aligned}
\hat{g} &= \overline{P}_{\mathcal{A}^R(\hat{\theta}_S^U)}(\hat{\theta}_S^U) - \overline{P}_{\mathcal{A}^U}(\hat{\theta}_S^U) \\
&= \overline{P}_{\mathcal{A}^R(\hat{\theta}_S^U)}(\hat{\theta}_S^U) - \overline{P}_{\mathcal{A}^R(\theta_S^0)}(\hat{\theta}_S^U) + \overline{P}_{\mathcal{A}^R(\theta_S^0)}(\hat{\theta}_S^U) - \overline{P}_{\mathcal{A}^U}(\hat{\theta}_S^U) \\
&= \Delta(\hat{\theta}_S^U;\theta_S^0) + \overline{P}_{\mathcal{A}^R(\theta_S^0)}(\hat{\theta}_S^U) - \overline{P}_{\mathcal{A}^U}(\hat{\theta}_S^U).
\end{aligned} \tag{35}$$

In this equation, the first term is the winner's curse, and the second term is 0 in expectation under $H_0$. Therefore, the set chosen at $\hat{\theta}_S^U$ appears, when evaluated at $\hat{\theta}_S^U$, to beat the set that is optimal at $\theta_S^0$ because $\mathcal{A}^R(\hat{\theta}_S^U)$ maximizes $\overline{P}_{\mathcal{A}}(\hat{\theta}_S^U)$. The curse is zero when $\mathcal{A}^R(\hat{\theta}_S^U) = \mathcal{A}^R(\theta_S^0)$; that is, when the estimates are close enough to the true parameters that the model selects the same applicants at both. As such, the winner's curse is a small-sample concern. In a very large sample, $\hat{\theta}_S^U$ is close to $\theta_S^0$, and because $\hat{\theta}_S^U$ converges to $\theta_S^0$ with probability one, the winner's curse term in Equation (35) converges to zero with probability one. Under $H_0$, the observed deviation gain therefore also converges to zero, so in a large enough sample, the winner's curse would not be a problem. However, as noted above, it is not negligible with 2,329 observations.

Now I analyze the perturbations version of the $z$ test in light of the winner's curse. The test draws perturbations $\tilde{\theta}_{S,1}^U, \ldots, \tilde{\theta}_{S,B}^U$ from $\mathcal{N}(\hat{\theta}_S^U, \hat{\Sigma})$, evaluates the two fixed sets at each perturbation via $\tilde{g}_b = g(\tilde{\theta}_{S,b}^U)$, and reports the fraction of perturbations with a negative gain as the p-value:

$$\hat{p} = \frac{1}{B} \sum_{b=1}^{B} \mathbb{1}(\tilde{g}_b \leq 0). \tag{36}$$

For two sets that do not depend on $\hat{\theta}_S^U$, this procedure is fine. The draws simulate the sampling distribution of the difference in average success probabilities, and $\hat{p}$ is a valid p-value by Theorem 4. But when the set is $\mathcal{A}^R(\hat{\theta}_S^U)$, the draws are centered about $\hat{\theta}_S^U$ instead of $\theta_S^0$:

$$\begin{aligned} \tilde{g}_b &= \overline{P}_{\mathcal{A}^R(\hat{\theta}_S^U)}(\tilde{\theta}_{S,b}^U) - \overline{P}_{\mathcal{A}^U}(\tilde{\theta}_{S,b}^U) \\ &= \hat{g} + \left[\overline{P}_{\mathcal{A}^R(\hat{\theta}_S^U)}(\tilde{\theta}_{S,b}^U) - \overline{P}_{\mathcal{A}^R(\hat{\theta}_S^U)}(\hat{\theta}_S^U)\right] - \left[\overline{P}_{\mathcal{A}^U}(\tilde{\theta}_{S,b}^U) - \overline{P}_{\mathcal{A}^U}(\hat{\theta}_S^U)\right], \end{aligned} \tag{37}$$

where to get from the first to the second line in this equation, I add and subtract $\hat{g}$.

The two bracketed terms in the second line are smooth functions of $\tilde{\theta}_{S,b}^U - \hat{\theta}_S^U$ with conditional mean approximately zero, so the perturbations are distributed around $\hat{g}$. The second term has conditional mean approximately zero because the committee's set was chosen *ex-ante*, meaning it should perform equally well on $\tilde{\theta}_{S,b}^U$ as on $\hat{\theta}_S^U$. The first term has conditional mean approximately zero because optimality at $\hat{\theta}_S^U$ constrains only which set scores highest at $\hat{\theta}_S^U$, not how the selected set's score moves with the parameter. As such, a set chosen to maximize the average success probability at $\hat{\theta}_S^U$ can do even better on $\tilde{\theta}_{S,b}^U$ than on $\hat{\theta}_S^U$. And because $\tilde{\theta}_{S,b}^U$, unlike the true $\theta_S^0$, is centered about $\hat{\theta}_S^U$, the first term will be positive for many draws. If the model is linear probability, then due to linearity of expectations, both bracketed terms will have conditional mean zero. The test therefore measures the sampling variance of the average success probabilities of two fixed sets around a center, namely $\hat{\theta}$, that contains the entire winner's curse $\Delta(\hat{\theta}_S^U; \theta_S^0)$. Its p-value is small whenever the curse is large relative to that variance, whether or not $H_0$ holds.

Given that this test does not correct for the winner's curse, a different correction is necessary. The true curse $\Delta(\hat{\theta}_S^U; \theta_S^0)$ cannot be computed since $\theta_S^0$ is unknown. Therefore, I implement a parametric bootstrap correction that estimates the winner's curse by applying the same construction as the curse one level down. The estimate $\hat{\theta}_S^U$ stands in the role of the unknown $\theta_S^0$, and a

perturbation $\tilde{\theta}^U_{S,b}$ stands in the role of the estimate $\hat{\theta}^U_S$. At each perturbation $\tilde{\theta}^U_{S,b}$, the model's decision rule is solved again, choosing the set $\mathcal{A}^R(\tilde{\theta}^U_{S,b})$ optimal at $\tilde{\theta}^U_{S,b}$, and that set is evaluated at $\tilde{\theta}^U_{S,b}$. Meanwhile, the set chosen at the original estimate, $\mathcal{A}^R(\hat{\theta}^U_S)$, is evaluated at the same perturbation $\tilde{\theta}^U_{S,b}$. The difference in the average success probabilities of those sets:

$$\Delta_b = \Delta(\tilde{\theta}^U_{S,b}; \hat{\theta}^U_S) = \overline{P}_{\mathcal{A}^R(\tilde{\theta}^U_{S,b})}(\tilde{\theta}^U_{S,b}) - \overline{P}_{\mathcal{A}^R(\hat{\theta}^U_S)}(\tilde{\theta}^U_{S,b}), \qquad b = 1, \ldots, B, \tag{38}$$

is the amount by which a set chosen and evaluated at the same perturbation appears to beat a set chosen elsewhere, when $\hat{\theta}^U_S$ is the true parameter. This difference is positive because $\mathcal{A}^R(\tilde{\theta}^U_{S,b})$ maximizes $\overline{P}_{\mathcal{A}}(\tilde{\theta}^U_{S,b})$.[24] Its average, $\overline{\Delta}_B = \frac{1}{B}\sum_b \Delta_b$, is the estimate of the winner's curse. The corrected gain at the estimates is $\hat{g} - \overline{\Delta}_B$, whereas the corrected gain at each perturbation is $\tilde{g}_b - \overline{\Delta}_B$. The fraction of perturbations where the corrected gain is weakly negative is:

$$\hat{p}^{\text{cor}} = \frac{1}{B}\sum_{b=1}^{B} \mathbb{1}(\tilde{g}_b - \overline{\Delta}_B \leq 0). \tag{39}$$

Two properties follow. First, conditional on the data, as $B \to \infty$, $\overline{\Delta}_B$ converges to $\mathbb{E}[\Delta(\tilde{\theta}^U_S; \hat{\theta}^U_S)]$ with $\tilde{\theta}^U_S \sim \mathcal{N}(\hat{\theta}^U_S, \hat{\Sigma})$. This is the expected winner's curse in the data-generating process in which: (1) $\hat{\theta}^U_S$ is the true parameter, (2) the committee is optimal, so its set is $\mathcal{A}^R(\hat{\theta}^U_S)$, i.e. the optimal set for $\hat{\theta}^U_S$, and (3) the estimate is $\hat{\theta}^U_S$ plus $\mathcal{N}(0, \hat{\Sigma})$ sampling error. In that process, the true gain of any decision rule over the committee is zero, while an econometrician with estimate $\tilde{\theta}^U_S$ who chooses $\mathcal{A}^R(\tilde{\theta}^U_S)$ and evaluates both sets at $\tilde{\theta}^U_S$ obtains the in-sample gain $\Delta(\tilde{\theta}^U_S; \hat{\theta}^U_S)$. The observed gain is the same construction one level up, with $(\theta^0_S, \hat{\theta}^U_S)$ in place of $(\hat{\theta}^U_S, \tilde{\theta}^U_S)$. Second, because $\overline{\Delta}_B \geq 0$, every corrected draw lies weakly below its uncorrected draw, so $\hat{p}^{\text{cor}} \geq \hat{p}$ in every sample.

However, while $\hat{p}^{\text{cor}}$ is the analogue to the $z$ test with perturbations's p-value, which is valid for any two exogenously chosen sets, I do not claim that $\hat{p}^{\text{cor}}$ is itself a p-value, which is the probability, if the committee is optimal, of a corrected gain at least as large as the one observed, $\hat{g} - \overline{\Delta}_B$. The reason is that if the committee is optimal, $\hat{g} - \overline{\Delta}_B$ varies across samples because different sets are selected in each sample. In contrast, $\tilde{g}_b - \overline{\Delta}_B$ varies across perturbations only because the same two sets are scored at different parameter values. That is, the perturbations capture the uncertainty in how well each set performs, but not the uncertainty in which set the model selects. Therefore, I treat $\hat{p}^{\text{cor}}$ as the summary statistic of the gains CDFs, namely the fraction of the corrected model CDF that lies to the left of the committee's CDF.

I now analyze the effectiveness of the correction for finite samples. For this analysis, suppose for simplicity that the success probability is parameterized linearly, i.e. $P_{y_n}(\theta_S) = f_S(x_n, z_n, t_n, \overline{k}_n)\theta_S$, rather than logistically. This approximation is reasonable for the admissions data, as the only applicants whose accept/reject decision changes with a perturbation are those near the cutoff. In addition, their success probabilities lie in a range of about

[24] This argument takes the transition sets as fixed. In reality, the model's decision rule chooses its own Round 1 transition set, which could change when the success probability parameters used in Round 1 are perturbed. Therefore, $\Delta_b$ could in principle be negative for a perturbation, though that would be very unlikely in practice.

0.4 to 0.5, where the logistic function is close to linear. Consider one year's Round 2 decision rule, which accepts $N_A$ of the $N_T > N_A$ transitioned applicants, with the transition set held fixed. Denote $\beta$ as the applicant-level parameters in $\theta_S$, i.e. those on $w = (x, z)$, and denote $\overline{w}_{\mathcal{A}}$ as the average over a feasible set $\mathcal{A}$ of the covariates. The remaining parameters (the intercept, year dummies, and program rank) add the same amount to every applicant's success probability in the cohort, so $\overline{P}_{\mathcal{A}}(\theta_S)$ equals $\overline{w}'_{\mathcal{A}}\beta$ plus a constant that is the same for any feasible set and cancels out in any difference of average success probabilities. Denote $\overline{P}^{\max}(\beta) = \max_{\mathcal{A}} \overline{w}'_{\mathcal{A}}\beta$ as the highest average success probability, up to that constant, that any feasible set attains at $\beta$. The model's chosen set attains $\overline{P}^{\max}(\hat{\beta}^U)$, and the true optimal set attains $\overline{P}^{\max}(\beta_0)$. Let the sampling error $\eta = \hat{\beta}^U - \beta_0$ be normally distributed with mean zero, and let the draw $\tilde{\beta}^U - \hat{\beta}^U$ be an independent draw from the same distribution.

**Theorem 5.** Under a linear probability model, it is the case that:

$$\begin{aligned} \mathbb{E}\big[\Delta(\hat{\theta}^U_S; \theta^0_S)\big] &= \mathbb{E}\big[\overline{P}^{\max}(\beta_0 + \eta)\big] - \overline{P}^{\max}(\beta_0) \qquad \textbf{and} \\ \mathbb{E}\big[\overline{\Delta}_B\big] &= \mathbb{E}\big[\overline{P}^{\max}(\beta_0 + \sqrt{2}\eta)\big] - \mathbb{E}\big[\overline{P}^{\max}(\beta_0 + \eta)\big], \end{aligned} \tag{40}$$

where the first expectation is over the sampling error and the second over the sampling error and the perturbation draws. The winner's curse and its estimate are successive increments of $\mathbb{E}[\overline{P}^{\max}(\beta_0 + \sqrt{\tau}\eta)]$, from $\tau = 0$ to 1 and from 1 to 2. The correction is exact when that function is linear in $\tau$ on $[0, 2]$, too small when it is concave in $\tau$, and too large when it is convex in $\tau$.

**Proof.** See Appendix E.2.

**Corollary 1.** If $\beta_0 = 0$, so that every applicant has the same true success probability, then:

$$\mathbb{E}\big[\hat{g}\big] = \mathbb{E}\big[\overline{P}^{\max}(\eta)\big] \qquad \textbf{and} \qquad \mathbb{E}\big[\Delta(\tilde{\theta}^U_S; \hat{\theta}^U_S)\big] = (\sqrt{2} - 1)\,\mathbb{E}\big[\overline{P}^{\max}(\eta)\big]. \tag{41}$$

Here, the estimated correction has expectation $(\sqrt{2} - 1) \approx 0.414$ times the winner's curse. This is the worst-case scenario, meaning that in every data-generating process,

$$\mathbb{E}[\overline{\Delta}_B] \geq (\sqrt{2} - 1)\,\mathbb{E}[\Delta(\hat{\theta}^U_S; \theta^0_S)]. \tag{42}$$

**Proof.** See Appendix E.2.

The correction is exact when the function $\mathbb{E}[\overline{P}^{\max}(\beta_0 + \sqrt{\tau}\eta)]$ is linear in $\tau$, and it has an error whose sign and size are set by the function's curvature on $[0, 2]$. The expectation in the function averages over the sampling error, so whether the function is linear is decided by the data-generating process, which consists of $\beta_0$ and the covariance of $\eta$. The theorem's exactness is therefore a statement about the average of $\overline{\Delta}_B$ across samples. In a given sample, $\overline{\Delta}_B$ is the winner's curse from a data-generating process whose true parameter is $\hat{\beta}^U$ rather than $\beta_0$. As such, $\overline{\Delta}_B$ equals the expected winner's curse regardless of the function's shape only in a sample where $\hat{\beta}^U = \beta_0$. But even if $\hat{\beta}^U \neq \beta_0$, $\overline{\Delta}_B$ equals the expected curse if the function is linear.

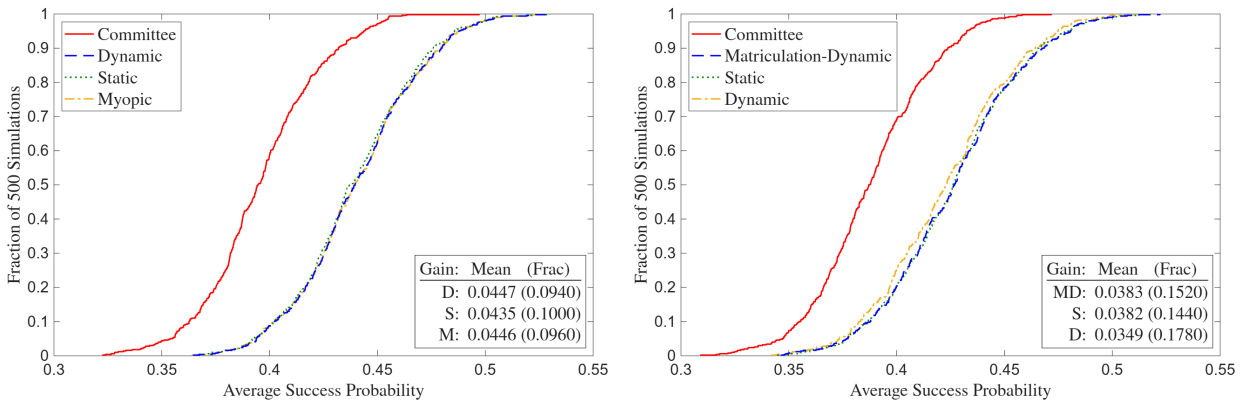
In practice, $\mathbb{E}[\overline{P}^{\max}(\beta_0 + \sqrt{\tau}\eta)]$ is close to linear when many applicants lie close to the cutoff, but most of the dispersion of the estimated success probabilities across applicants is real. In that

case, the bootstrap data-generating process resembles the true one, and the correction removes the winner's curse. In contrast, when the estimated dispersion is mostly sampling error, $\mathbb{E}[\overline{P}^{\max}(\beta_0 + \sqrt{\tau}\eta)]$ is concave, the bootstrap data-generating process starts with more dispersion than the true process did, and the correction removes only part of the curse ($\sqrt{2}-1$ of it in the worst-case scenario). This is the inconsistency of the bootstrap at the boundary of the parameter space in D. W. K. Andrews (2000). Finally, when the applicants nearest the cutoff are separated by moderate gaps, $\mathbb{E}[\overline{P}^{\max}(\beta_0 + \sqrt{\tau}\eta)]$ is convex. One unit of $\eta$ rarely reorders the applicants, whereas $\sqrt{2}$ units of $\eta$ does, so the correction is larger than the curse. The corollary bounds the ratio of the estimated correction to the curse from below only. In the convex case, the ratio can be large, but the curse itself is typically small, so the overstatement is also small in absolute terms.

In general, the linear case, where the observed gain is biased but the correction is exact, is plausible for admissions data. While the application file predicts placement, the marginal acceptances are contested by applicants whose true success probabilities differ slightly. The model awards those acceptances to whomever the sampling error favors, so evaluating the model's average success probability at the same estimates it was trained on biases upward the model's set's average success probability. But the correction accounts for that bias because perturbing the estimates reorders the marginal applicants as a holdout set would. Still, in Section 5.3, I implement a calibration check that provides evidence supporting the robustness of the corrected gains.

## 5.2 Corrected Deviation Gains

Figure 4 reveals that the corrected average success probabilities of the model's acceptance and matriculation sets are about 4 percentage points higher than those of the committee's. There is little variation in the gains across models, though as expected, the dynamic model performs slightly worse than the matriculation-dynamic model for the matriculation set. Excluding the dynamic model for the matriculation set, the fraction of perturbations with a negative gain is in the 10–15% range, though as I explain in the previous section, this fraction is not a p-value. Figure 4 stands in contrast to the uncorrected Figure 8 in Appendix C. There, the magnitude of the gains is about 9–10 percentage points, and the gain is negative in fewer than one percent of the perturbations. If the model's sets were exogenously chosen, the gains would be statistically significant at the 1% level. But the sets were not exogenously chosen, and as the correction makes clear, the winner's curse in this dataset is non-negligible. In turn, while the corrected gains are about 4 percentage points, I cannot establish that they are statistically significant, nor can I rule out the possibility that the committee is optimal. In particular, the data do not reject, at conventional significance levels, a process in which the application file carries no information about success (a likelihood-ratio test in Section 5.3 has a p-value of 0.1166), and in that process, every decision rule, including the committee's, is optimal. But that is not to say that the uncorrected gains are entirely spurious, as a calibration exercise I perform in Section 5.3 suggests that the corrected gains are close to the true gains in a data-generating process resembling the data.




(a) Acceptance Set

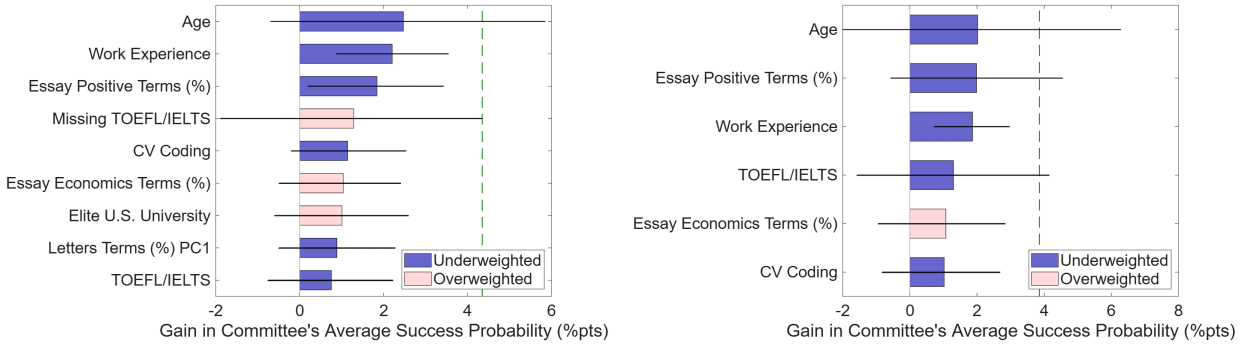
(b) Matriculation Set

Figure 4: Deviation Gains

As such, I determine which variables may be responsible for the gains. For simplicity, I use the static models for this attribution. I start with the Rounds 1 and 2 conditional choice probability parameter weights for each variable. One variable at a time, I hold all the other variables' weights fixed and choose the Rounds 1 and 2 weights that maximize the average success probability of the acceptance or matriculation set. Once those weights are chosen, I recalculate the corrected average success probability using the 500 perturbations and plot the results in Figure 5. The bar shows the average of those average success probabilities across the perturbations, and the whiskers extend to the 5th and 95th percentiles.[25] Also, the bars are dark blue for previously-underweighted variables where the reweighting increases their average in the acceptance or matriculation set. In contrast, the bars are light red for previously-overweighted variables where the reweighting decreases their average. An objective or subjective variable qualifies for inclusion in the figure if its bar reaches at least 0.5 percentage points. For each of the variables in Figure 5, Table 21 of Appendix C contains their average values for the full sample, the committee's acceptance or matriculation set, the single-variable optimized set, and the joint optimized set.

The bars are similar for both the acceptance and matriculation sets; all bars for the matriculation set are present for the acceptance set, only three bars are present for the acceptance set only, and no bars change colors across sets. Starting with the matriculation set, the bars fall into four categories. The first and most pronounced category is experience; older applicants and those with work experience are more likely to succeed, and the committee's set may not give them enough credit. The second category is English proficiency, as applicants who score low on the TOEFL/IELTS are less likely to succeed. The third category is the subtext of the application essays. Applicants with a high percentage of positive words in their essays are more likely to succeed, and those with a high percentage of economics words (i.e. words like microeconomics,

[25] Unlike a 95% confidence interval, the whiskers do not extend to the 2.5th and 97.5th percentiles, since optimality, namely the committee's average success probability being at least as high as the model's, is one-sided.




(a) Acceptance Set

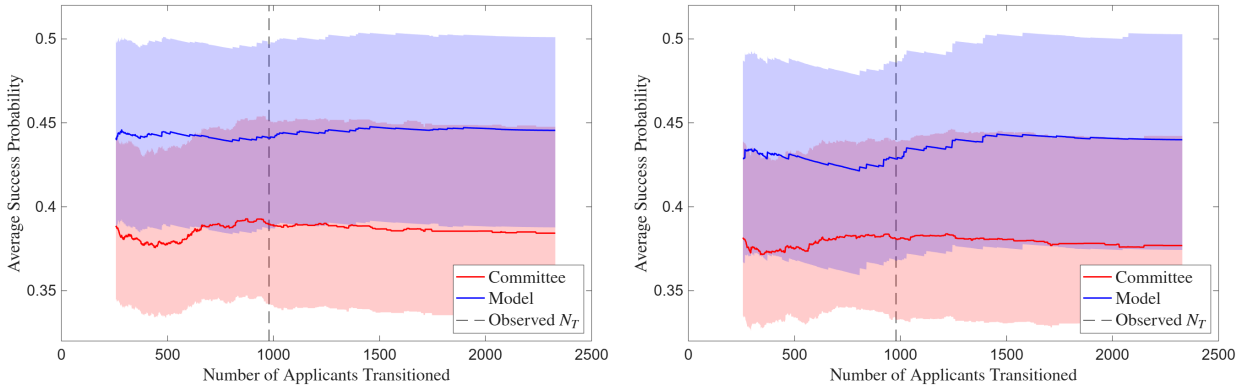
(b) Matriculation Set

Figure 5: Deviation Gains Attribution

macroeconomics, econometrics, etc.) are less likely to succeed. The fourth category relates to applicants with more of an industry background. In particular, applicants with a high number of coding languages on their CVs are less likely to be accepted despite not being less likely to succeed. Finally, *Elite U.S. University* shows up for the acceptance set but not the matriculation set. Applicants from an elite U.S. university are more likely to be accepted and not more likely to succeed.[26] However, they are also less likely to matriculate if accepted, so accepting them in greater numbers does not meaningfully decrease the matriculation set's average success probability. As a caveat, the whiskers show that for most of the variables, the gain turns negative for a minority of perturbations, meaning that these gains are small relative to the sampling error.

There are two additional points worth making. First, as expected, there is substantial overlap between the variables that survive the lasso in the sixth column of Table 12 in Appendix B and the variables that show up in Figure 5. Second, the AMEs in Table 2 of Section 3 hint at the attribution results. For example, *Work Experience* has an AME of 0.0140 for *Accept* vs. 0.0358 for *Success*, and Figure 5 shows it as underweighted. However, there are other variables in Table 2 whose *Accept* AMEs do not line up with their *Success* AMEs, yet are neither under nor overweighted. For example, *Female* has a positive *Accept* AME and *International Asian* has a negative *Accept* AME, both variables have insignificant *Success* AMEs, and neither variable shows up in Figure 5. The reason is that there are enough strong applicants with *Female* $= 1$ and *International Asian* $= 0$ that the committee's decision rule does not affect the average success probability of the acceptance or matriculation sets. So if these two *Accept* AMEs resulted from a diversity objective, that objective was nonetheless also consistent with a success probability-maximizing objective. In contrast, *CV Coding* has a negative *Accept* AME, an insignificant *Success* AME, and it shows up as underweighted in Figure 5. There is no way to know from reading Table 2 that the results would work out that way, which demonstrates why a model is necessary to relate acceptance AMEs to success AMEs when drawing any conclusions about optimality.

[26] Bai et al. (2022) find that attending an Ivy Plus undergraduate institution predicts success. However, unlike their university, this paper's university is less likely to receive applications from the strongest Ivy Plus students.




(a) Acceptance Set

(b) Matriculation Set

Figure 6: Benefit of Reading More Files

Table 2 also shows that *GRE Quantitative Percentile*'s AME is positive and significant until I control for *Program Rank* and *Missing Program Rank*, at which point it is no longer significant. Controlling for *Program Rank* guards against the possibility that the quantitative GRE is only positively correlated with success because a high percentile is often required to get into a good PhD program, which in turn is often required to attain a good placement. In that case, it would be incorrect to recommend selecting based on the quantitative GRE. However, *GRE Quantitative Percentile*'s insignificant AME does not necessarily mean that selecting on it is incorrect. For instance, this university screens out applicants with low percentiles. It is possible that despite the insignificant AME, such applicants would not do well at this university. The reason is that there are very few applicants at top PhD programs with low percentiles, so the data cannot identify well what would happen if someone with a low percentile attended a top program. Also, Figure 5 shows that there is no evidence that the committee is overweighting *GRE Quantitative Percentile*.

Finally, in Figure 6, I present the results from a counterfactual in which I allow $N_T$, which is the number of files the committee chooses to transition to Round 2, to vary within each year. Assuming the committee reads every accepted applicant's file, the minimum value of $N_T$ is $N_A$. And the maximum value of $N_T$ is the total number of applicants $N$. This counterfactual seeks to answer the question of whether reading more files would have helped the committee attain an acceptance or matriculation set with a higher average success probability. I find little evidence that reading more files would have helped. The committee's line is effectively flat across files, and while parts of the model's line are increasing and concave (indicating positive but diminishing returns to reading more files), the increase in files read all the way from $N_A$ to $N$ barely changes the average success probabilities. Therefore, reading more files appears less effective than reweighting the information attained from the objective variables and the files already read.

However, in a different setting where the curve is increasing and concave, the same counterfactual would support a revealed-preference estimate of the dollar value the committee places on success, in the spirit of the value of a statistical life literature, which infers the value of a life

from the wage premium that workers demand for bearing risk. Let $\overline{P}(n)$ be the average success probability of the acceptance set in a given year when $n$ files are transitioned. Since the number of acceptances $N_A$ is fixed, the expected number of successes is $N_A\overline{P}(n)$. For the matriculation set version of this counterfactual, $N_A$ in the objective becomes $N_M$. One committee member reads each file, so if they spend $h$ hours on it at an hourly wage of $w$, then transitioning one more file costs $hw$. If the committee values each success at $V$ dollars (in present value, since success is realized years after admission) and chooses $N_T$ to maximize $VN_A\overline{P}(n) - hwn$, then at an interior optimum, the last file read must just pay for itself. Namely, $VN_A\overline{P}'(N_T) = hw$, or $V = hw/(N_A\overline{P}'(N_T))$. The denominator is the number of additional expected successes the last file bought, so $V$ is the cost the committee was willing to incur per additional expected success. Also, the cost's wage-free counterpart $h/(N_A\overline{P}'(N_T))$ is in hours per additional expected success. Because this is a marginal calculation, I must calculate the slope of the curve at $N_T$. Since the curve is a step function, I would fit it to a smooth increasing and concave function before differentiating. I could also run the calculation in reverse. Given an outside estimate of $V$, such as the value of an additional top placement to a department's ranking, comparing $VN_A\overline{P}'(N_T)$ to $hw$ tests whether the committee reads too few or too many files without assuming $N_T$ is optimal.

## 5.3 Robustness of the Correction

In the proof of Theorem 5, I show that if the function $\mathbb{E}[\overline{P}^{\max}(\beta_0 + \sqrt{\tau}\eta)]$ is linear in $\tau$, then the bootstrap correction removes all of the winner's curse. However, the possibility remains that this function is concave in $\tau$, in which case due to a potential lack of information in the application files predicting success, the correction might only be removing part of the curse. There is no way to know for sure without knowing the true parameters, but there is a way, as follows, to find evidence of whether there is enough information in the files for the correction to be effective.

In the no-information process where $\beta_0 = 0$, every applicant in a year has the same success probability at this university's rank. Therefore, every feasible set has the same true average success probability, and the committee's decision rule is optimal, as is every other rule. Between that process and the estimated one lies the family of processes $\theta_S(\tau)$ for signal scale $\tau \in [0,1]$. Under $\theta_S(\tau)$, applicant $n$'s success probability is the logistic function of the sum of two terms: the applicant's index under the intercept, year, and program-rank parameters estimated with $\beta = 0$, and $\tau$ times the applicant-level index $w_n'\hat{\beta}^U$ centered at its sample mean, which holds the applicant pool's average success probability approximately fixed. At $\tau = 0$, the application file carries no information, whereas at $\tau = 1$, its applicant-level parameters are the estimates.

Under every $\theta_S(\tau)$, the true deviation gain, $\overline{P}_{\mathcal{A}^R(\theta_S(\tau))}(\theta_S(\tau)) - \overline{P}_{\mathcal{A}^U}(\theta_S(\tau))$, is known. At every $\tau \in \{0, 0.25, 0.5, 0.75, 1\}$, I draw 500 Bernoulli random outcome vectors from $\theta_S(\tau)$ and refit the success model on each outcome vector, once with all parameters free and once with $\beta$ restricted to zero. I then record each outcome vector's likelihood-ratio (LR) statistic for the null hypothesis that the applicant-level parameters are zero; the average LR statistic rises with $\tau$ from

Table 6: Calibrated Deviation Gains

| | Acceptance Set | Matriculation Set |
|---|---|---|
| Observed Gain, Uncorrected | 0.1015 | 0.1124 |
| Observed Gain, Corrected | 0.0514 | 0.0475 |
| Perturbed Gain, Uncorrected | 0.0937 | 0.1031 |
| Perturbed Gain, Corrected | 0.0435 | 0.0382 |
| Likelihood-Ratio P-Value | 0.1166 | 0.1166 |
| Calibrated Signal Scale | 0.4197 | 0.4197 |
| Calibrated In-Sample Gain | 0.1242 | 0.1481 |
| Calibrated True Gain | 0.0491 | 0.0560 |

its null value. The idea is to see which $\tau$ generates a dataset most similar to the actual dataset.

The LR statistic in the data is 46.3094 under 36 degrees of freedom, which per Table 6 yields a p-value of 0.1166. Linearly interpolating the simulated average LR statistics in $\tau^2$, I find that the scale at which they equal 46.3094 is $\tau = 0.4197$. At that scale, the true optimal gains are 0.0491 and 0.0560 for the acceptance and matriculation sets, and the corresponding winner's curses are 0.0751 and 0.0922. The bootstrap estimates of the curses, which are the differences between the first two rows of Table 6 (about 0.0502 and 0.0649), recover about two-thirds of the curses. For that reason, the corrected observed gains of 0.0514 and 0.0475 might exceed the true gains. However, the corrected observed gains are close to the calibrated true gains. The reason is that while the corrections are estimated to leave about one-third of the curses, the observed in-sample gains of 0.1015 and 0.1124 fall below the 0.1242 and 0.1481 that the calibrated process produces on average, a discrepancy that lies within the ordinary range of this process.

Meanwhile, the perturbed gains are lower than the observed gains because of Jensen's inequality. Perturbing the parameters moves a set's average success probability upward where the logistic function is convex, which occurs when the success probabilities in that average are below one half. The further below one half the success probabilities are, the greater the effect of the perturbations. Therefore, in Equation (37), the committee's acceptance set gains more from the perturbations than the model's set, the second bracketed term exceeds the first on average, and the perturbed gain falls below the observed gain. In turn, the corrected perturbed gains in Table 6, which are also those in Figure 4, end up as the most conservative estimates of the true gains.

My overall interpretation of these results is that while some of the uncorrected gains are a result of the winner's curse, and I cannot rule out the possibility that the committee was optimal, the evidence suggests that the committee was leaving a few percentage points of gains on the table. Still, these gains are small enough in magnitude that I would not recommend overhauling the committee's admissions process based on this model, especially since the committee may have objectives beyond selecting on success probabilities, such as diversity. Rather, I would give the committee the breakdown of the results and let them use that information as they see fit.

# 6 Conclusion

This paper studies how decision-makers select among options whose true quality is uncertain and costly to evaluate. Graduate admissions is a natural setting for such a study, as it features a low-cost screening process followed by a high-cost review of the application files that survive the screen. Because not all accepted applicants matriculate, the committee must also consider matriculation probabilities when making admissions decisions. By structurally modeling the committee's decision problem, I develop the first framework in the economics graduate admissions literature that links application information to admissions decisions and post-program success. This framework makes optimality testable if the model's set is exogenously chosen via a holdout set. With a parametric bootstrap correction against overfitting, it also makes gains from deviating to the optimal decision rule quantifiable regardless of whether a holdout set is feasible.

Before estimating structural models, I start with regressions of acceptance, matriculation, and success dependent variables on information in application files from a research university's economics PhD program in 2015–2019. I find that different independent variables best predict each dependent variable. The score the committee gives every applicant whose file it read best predicts acceptance, and attending the university's open house best predicts matriculation if accepted. I use program completion and placement rank to measure success, and my primary success measure is a binary variable that equals 1 if the applicant completed a PhD program and attained a top 100 or top 100-equivalent placement. The best predictor of success is the rank of the PhD program the applicant attended, though other variables like age, work experience, and TOEFL/IELTS scores also predict success, with program rank held equal. To avoid selection bias, I use publicly-available information from the Internet to track PhD program and placement outcomes for all applicants, not just those who attended this university. I also regress acceptance on application file variables using a second dataset of this university's economics, government, history, linguistics, and psychology PhD programs from 2020–2023, and I find that economics admissions is the most data-driven, based on variables such as GRE scores, of the five programs.

There are papers in the graduate admissions literature dating back to Krueger and Wu (2000) that qualitatively compare the determinants of acceptance to those of success, but I show that while such comparisons can provide useful information, they are insufficient to determine whether an admissions committee is optimizing. If I were to test whether the average marginal effects (AMEs) of variables in application files, such as GRE scores, are the same in an acceptance regression vs. a success regression, I would not be testing optimality. As I show at the end of Section 3, equal AMEs are neither necessary nor sufficient for optimization. Therefore, any assessments of optimality must be based on models of the committee's selection problem that account for evaluation costs and multi-stage decision-making. For this reason, I develop and estimate such models of the committee's problem in order of increasing complexity.

I start with two-round dynamic, static, and myopic models based on cutoff rules. In these models, there is a Round 1 screen based on only objective variables like GRE scores, followed by

a Round 2 final decision based on objective and subjective variables, such as those constructed by textual analysis of recommendation letters. I then develop a two-round matriculation-dynamic model, in which not every accepted applicant matriculates, and the committee considers matriculation probabilities in Round 1. The reason for doing so is to ensure that the committee can fill its cohort with applicants who do not have particularly weak subjective variables. I derive all these models from dynamic portfolio choice problems that the committee faces, and in Section 4 and Appendix D, I solve for the cutoff rule solutions to those problems. I estimate the models on this university's data and back out the committee's Rounds 1 and 2 cutoffs in each year.

After estimating the structural models, I use the estimates to compare each two-round model's chosen set to the committee's chosen set. For each model, I simulate the acceptance and matriculation processes, the latter based on estimated matriculation probabilities, to attain two acceptance sets and two matriculation sets: one each for the committee and one each for that particular model. I select the model's acceptance and matriculation sets using the unrestricted success probability parameter estimates, but I calculate the average success probability of each set using perturbations of the estimates about their asymptotic distribution. I also prove that if the model's set is exogenously chosen, the fraction of perturbations where the model's set's average success probability does not exceed the committee's is a p-value for the null hypothesis that the model's acceptance or matriculation set's average success probability is no greater than that of the committee. In a setting with many observations, such as law school admissions with tens of thousands of applicants and hundreds of matriculants per year, a holdout set can be used to make the model's set exogenously chosen. But for economics PhD admissions with much smaller cohorts, a holdout set is not feasible, and I stick to estimating the magnitude of the deviation gains.

I find that the models' decision rules exhibit deviation gains of 4 percentage points relative to the committee's rule, though I cannot establish that these gains are statistically significant, nor can I rule out the possibility that the committee is optimal. For example, the average success probability of the committee's matriculation set is 39%, whereas that of the model's set is 43%. These gains may be attributed to experience (in the form of *Age* and *Work Experience*), English proficiency (*TOEFL/IELTS* scores), application essay subtext (a higher rate of positive terms predicts success, and a higher rate of economics jargon terms predicts failure), and industry backgrounds (*CV Coding* predicts rejection but not failure). These gains are corrected for the "winner's curse" in I. Andrews et al. (2024), which occurs when the estimates used to make selections are the same estimates used to evaluate those selections. That said, a 4 percentage point gain still may not be possible if today's committee were to use the 2015–2019 model's success probability parameter weights, as the determinants of success may have changed since 2019. For example, the rate of positive words in essays may no longer predict success if applicants are now using AI to write their essays. I also run a counterfactual in which I allow the number of files that the committee transitions to Round 2 to vary. The goal is to determine whether reading more files would increase the average success probability of the acceptance or matriculation set, and if so, by how much. But I find no evidence that it would increase the average success probability of

either set, as the committee's curve is essentially flat in the number of files transitioned.

In addition to its contributions to the graduate admissions literature and to literatures on dynamic selection and costly evaluation, this paper contributes to the literature on synthetic data and privacy protection. Recent economics papers use synthetic data provided by the U.S. Census Bureau, but this paper is the first to design and implement a comprehensive synthetic data protocol from the construction of the synthetic data to the final estimation results. As a result, it serves as a template for researchers who wish to work with confidential data that they are not allowed to access directly, which can help them make discoveries they could not have otherwise.

More broadly, this paper provides a tractable framework for studying selection under costly evaluation. By doing so, it advances the graduate admissions literature from identifying variables that predict acceptance and success to a framework for testing whether decisions were optimal conditional on time and effort constraints. This study is based on one institution, so the specific empirical results reflect that setting. However, the framework applies broadly to environments where due to evaluation costs, decision-makers must employ a multi-stage process to select among their available options. Such environments range from hiring committees to clinical trials of medicines. In particular, in a setting with more observations and/or less effective decision-makers, such as hiring at a large company with bias-prone hiring managers, the framework may be more likely to find evidence in favor of large deviation gains. Moreover, admissions committees may pursue objectives beyond maximizing success probabilities, such as diversity or research field balance. The framework can be extended to incorporate such objectives into the committee's problem. Also, a worthwhile avenue for future research would be collecting data from multiple universities to model the applicant's decision problem along with the committee's and compute equilibrium admissions outcomes for applicants across universities.

Ultimately, the methodology in this paper can help decision-makers make better selection decisions based on *ex-ante* information when faced with uncertainty and costly evaluation. Crucially, decision-makers may be able to improve selection outcomes without having to increase evaluation effort. Instead, they can use the paper's framework to refine how they use that information, just as data analytics has helped organizations, such as the Oakland Athletics Major League Baseball team as described in the book *Moneyball* (Lewis 2004), with their decision-making. That said, predicting success remains difficult. Like Athey et al. (2007), I find that variables in application files have limited explanatory power for placement outcomes, and traits such as motivation are hard to measure even with proxies, such as grindstone terms in recommendation letters. Moreover, my results are relative to the information that my variables capture, so the files may contain information that neither the committee nor the model uses. A large language model run locally, keeping applicant information confidential, could read each file in full and reveal whether such information exists. If so, then that information could make larger gains possible. Nonetheless, modest improvements in selection can still be worthwhile. And while any individual university's gains may shrink if other universities adopt similar methods, much as other baseball teams eroded the Athletics' advantage by also adopting analytics, better selection can improve how students are matched to programs, making its value positive-sum.

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# Appendices

## A Variables, Rankings, and Dictionaries

This table contains a list of the variables in this paper. Variables available only for the economics program are bold, and those available for all but economics are italic. *#Math Courses* and *Advanced Math* became objective variables starting in 2020 due to a change in the university's online application system. PhD program rankings are from U.S. News, and placement rankings are from IDEAS/RePEc. The dictionaries for the subjective independent variables are from Madera, Hebl, and Martin (2009), Young and Soroka (2012), and Bai et al. (2022). These dictionaries list word roots (e.g. diligen*), and I match words against the roots directly. A word counts for a dictionary if it begins with the same letter as, and contains, one of that dictionary's roots, which credits inflected and derived forms (e.g. diligent, diligence) to their root. Each dictionary is counted independently against the full document, a word counts at most once per dictionary, and a word containing roots from several dictionaries counts in each of those dictionaries.

Table 7: Variables

(a) Dependent

| Admission | Success |
|---|---|
| **𝟙(Transition)** | **𝟙(Complete Program)** |
| 𝟙(Accept) | **𝟙(Academic Placement)** |
| **𝟙(Waitlist)** | **𝟙(Government Placement)** |
| **𝟙(Open House)** | **𝟙(Consulting Placement)** |
| 𝟙(Matriculate) | **𝟙(Tech Placement)** |
| **𝟙(Attend Program)** | **𝟙(Other Placement)** |
| **Program Rank** | **𝟙(No Placement)** |
| | **Placement Rank** |
| | **𝟙(Top 100 Placement)** |

(b) Objective Independent

| Demographic | Performance |
|---|---|
| **Year** | GRE Verbal Percentile |
| *𝟙(Covid Year)* | GRE Quantitative Percentile |
| Age | GRE Analytic Percentile |
| 𝟙(Female) | Undergraduate GPA |
| **𝟙(U.S.)** | TOEFL/IELTS |
| *𝟙(U.S. African/Hispanic/Native)* | 𝟙(Elite U.S. University) |
| *𝟙(U.S. Asian)* | 𝟙(Elite U.S. Liberal Arts College) |
| *𝟙(U.S. Other)* | 𝟙(Other U.S. University/College) |
| 𝟙(International Asian) | 𝟙(Elite Foreign University) |
| 𝟙(International Other) | 𝟙(Other Foreign University) |
| | 𝟙(Graduate Degree) |
| | 𝟙(Work Experience) |
| | #Professor Recommenders |
| | #Prolific Recommenders |
| | *Concentration Dummy Variables* |

(c) Subjective Independent

| Score/Recommendation Letters | Supplement/CV/Essay |
|---|---|
| **Committee Score** | **#Math Courses** |
| **Letters Length** | **𝟙(Advanced Math)** |
| **Positive Terms (%) (YS 2012)** | **CV Length** |
| **Negative Terms (%) (YS 2012)** | **CV Coding** |
| **Standout Terms (%) (BEMS 2022)** | **CV 𝟙(Publication/Working Paper)** |
| **Ability Terms (%) (BEMS 2022)** | **Essay Length** |
| **Research Terms (%) (BEMS 2022)** | **Essay 𝟙(Specific Professor)** |
| **Grindstone Terms (%) (BEMS 2022)** | **Essay Economics Terms (%)** |
| **Teaching Terms (%) (BEMS 2022)** | **Essay Research Terms (%)** |
| **Communal Terms (%) (BEMS 2022)** | **Essay Positive Terms (%)** |
| **Agentic Terms (%) (MHM 2009)** | **Essay Negative Terms (%)** |
| | **Field Dummy Variables** |

## Academia-Equivalent Rankings (Government)

- **European Institutions:** European Central Bank (40), HM Treasury (80)
- **Foreign Central Banks:** Banco Central do Brasil (93), Banco de México (93), Bank of Canada (80), Bank of England (67), Bank of Israel (93), Bank of Italy (80), Bank of Japan (80), Bank of Spain (80), Banque de France (80), Central Bank of Chile (93), Central Bank of Colombia (93), Central Bank of Korea (93), Central Bank of Turkey (93), Norges Bank (80), Reserve Bank of Australia (80), Reserve Bank of India (93), Reserve Bank of New Zealand (93), Sveriges Riksbank (80), Swiss National Bank (80)
- **Multilaterals/International:** African Development Bank (93), Asian Development Bank (93), Bank for International Settlements (53), European Bank for Reconstruction and Development (93), European Investment Bank (93), Inter-American Development Bank (93), International Monetary Fund (40), Statistics Canada (107), World Bank (40)
- **Policy Research Institutes:** American Enterprise Institute (93), American Institutes for Research (107), Becker Friedman Institute (80), Brookings Institution (80), Center for Global Development (93), Center for Migration Studies in New York (112), Center for Strategic and International Studies (107), Cowles Foundation (80), Harris School Policy Labs (93), Hoover Institution (80), IDinsight (112), Innovations for Poverty Action (93), J-PAL (80), MDRC (107), National Bureau of Economic Research (67), Peterson Institute for International Economics (80), Pew Research Center (107), RAND Corporation (80), Resources for the Future (93), Urban Institute (107)
- **U.S. Executive/Cabinet:** Department of Commerce (107), Department of Justice (80), Department of Labor (107), Department of the Treasury (80)
- **U.S. Executive Policy Offices:** Council of Economic Advisers (80), Office of Management and Budget (93)
- **U.S. Federal Reserve System:** Board of Governors (40), Atlanta (80), Boston (80), Chicago (80), Cleveland (80), Dallas (80), Kansas City (80), Minneapolis (80), New York (67), Philadelphia (80), Richmond (80), San Francisco (80), St. Louis (80)
- **U.S. GSE/Housing Finance:** Fannie Mae (120), Freddie Mac (120)
- **U.S. Independent Agencies:** Commodity Futures Trading Commission (93), Federal Deposit Insurance Corporation (93), Federal Energy Regulatory Commission (93), Federal Housing Finance Agency (93), Federal Trade Commission (80), Office of Financial Research (93), Securities and Exchange Commission (93)
- **U.S. Legislative Committees:** Congressional Budget Office (80), Congressional Research Service (107), Joint Committee on Taxation (93), Joint Economic Committee (107)

- **U.S. Science and Research:** National Science Foundation (107)
- **U.S. Statistical Agencies:** Bureau of Economic Analysis (93), Bureau of Labor Statistics (93), Census Bureau (93)

## Academia-Equivalent Rankings (Consulting/Technology/Other)

- **Consulting:** Aecon Consulting (130), AlixPartners (125), Analysis Group (120), Bain & Company (120), Bates White (120), Berkeley Research Group (120), Boston Consulting Group (120), Brattle Group (120), Charles River Associates (120), Coleman Research (130), Compass Lexecon (120), Cornerstone Research (120), Deloitte Economic and Financial Advisory (125), Econic Partners (130), Economists Incorporated (120), Edgeworth Economics (120), Ernst & Young Economic Advisory (125), Frontier Economics (120), FTI Consulting (120), Guidehouse (130), KPMG Economics & Valuation (125), Matrix Economics (130), McKinsey & Company (120), Mercer (135), NERA Economic Consulting (120), Oliver Wyman (120), PwC Economics (125), Roland Berger (125), Willis Towers Watson (135)
- **Tech (Athey & Luca 2019):** Airbnb (107), Alibaba (120), Amazon (93), AppNexus (135), Apple (107), Block/Square (120), ByteDance/TikTok (120), CoreLogic (135), Coursera (120), DStillery (135), Didichuxing (125), Digonex (135), DoorDash (107), eBay (107), ECONorthwest (125), Expedia (120), Forkcast (135), Glassdoor (120), Google (93), Granular (135), Groupon (125), Houzz (130), Huawei (120), IBM (120), Indeed (120), ING (130), Instacart (120), Intel (120), Kensho (125), Lending Club (135), LinkedIn (107), Lyft (120), Meta/Facebook (93), Microsoft (93), Netflix (93), Nuna (135), Nvidia (120), Pandora (135), PayPal (120), Pinterest (120), Prattle (135), Quantco (130), Quora (130), Redfin (120), Ripple (135), Roblox (120), Rover (135), Salesforce (120), Shopify (120), Spotify (120), Stripe (107), Trulia (125), Uber (107), Upwork (135), Visa (125), Walmart (125), Wealthfront (135), Yahoo (125), Yelp (125), Zillow (120)
- **Tech Labs:** Amazon Science (40), Apple Machine Learning Research (80), DeepMind (80), Google Research (40), Meta Research (80), Microsoft Research (40), Netflix Research (93), OpenAI Research (80), Uber Research (93)
- **Other:** AIG (135), Bank of America (125), Bloomberg (125), Boeing (135), Capital One (125), Exelon (135), ExxonMobil (135), Ford (135), General Electric (135), General Motors (135), Goldman Sachs (125), JPMorgan Chase (125), Liberty Mutual (135), Moody's Analytics (125), Morningstar (125), Nielsen (135), PNC (135), Progressive (135), S&P Global (125), Shell (135), Swiss Re (130), Toyota (130), Other (135), No Placement (150)

## Dictionaries (Recommendation Letters)

- **Positive/Negative Words:** Young and Soroka (2012)'s Lexicoder Sentiment Dictionary
- **Standout, Ability, Research, Grindstone, Communal Words:** Bai et al. (2022)
- **Agentic Words:** assertive, confident, aggressive, ambitious, dominant, forceful, independent, daring, outspoken, intellectual (Madera et al. 2009); **AND** audacious, bold, command, compete, decisive, determine, drive, enterprise, fearless, influential, leader, pioneer, powerful, proactive, resilient, resourceful, self-assured, strategic, tenacious, vision
- **Teaching Words:** academic, assess, clarity, coach, collaborate, communicate, cultivate, curriculum, demonstrate, develop, educate, encourage, engage, enlighten, evaluate, explain, facilitate, foster, guide, impart, inspire, insight, instruct, knowledge, learn, lesson, mentor, nurture, participate, pedagogy, present, scholar, support, teach, train, tutor, workshop. (Note: I constructed this dictionary because it is unavailable in Bai et al. 2022.)

## Dictionaries (Application Essays/CVs)

- **Coding:** c/c++, eviews, fortran, java, jax, julia, latex, matlab, python, r, sas, stata, webscrape
- **Publication:** Journal names from IDEAS/RePEc Aggregate Rankings for Journals
- **Working Paper:** arxiv, center for economic policy research, cepr, econpapers, national bureau of economic research, nber, research papers in economics, repec, social science research network, ssrn, working paper
- **Economics:** applied, asset, behavior, climate, computation, data, development, dynamic programming, economics, education, empirical, environment, experiment, finance, fiscal, game theory, health, immigration, industrial, inequality, international, io, labor, linear programming, macro, math, metrics, micro, monetary, policy, political, poverty, pricing, probability, public, sports, statistic, theory, trade, urban

## A.1 Academia-Equivalent Prompt and Link to Rankings

*I am writing a research paper that entails using data in application files to predict success in economics PhD programs, where success is defined as the quality of the first post-PhD job placement. For academic jobs, I use RePEc rankings. For non-academic jobs, I will follow the same procedure as Krueger and Wu (2000):*

*"Another issue concerns the ranking of the various job placements. For our analysis, we assumed that one goal of admissions and graduate training in economics is to place students in leading research jobs, although we recognize that different departments have different objectives. In ranking academic institutions, we relied primarily on the ranking of the top 100 economics departments based on faculty publications in elite journals produced by Scott and Mitias (1996). Students who were placed in business school jobs were given a rank equal to their university's economics department ranking plus five. (A lower rank signified a more prestigious job placement.) The World Bank, IMF, and Federal Reserve Board received a ranking equivalent to the 40th best economics department. Consulting jobs (e.g., NERA, Abt, DRI) were given a rank equal to the 120th best economics department. Finally, applicants who could not be found were assigned a rank of 150, the worst rank we gave, and treated as censored observations in much of the analysis."*

*I am attaching an Excel file with the RePEc rankings, as well as job titles for Government, Consulting, Tech, and Other Non-Academia. In each of the latter four sheets, please fill in the "Academia-Equivalent Ranking" column with the ranking you deem most appropriate. Keep it as consistent as possible with Krueger and Wu, meaning that the best non-academic placements should receive a ranking of 40. Krueger and Wu deem those placements to be the World Bank, IMF, and Federal Reserve Board (that is, the Board of Governors and not the regional banks). However, because the economics profession has changed since 2000, you can give additional placements a ranking of 40. Limit any rankings of 40 to the Government sheet and the "Tech Research Labs" part of the Tech sheet. Be very strict about which, if any, additional placements you give a 40, since Krueger and Wu gave only three 40s, each of which were to very prestigious placements. You can give at most three additional 40s in Government, and you can give at most three 40s in Tech.*

*Regarding rankings other than 40 that you can give, you can give 80, 120, and 135. The logic behind 80 is that it is halfway between 40 and 120, and the logic behind 135 is that it is halfway between 120 and 150. Do not give any placements a rank of 150; 150 is reserved for applicants who cannot be found. 40 is for placements in the World Bank tier, 80 is for placements between the World Bank and NERA tiers, 120 is for placements in the NERA tier, and 135 is for placements that are below the NERA tier. If you ever give a Consulting or Other Non-Academia placement a ranking better than 120, make sure you have a very sound reason for doing so because Krueger and Wu capped rankings for consulting placements at 120. Please post all your academia-equivalent rankings in the chat as well, as I will save a screenshot of the chat.*

## A.2 Synthetic Data Example

As a demonstration of synthetic data, I use a Gaussian copula to simulate 1,000 observations of the following independent variables: *GRE Quantitative Percentile* (continuous), *Gender* (binary), *Demographic* (categorical with three demographics), *Advanced Math* (binary), and *Committee Score* (continuous). Then I use the standard logistic function to simulate binary dependent variables for transitioning past the first-round screen, getting waitlisted, getting accepted into the program, matriculating at the program if accepted, and obtaining a successful placement above a threshold ranking. I also generate the (continuous) rank of the PhD program each applicant attends, which is constant for the applicants with $Matriculate = 1$ and equals 150 for applicants who do not attend a PhD program (for these applicants, $Success = 0$). The top frame of Figure 7 contains the first ten observations of this simulated dataset. *Committee Score* is missing for all applicants who did not survive the first-round screen, which is a feature of the actual data.

I then pass this dataset through another Gaussian copula and use that copula to generate a synthetic dataset, the first ten observations of which are in the bottom frame of the figure. The means, variances, and correlations of the variables are similar in both datasets, with only the correlations in the synthetic dataset slightly attenuated. However, none of the rows of the simulated dataset are present in the synthetic dataset. In addition, important statistical properties; such as *Accept* equaling 1 only when *Transition* equals 1, *Demographic 1* and *Demographic 2* never both equaling 1, and *Committee Score* missing when *Transition* equals 0; are preserved.

| Transition | Waitlist | Accept | Matriculate | Success | GRE Quantitative Percentile | Gender | Demographic 1 | Demographic 2 | Advanced Math | Committee Score | Program Rank |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 70 | 1 | 0 | 1 | 0 | · | 27 |
| 0 | 0 | 0 | 0 | 0 | 92 | 1 | 1 | 0 | 1 | · | 150 |
| 0 | 0 | 0 | 0 | 0 | 85 | 1 | 1 | 0 | 0 | · | 80 |
| 0 | 0 | 0 | 0 | 0 | 72 | 1 | 0 | 0 | 1 | · | 51 |
| 0 | 0 | 0 | 0 | 0 | 74 | 0 | 1 | 0 | 0 | · | 150 |
| 0 | 0 | 0 | 0 | 0 | 79 | 0 | 0 | 0 | 1 | · | 26 |
| 1 | 1 | 0 | 0 | 1 | 97 | 0 | 0 | 1 | 1 | 8 | 43 |
| 0 | 0 | 0 | 0 | 0 | 76 | 1 | 0 | 0 | 0 | · | 17 |
| 1 | 1 | 0 | 0 | 0 | 75 | 0 | 0 | 0 | 0 | 3 | 70 |
| 0 | 0 | 0 | 0 | 0 | 75 | 0 | 0 | 1 | 0 | · | 150 |

⇓

| Transition | Waitlist | Accept | Matriculate | Success | GRE Quantitative Percentile | Gender | Demographic 1 | Demographic 2 | Advanced Math | Committee Score | Program Rank |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 0 | 92 | 0 | 0 | 1 | 1 | 8 | 31 |
| 0 | 0 | 0 | 0 | 0 | 83 | 0 | 0 | 0 | 1 | · | 60 |
| 0 | 0 | 0 | 0 | 1 | 94 | 0 | 1 | 0 | 1 | · | 19 |
| 1 | 1 | 1 | 1 | 0 | 90 | 0 | 0 | 1 | 1 | 5 | 37 |
| 0 | 0 | 0 | 0 | 1 | 86 | 0 | 0 | 1 | 0 | · | 52 |
| 1 | 0 | 1 | 0 | 1 | 91 | 1 | 1 | 0 | 0 | 2 | 29 |
| 1 | 1 | 0 | 0 | 0 | 86 | 0 | 0 | 0 | 1 | 5 | 71 |
| 1 | 1 | 0 | 0 | 0 | 98 | 0 | 1 | 0 | 0 | 5 | 21 |
| 1 | 1 | 0 | 0 | 1 | 88 | 1 | 1 | 0 | 1 | 4 | 37 |
| 0 | 0 | 0 | 0 | 0 | 92 | 1 | 1 | 0 | 0 | · | 150 |

Figure 7: Synthetic Data Example

# B Additional Regression Results

Table 8: Summary Statistics

| | Full Sample | | Transition = 1 | | Accept = 1 | | Matriculate = 1 | |
|---|---|---|---|---|---|---|---|---|
| | Mean | SD | Mean | SD | Mean | SD | Mean | SD |
| Transition | 0.4204 | 0.4937 | 1.0000 | 0.0000 | 1.0000 | 0.0000 | 1.0000 | 0.0000 |
| Waitlist | 0.3324 | 0.4712 | 0.7686 | 0.4220 | 0.0743 | 0.2628 | 0.0896 | 0.2877 |
| Accept | 0.1103 | 0.3134 | 0.2625 | 0.4402 | 1.0000 | 0.0000 | 1.0000 | 0.0000 |
| Matriculate | 0.0339 | 0.1811 | 0.0807 | 0.2725 | 0.3074 | 0.4623 | 1.0000 | 0.0000 |
| Open House | 0.0353 | 0.1847 | 0.0817 | 0.2741 | 0.3267 | 0.4702 | 0.5522 | 0.5010 |
| Attend Program | 0.6024 | 0.4895 | 0.6752 | 0.4685 | 0.7704 | 0.4214 | 1.0000 | 0.0000 |
| Complete Program | 0.5578 | 0.4968 | 0.6210 | 0.4854 | 0.6654 | 0.4728 | 0.6709 | 0.4729 |
| Top 100 Placement | 0.1971 | 0.3979 | 0.2472 | 0.4316 | 0.3074 | 0.4623 | 0.3418 | 0.4773 |
| Program Rank | 88.9133 | 57.4823 | 78.5844 | 57.1361 | 63.3828 | 54.1845 | 37.0000 | 0.0000 |
| Missing Program Rank | 0.4199 | 0.4937 | 0.3422 | 0.4747 | 0.2490 | 0.4333 | 0.0000 | 0.0000 |
| Placement Rank | 127.7505 | 34.9306 | 123.0222 | 37.2570 | 117.2696 | 40.5746 | 117.4430 | 37.5728 |
| Missing Placement Rank | 0.4444 | 0.4970 | 0.3830 | 0.4864 | 0.3346 | 0.4728 | 0.3291 | 0.4729 |
| 2016 | 0.2052 | 0.4040 | 0.1696 | 0.3754 | 0.1790 | 0.3841 | 0.2025 | 0.4045 |
| 2017 | 0.2014 | 0.4011 | 0.1788 | 0.3833 | 0.1829 | 0.3873 | 0.1772 | 0.3843 |
| 2018 | 0.1928 | 0.3946 | 0.2686 | 0.4435 | 0.2062 | 0.4054 | 0.2532 | 0.4376 |
| 2019 | 0.2027 | 0.4021 | 0.2084 | 0.4064 | 0.2179 | 0.4136 | 0.2152 | 0.4136 |
| Age | 25.9910 | 3.1479 | 26.2635 | 3.1823 | 26.0817 | 3.0652 | 26.3291 | 3.2806 |
| Female | 0.3912 | 0.4881 | 0.4147 | 0.4929 | 0.4397 | 0.4973 | 0.4177 | 0.4963 |
| U.S. | 0.2357 | 0.4245 | 0.2380 | 0.4261 | 0.2996 | 0.4590 | 0.3038 | 0.4628 |
| International Asian | 0.4809 | 0.4997 | 0.5230 | 0.4997 | 0.3424 | 0.4754 | 0.4304 | 0.4983 |
| GRE Verbal Percentile | 68.0129 | 22.5869 | 76.8421 | 18.3703 | 79.5703 | 18.3047 | 76.9744 | 19.4041 |
| GRE Quantitative Percentile | 88.4550 | 11.6330 | 93.1362 | 5.4135 | 93.5391 | 4.4614 | 92.3077 | 5.7554 |
| GRE Analytic Percentile | 50.0613 | 27.5744 | 56.3653 | 26.5466 | 60.3398 | 26.0880 | 57.4615 | 26.6481 |
| Undergraduate GPA | 3.6307 | 0.3026 | 3.6645 | 0.3003 | 3.7227 | 0.2610 | 3.7068 | 0.2880 |
| TOEFL/IELTS | 7.4376 | 0.5587 | 7.5641 | 0.5834 | 7.7011 | 0.6667 | 7.4464 | 0.6137 |
| Elite U.S. University | 0.1267 | 0.3327 | 0.1736 | 0.3790 | 0.2257 | 0.4188 | 0.1772 | 0.3843 |
| Elite U.S. Liberal Arts College | 0.0296 | 0.1696 | 0.0347 | 0.1832 | 0.0428 | 0.2028 | 0.0633 | 0.2450 |
| Elite Foreign University | 0.0386 | 0.1928 | 0.0490 | 0.2160 | 0.0350 | 0.1842 | 0.0253 | 0.1581 |
| Other Foreign University | 0.4049 | 0.4910 | 0.4188 | 0.4936 | 0.3619 | 0.4815 | 0.4430 | 0.4999 |
| Missing Undergraduate Institution | 0.2203 | 0.4145 | 0.1910 | 0.3933 | 0.1984 | 0.3996 | 0.1266 | 0.3346 |
| Graduate Degree | 0.5067 | 0.5001 | 0.5730 | 0.4949 | 0.5136 | 0.5008 | 0.5190 | 0.5028 |
| Work Experience | 0.5247 | 0.4995 | 0.5894 | 0.4922 | 0.6770 | 0.4685 | 0.7468 | 0.4376 |
| #Professor Recommenders | 2.6230 | 0.7371 | 2.6302 | 0.7165 | 2.5992 | 0.7118 | 2.5696 | 0.7104 |
| #Prolific Recommenders | 1.2602 | 1.0686 | 1.3912 | 1.0840 | 1.4825 | 1.0424 | 1.3038 | 1.0664 |
| Missing GRE | 0.0335 | 0.1800 | 0.0102 | 0.1006 | 0.0039 | 0.0624 | 0.0127 | 0.1125 |
| Missing Undergraduate GPA | 0.7853 | 0.4107 | 0.7487 | 0.4340 | 0.6770 | 0.4685 | 0.6835 | 0.4681 |
| Missing TOEFL/IELTS | 0.6076 | 0.4884 | 0.5935 | 0.4914 | 0.6615 | 0.4741 | 0.6456 | 0.4814 |
| Missing Age | 0.0009 | 0.0293 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| Committee Score | 6.4070 | 1.8701 | 6.4070 | 1.8701 | 8.5003 | 0.8151 | 8.4854 | 0.7157 |
| #Math Courses | 5.7637 | 1.8154 | 5.8173 | 1.7904 | 5.6992 | 1.8640 | 5.6585 | 1.6219 |
| Advanced Math | 0.4860 | 0.5004 | 0.4909 | 0.5007 | 0.5680 | 0.4973 | 0.4651 | 0.5047 |
| Letters Length/100 | 8.3474 | 3.5749 | 8.5333 | 3.8263 | 10.0514 | 4.2864 | 9.7088 | 3.4876 |
| Letters Positive Terms (%) | 11.9770 | 2.2993 | 11.9976 | 2.2077 | 12.2167 | 1.9255 | 12.5110 | 2.0904 |
| Letters Negative Terms (%) | 5.7286 | 1.3427 | 5.7216 | 1.3318 | 5.7446 | 1.2313 | 5.8938 | 1.2871 |
| Letters Standout Terms (%) | 0.6241 | 0.4020 | 0.6241 | 0.4029 | 0.7151 | 0.4509 | 0.7077 | 0.5109 |
| Letters Ability Terms (%) | 1.3326 | 0.6239 | 1.3457 | 0.5964 | 1.3796 | 0.5620 | 1.4334 | 0.6037 |
| Letters Research Terms (%) | 3.4837 | 1.3319 | 3.5208 | 1.3423 | 3.8548 | 1.1984 | 3.8395 | 1.2121 |
| Letters Grindstone Terms (%) | 1.8189 | 0.7734 | 1.8674 | 0.7808 | 1.9975 | 0.7588 | 2.1600 | 0.8093 |
| Letters Teaching Terms (%) | 1.7379 | 0.7710 | 1.7489 | 0.7759 | 1.7275 | 0.7536 | 1.8976 | 0.7577 |
| Letters Communal Terms (%) | 0.5946 | 0.3890 | 0.6019 | 0.3954 | 0.5494 | 0.3234 | 0.5630 | 0.3875 |
| Letters Agentic Terms (%) | 0.4768 | 0.3150 | 0.4842 | 0.3229 | 0.5039 | 0.3222 | 0.5597 | 0.3314 |
| Letters Terms (%) PC1 | -0.0579 | 1.6183 | 0.0000 | 1.5768 | 0.2176 | 1.3817 | 0.5697 | 1.5386 |
| Letters Terms (%) PC2 | -0.0271 | 1.1177 | 0.0000 | 1.1446 | 0.1002 | 1.1774 | 0.1699 | 1.2823 |
| CV Length/100 | 4.3532 | 2.4237 | 4.3982 | 2.4758 | 4.9641 | 2.7175 | 4.8261 | 2.3473 |
| CV Coding | 2.8321 | 2.1135 | 2.7931 | 2.1000 | 2.8618 | 2.1129 | 2.5122 | 1.6450 |
| CV Publication/Working Paper | 0.2518 | 0.4346 | 0.2571 | 0.4377 | 0.3659 | 0.4836 | 0.3902 | 0.4939 |
| Essay Length/100 | 4.5099 | 1.5741 | 4.5173 | 1.5805 | 4.7488 | 1.6373 | 4.9816 | 1.7932 |
| Essay Specific Professor | 0.4822 | 0.5003 | 0.5219 | 0.5003 | 0.5691 | 0.4972 | 0.6512 | 0.4822 |
| Essay Economics Terms (%) | 9.7284 | 2.6219 | 9.7216 | 2.6635 | 9.5303 | 2.7324 | 9.8941 | 3.2608 |
| Essay Research Terms (%) | 5.7975 | 2.0501 | 5.7402 | 2.0168 | 5.7313 | 1.8797 | 5.5231 | 1.8432 |
| Essay Positive Terms (%) | 11.8293 | 2.6518 | 11.7324 | 2.6650 | 11.4734 | 2.4592 | 11.3169 | 2.4455 |
| Essay Negative Terms (%) | 6.2844 | 1.9056 | 6.3233 | 1.9712 | 6.2889 | 2.0860 | 6.2651 | 1.9436 |
| Missing Committee Score | 0.5844 | 0.4929 | 0.0112 | 0.1055 | 0.0195 | 0.1384 | 0.0633 | 0.2450 |
| Missing Application Pdf | 0.8154 | 0.3881 | 0.6650 | 0.4722 | 0.5136 | 0.5008 | 0.4557 | 0.5012 |
| Missing #Math Courses | 0.8201 | 0.3842 | 0.6701 | 0.4704 | 0.5214 | 0.5005 | 0.4810 | 0.5028 |
| Missing Letters | 0.8175 | 0.3863 | 0.6691 | 0.4708 | 0.5136 | 0.5008 | 0.4557 | 0.5012 |
| Missing CV | 0.8210 | 0.3835 | 0.6742 | 0.4689 | 0.5214 | 0.5005 | 0.4810 | 0.5028 |
| Missing Essay | 0.8192 | 0.3849 | 0.6731 | 0.4693 | 0.5214 | 0.5005 | 0.4557 | 0.5012 |
| Observations | 2329 | | 979 | | 257 | | 79 | |

# Table 9: Correlation Matrix

# Table 10a: Logistic Regression Results

| Dependent Variable | Accept | Accept | Matriculate | T100 Placement | T100 Placement | T100 Placement | Placement Rank | Placement Rank | Placement Rank |
|---|---|---|---|---|---|---|---|---|---|
| 2016 | -0.3826 | -0.5049 | -0.2577 | 0.0230 | 0.0301 | 0.0885 | 1.8038 | 1.7607 | -0.6618 |
| | (0.2365) | (0.4703) | (0.6853) | (0.1708) | (0.1711) | (0.1887) | (4.2583) | (4.2504) | (3.6266) |
| 2017 | -0.0954 | -1.3247*** | 0.1357 | 0.0829 | 0.0711 | 0.1460 | 2.7746 | 3.0235 | 0.7473 |
| | (0.2564) | (0.4881) | (0.6732) | (0.1707) | (0.1721) | (0.1934) | (4.2295) | (4.2188) | (3.7157) |
| 2018 | -0.2206 | -3.7609*** | 0.9476 | 0.0624 | -0.0392 | 0.0142 | 2.7265 | 5.2685 | 4.9770 |
| | (0.2648) | (0.6518) | (0.6200) | (0.1761) | (0.1807) | (0.2011) | (4.4414) | (4.5011) | (3.8786) |
| 2019 | -0.1059 | -3.1670*** | 0.5909 | 0.0594 | 0.0144 | 0.0366 | 5.7085 | 6.6750 | 5.2241 |
| | (0.2653) | (0.7114) | (0.6855) | (0.1821) | (0.1830) | (0.2052) | (4.6400) | (4.6266) | (4.1424) |
| Age | -0.0218 | -0.0080 | -0.1567** | 0.0240 | 0.0242 | 0.0420* | 0.3596 | 0.3632 | -0.1277 |
| | (0.0342) | (0.0592) | (0.0778) | (0.0207) | (0.0205) | (0.0236) | (0.5457) | (0.5410) | (0.4839) |
| Female | 0.3759** | 0.8784** | -0.3445 | -0.0880 | -0.1026 | -0.0256 | 1.8085 | 2.0921 | -1.6375 |
| | (0.1657) | (0.3813) | (0.3794) | (0.1126) | (0.1130) | (0.1252) | (2.8776) | (2.8750) | (2.5117) |
| U.S. | -0.4982* | -0.0681 | 1.0348* | 0.1167 | 0.1396 | 0.1520 | -0.6372 | -0.9759 | -0.7059 |
| | (0.2624) | (0.4538) | (0.6098) | (0.1903) | (0.1909) | (0.2236) | (4.7990) | (4.7913) | (4.3464) |
| International Asian | -1.1569*** | -1.1507*** | 1.1559** | 0.1695 | 0.1949 | -0.1005 | -7.8212** | -8.2008** | 3.3007 |
| | (0.2129) | (0.4146) | (0.4904) | (0.1379) | (0.1395) | (0.1546) | (3.5528) | (3.5708) | (3.1577) |
| GRE Verbal Percentile | 0.0264*** | 0.0079 | -0.0107 | 0.0049 | 0.0030 | 0.0007 | -0.1958** | -0.1470* | -0.0330 |
| | (0.0054) | (0.0130) | (0.0118) | (0.0031) | (0.0032) | (0.0036) | (0.0787) | (0.0806) | (0.0735) |
| GRE Quantitative Percentile | 0.1329*** | 0.0589** | -0.1033** | 0.0245*** | 0.0184** | 0.0047 | -0.8498*** | -0.7196*** | -0.2428* |
| | (0.0202) | (0.0275) | (0.0464) | (0.0077) | (0.0075) | (0.0084) | (0.1567) | (0.1578) | (0.1460) |
| GRE Analytic Percentile | 0.0076** | 0.0094 | -0.0119 | -0.0020 | -0.0027 | -0.0021 | 0.0656 | 0.0821 | 0.0160 |
| | (0.0039) | (0.0085) | (0.0095) | (0.0025) | (0.0025) | (0.0029) | (0.0669) | (0.0671) | (0.0618) |
| Undergraduate GPA | 1.2726* | 1.0509 | -0.7209 | -0.3976 | -0.4722 | -0.8687* | 4.8040 | 6.9311 | 17.2597** |
| | (0.6975) | (1.2453) | (1.2760) | (0.3912) | (0.3848) | (0.4442) | (10.1461) | (10.0615) | (7.7507) |
| TOEFL/IELTS | 0.4236 | -0.1165 | -0.4434 | 0.1778 | 0.1437 | 0.3240* | -2.2504 | -1.2960 | -7.6837* |
| | (0.2924) | (0.5976) | (0.5525) | (0.1571) | (0.1559) | (0.1757) | (4.3852) | (4.3431) | (3.9285) |
| Elite U.S. University | 0.7328*** | -0.1822 | -0.4980 | -0.0159 | -0.0853 | -0.1523 | -2.3790 | -0.3552 | -0.3450 |
| | (0.2774) | (0.5918) | (0.5799) | (0.2133) | (0.2133) | (0.2344) | (5.2934) | (5.2869) | (4.4943) |
| Elite U.S. Liberal Arts College | 0.0167 | 0.5446 | 0.9907 | 0.0731 | 0.0803 | -0.1010 | -3.9470 | -3.8047 | -3.4006 |
| | (0.4568) | (0.6896) | (0.8447) | (0.3370) | (0.3361) | (0.3936) | (8.8571) | (8.7916) | (8.2332) |
| Elite Foreign University | 0.1244 | -0.5507 | -0.0673 | 0.1210 | 0.0751 | 0.0969 | -2.7705 | -1.5414 | -2.2142 |
| | (0.5319) | (0.9725) | (1.1224) | (0.3239) | (0.3259) | (0.3526) | (7.4748) | (7.4679) | (6.1380) |
| Other Foreign University | 0.5323* | 0.6328 | 0.7373 | 0.1483 | 0.1091 | 0.1587 | -6.7584 | -5.7186 | -9.1260** |
| | (0.3022) | (0.6505) | (0.6347) | (0.1964) | (0.1970) | (0.2216) | (4.9679) | (4.9620) | (4.3486) |
| Missing Undergraduate Institution | 0.3260 | 0.2482 | -0.6851 | 0.2292 | 0.2082 | 0.1770 | -7.9862 | -7.5382 | -8.1758* |
| | (0.3111) | (0.6016) | (0.7260) | (0.2017) | (0.2022) | (0.2345) | (5.1133) | (5.1129) | (4.5085) |
| Graduate Degree | 0.1726 | -0.2058 | -0.2954 | -0.0406 | -0.0921 | -0.1835 | -3.2803 | -2.0242 | 2.6215 |
| | (0.1788) | (0.3845) | (0.5017) | (0.1231) | (0.1273) | (0.1370) | (3.1662) | (3.2087) | (2.7452) |
| Work Experience | 0.7785*** | 0.7519** | 1.1183** | 0.2298* | 0.1772 | 0.2839** | -3.5942 | -2.2559 | -6.4380** |
| | (0.1915) | (0.3400) | (0.5291) | (0.1253) | (0.1270) | (0.1422) | (3.2005) | (3.2301) | (2.9093) |
| #Professor Recommenders | -0.0377 | -0.1797 | 0.3587 | -0.0472 | -0.0513 | -0.0547 | 0.8765 | 0.9972 | 0.7371 |
| | (0.1079) | (0.2434) | (0.3115) | (0.0749) | (0.0754) | (0.0862) | (2.0492) | (2.0495) | (1.7160) |
| #Prolific Recommenders | 0.2208*** | 0.2081 | -0.2993 | 0.1011* | 0.0899* | 0.0600 | -1.4623 | -1.1967 | 0.1726 |
| | (0.0811) | (0.1721) | (0.2060) | (0.0528) | (0.0533) | (0.0585) | (1.3680) | (1.3670) | (1.1704) |
| Missing GRE | -1.5399 | 1.0141 | | -0.4515 | -0.3258 | 0.0174 | 11.5202 | 8.5434 | -3.7106 |
| | (0.9821) | (1.1356) | | (0.3510) | (0.3544) | (0.3797) | (8.6385) | (8.6602) | (6.1802) |
| Missing Undergraduate GPA | -0.3432 | -0.2798 | 0.0798 | -0.1010 | -0.0639 | -0.0513 | 2.4296 | 1.5256 | -0.8710 |
| | (0.2210) | (0.4413) | (0.4943) | (0.1565) | (0.1585) | (0.1756) | (3.9808) | (3.9842) | (3.3572) |
| Missing TOEFL/IELTS | 0.4719** | 0.3743 | -0.0734 | -0.1261 | -0.1447 | -0.0065 | 4.0029 | 4.3290 | -2.3792 |
| | (0.2062) | (0.4223) | (0.5891) | (0.1267) | (0.1282) | (0.1443) | (3.3209) | (3.3244) | (2.9657) |
| Committee Score | | 4.2419*** | -0.2608 | | 0.0943** | 0.0059 | | -2.5088** | 0.6884 |
| | | (0.4534) | (0.2250) | | (0.0456) | (0.0488) | | (1.1860) | (1.0132) |
| #Math Courses | -0.0949 | -0.2706 | 0.0755 | 0.0626 | 0.0662 | 0.0008 | -2.3828 | -2.4550 | 0.9116 |
| | (0.0773) | (0.2829) | (0.1680) | (0.0759) | (0.0766) | (0.0885) | (1.8033) | (1.8149) | (1.5082) |
| Advanced Math | 0.6333** | 1.3229 | -1.3706** | -0.1932 | -0.2078 | -0.2275 | 3.0400 | 3.4570 | 1.9957 |
| | (0.3036) | (0.9203) | (0.5621) | (0.2605) | (0.2639) | (0.2949) | (6.4446) | (6.4377) | (5.5819) |
| Letters Length/100 | 0.1358*** | 0.1084 | -0.0910 | 0.0067 | -0.0018 | -0.0150 | -0.2593 | -0.0078 | 0.7206 |
| | (0.0518) | (0.1424) | (0.0712) | (0.0335) | (0.0339) | (0.0388) | (0.8254) | (0.8325) | (0.7084) |
| Letters Terms (%) PC1 | 0.1507 | 0.2939 | 0.4716** | 0.0362 | 0.0251 | 0.1260 | 1.6082 | 1.8374 | -0.5612 |
| | (0.0923) | (0.2912) | (0.2051) | (0.0732) | (0.0748) | (0.0915) | (2.0561) | (2.0540) | (1.8186) |
| Letters Terms (%) PC2 | -0.0913 | -0.4980 | 0.2724 | -0.1226 | -0.1199 | -0.1420 | 0.7118 | 0.5616 | 0.1155 |
| | (0.1464) | (0.4143) | (0.2604) | (0.1162) | (0.1177) | (0.1272) | (2.7992) | (2.8081) | (2.3450) |
| CV Length/100 | 0.0361 | 0.0559 | -0.1986 | 0.0301 | 0.0274 | 0.0352 | -1.0456 | -1.0120 | -0.8208 |
| | (0.0617) | (0.1186) | (0.1337) | (0.0562) | (0.0563) | (0.0593) | (1.3224) | (1.3130) | (1.0252) |
| CV Coding | -0.0439 | -0.2369* | -0.2911** | 0.0716 | 0.0710 | 0.0992 | -1.9601 | -1.8941 | -2.0142 |
| | (0.0765) | (0.1381) | (0.1390) | (0.0532) | (0.0532) | (0.0654) | (1.5459) | (1.5358) | (1.3384) |
| CV Publication/Working Paper | 0.6162* | 0.8309 | -0.1668 | 0.2940 | 0.2834 | 0.1310 | -11.5166 | -10.8849 | -4.6204 |
| | (0.3421) | (0.9326) | (0.5756) | (0.2998) | (0.3036) | (0.3383) | (7.2873) | (7.3131) | (6.7001) |
| Essay Length/100 | 0.1747* | 0.2225 | 0.1776 | -0.0008 | -0.0008 | -0.0711 | -0.4758 | -0.5351 | 1.8425 |
| | (0.0933) | (0.3920) | (0.1536) | (0.0751) | (0.0764) | (0.0821) | (1.8648) | (1.8721) | (1.6240) |
| Essay Specific Professor | 0.3842 | 0.4170 | 0.6845 | 0.2144 | 0.1774 | 0.2872 | -6.8688 | -5.9104 | -7.5240 |
| | (0.2908) | (0.7014) | (0.5119) | (0.2407) | (0.2410) | (0.2789) | (6.0223) | (5.9982) | (5.2718) |
| Essay Economics Terms (%) | -0.0169 | 0.0070 | 0.2576** | -0.1126** | -0.1150** | -0.0822 | 3.6540*** | 3.7277*** | 2.1702* |
| | (0.0547) | (0.0866) | (0.1159) | (0.0533) | (0.0531) | (0.0618) | (1.2540) | (1.2496) | (1.1179) |
| Essay Research Terms (%) | -0.0018 | -0.0459 | -0.1868 | 0.0697 | 0.0642 | 0.0150 | -3.7084** | -3.6795** | -1.0575 |
| | (0.0736) | (0.1690) | (0.1451) | (0.0633) | (0.0639) | (0.0674) | (1.6220) | (1.6209) | (1.3903) |
| Essay Positive Terms (%) | -0.0202 | -0.0700 | -0.0504 | 0.0724 | 0.0772* | 0.0977* | -0.9256 | -0.9929 | -0.7266 |
| | (0.0512) | (0.1044) | (0.1331) | (0.0453) | (0.0458) | (0.0542) | (1.1634) | (1.1496) | (0.9752) |
| Essay Negative Terms (%) | -0.0151 | -0.0886 | 0.0019 | -0.0386 | -0.0454 | -0.0472 | 0.8164 | 0.9436 | 0.9507 |
| | (0.0770) | (0.1557) | (0.1071) | (0.0663) | (0.0663) | (0.0748) | (1.6317) | (1.6259) | (1.3985) |
| Missing Committee Score | | -0.5168 | | | -0.3458** | -0.2199 | | 9.2234*** | 2.8589 |
| | | (0.6383) | | | (0.1397) | (0.1519) | | (3.4582) | (3.0258) |
| Missing Application Pdf | -1.0107*** | 0.1926 | -0.7407 | 0.1461 | 0.2686 | 0.2416 | -5.1611 | -8.4497 | -6.8690 |
| | (0.3188) | (0.8088) | (0.6084) | (0.2442) | (0.2479) | (0.2763) | (6.0328) | (6.1404) | (5.4325) |
| Open House | | | 1.8384*** | | | | | | |
| | | | (0.4115) | | | | | | |
| Program Rank | | | | | | -0.0087*** | | | 0.1813*** |
| | | | | | | (0.0021) | | | (0.0404) |
| Missing Program Rank | | | | | | -3.1432*** | | | 89.4721*** |
| | | | | | | (0.4077) | | | (5.6947) |
| Sample | All | All | Accept = 1 | All | All | All | All | All | All |
| Observations | 2329 | 2329 | 257 | 2329 | 2329 | 2329 | 2329 | 2329 | 2329 |
| Pseudo-$R^2$ | 0.2957 | 0.8108 | 0.2877 | 0.0308 | 0.0353 | 0.2430 | 0.0089 | 0.0097 | 0.1065 |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

# Table 10b: Logistic Regression AMEs

| Dependent Variable | Accept | Accept | Matriculate | T100 Placement | T100 Placement | T100 Placement | Placement Rank | Placement Rank | Placement Rank |
|---|---|---|---|---|---|---|---|---|---|
| 2016 | -0.0274 | -0.0094 | -0.0367 | 0.0035 | 0.0046 | 0.0112 | 0.8694 | 0.8485 | -0.3038 |
| | (0.0171) | (0.0085) | (0.0976) | (0.0262) | (0.0261) | (0.0238) | (2.0523) | (2.0482) | (1.6650) |
| 2017 | -0.0068 | -0.0246*** | 0.0193 | 0.0127 | 0.0108 | 0.0184 | 1.3374 | 1.4571 | 0.3431 |
| | (0.0184) | (0.0090) | (0.0958) | (0.0262) | (0.0263) | (0.0244) | (2.0386) | (2.0331) | (1.7054) |
| 2018 | -0.0158 | -0.0699*** | 0.1348 | 0.0096 | -0.0060 | 0.0018 | 1.3142 | 2.5391 | 2.2847 |
| | (0.0189) | (0.0086) | (0.0857) | (0.0270) | (0.0276) | (0.0254) | (2.1405) | (2.1686) | (1.7791) |
| 2019 | -0.0076 | -0.0589*** | 0.0841 | 0.0091 | 0.0022 | 0.0046 | 2.7516 | 3.2169 | 2.3982 |
| | (0.0190) | (0.0109) | (0.0970) | (0.0279) | (0.0279) | (0.0259) | (2.2355) | (2.2284) | (1.8977) |
| Age | -0.0016 | -0.0001 | -0.0223** | 0.0037 | 0.0037 | 0.0053* | 0.1734 | 0.1751 | -0.0586 |
| | (0.0024) | (0.0011) | (0.0108) | (0.0032) | (0.0031) | (0.0030) | (0.2629) | (0.2606) | (0.2222) |
| Female | 0.0269** | 0.0163** | -0.0490 | -0.0135 | -0.0157 | -0.0032 | 0.8717 | 1.0082 | -0.7517 |
| | (0.0118) | (0.0068) | (0.0534) | (0.0173) | (0.0172) | (0.0158) | (1.3867) | (1.3851) | (1.1534) |
| U.S. | -0.0356* | -0.0013 | 0.1472* | 0.0179 | 0.0213 | 0.0192 | -0.3071 | -0.4703 | -0.3241 |
| | (0.0188) | (0.0084) | (0.0864) | (0.0292) | (0.0291) | (0.0282) | (2.3132) | (2.3090) | (1.9951) |
| International Asian | -0.0828*** | -0.0214*** | 0.1644** | 0.0260 | 0.0298 | -0.0127 | -3.7699** | -3.9523** | 1.5152 |
| | (0.0148) | (0.0073) | (0.0691) | (0.0212) | (0.0213) | (0.0195) | (1.7114) | (1.7195) | (1.4490) |
| GRE Verbal Percentile | 0.0019*** | 0.0001 | -0.0015 | 0.0008 | 0.0005 | 0.0001 | -0.0944** | -0.0708* | -0.0151 |
| | (0.0004) | (0.0002) | (0.0017) | (0.0005) | (0.0005) | (0.0005) | (0.0379) | (0.0388) | (0.0337) |
| GRE Quantitative Percentile | 0.0095*** | 0.0011** | -0.0147** | 0.0038*** | 0.0028** | 0.0006 | -0.4096*** | -0.3468*** | -0.1114* |
| | (0.0014) | (0.0005) | (0.0063) | (0.0012) | (0.0011) | (0.0011) | (0.0757) | (0.0762) | (0.0670) |
| GRE Analytic Percentile | 0.0005** | 0.0002 | -0.0017 | -0.0003 | -0.0004 | -0.0003 | 0.0316 | 0.0395 | 0.0073 |
| | (0.0003) | (0.0002) | (0.0013) | (0.0004) | (0.0004) | (0.0004) | (0.0322) | (0.0323) | (0.0283) |
| Undergraduate GPA | 0.0910* | 0.0195 | -0.1026 | -0.0610 | -0.0721 | -0.1096** | 2.3156 | 3.3404 | 7.9232** |
| | (0.0499) | (0.0235) | (0.1820) | (0.0600) | (0.0587) | (0.0558) | (4.8911) | (4.8499) | (3.5564) |
| TOEFL/IELTS | 0.0303 | -0.0022 | -0.0631 | 0.0273 | 0.0219 | 0.0409* | -1.0847 | -0.6246 | -3.5273* |
| | (0.0209) | (0.0111) | (0.0776) | (0.0241) | (0.0238) | (0.0221) | (2.1145) | (2.0935) | (1.8033) |
| Elite U.S. University | 0.0524*** | -0.0034 | -0.0708 | -0.0024 | -0.0130 | -0.0192 | -1.1467 | -0.1712 | -0.1584 |
| | (0.0198) | (0.0111) | (0.0816) | (0.0327) | (0.0326) | (0.0296) | (2.5514) | (2.5479) | (2.0632) |
| Elite U.S. Liberal Arts College | 0.0012 | 0.0101 | 0.1409 | 0.0112 | 0.0123 | -0.0127 | -1.9025 | -1.8336 | -1.5611 |
| | (0.0327) | (0.0126) | (0.1197) | (0.0517) | (0.0513) | (0.0496) | (4.2692) | (4.2369) | (3.7802) |
| Elite Foreign University | 0.0089 | -0.0102 | -0.0096 | 0.0186 | 0.0115 | 0.0122 | -1.3354 | -0.7428 | -1.0164 |
| | (0.0381) | (0.0182) | (0.1597) | (0.0497) | (0.0497) | (0.0445) | (3.6022) | (3.5986) | (2.8173) |
| Other Foreign University | 0.0381* | 0.0118 | 0.1049 | 0.0228 | 0.0167 | 0.0200 | -3.2576 | -2.7560 | -4.1894** |
| | (0.0216) | (0.0119) | (0.0890) | (0.0301) | (0.0301) | (0.0279) | (2.3945) | (2.3912) | (1.9955) |
| Missing Undergraduate Institution | 0.0233 | 0.0046 | -0.0975 | 0.0352 | 0.0318 | 0.0223 | -3.8495 | -3.6329 | -3.7531* |
| | (0.0222) | (0.0111) | (0.1014) | (0.0309) | (0.0308) | (0.0296) | (2.4639) | (2.4635) | (2.0698) |
| Graduate Degree | 0.0123 | -0.0038 | -0.0420 | -0.0062 | -0.0141 | -0.0231 | -1.5811 | -0.9755 | 1.2034 |
| | (0.0128) | (0.0071) | (0.0712) | (0.0189) | (0.0194) | (0.0172) | (1.5257) | (1.5461) | (1.2606) |
| Work Experience | 0.0557*** | 0.0140** | 0.1591** | 0.0353* | 0.0271 | 0.0358** | -1.7325 | -1.0872 | -2.9554** |
| | (0.0135) | (0.0064) | (0.0700) | (0.0192) | (0.0194) | (0.0178) | (1.5432) | (1.5570) | (1.3344) |
| #Professor Recommenders | -0.0027 | -0.0033 | 0.0510 | -0.0072 | -0.0078 | -0.0069 | 0.4225 | 0.4806 | 0.3384 |
| | (0.0077) | (0.0045) | (0.0437) | (0.0115) | (0.0115) | (0.0109) | (0.9881) | (0.9881) | (0.7878) |
| #Prolific Recommenders | 0.0158*** | 0.0039 | -0.0426 | 0.0155* | 0.0137* | 0.0076 | -0.7048 | -0.5767 | 0.0792 |
| | (0.0058) | (0.0031) | (0.0285) | (0.0081) | (0.0081) | (0.0074) | (0.6595) | (0.6589) | (0.5373) |
| Missing GRE | -0.1102 | 0.0188 | | -0.0693 | -0.0497 | 0.0022 | 5.5529 | 4.1174 | -1.7034 |
| | (0.0706) | (0.0212) | | (0.0538) | (0.0541) | (0.0479) | (4.1627) | (4.1727) | (2.8358) |
| Missing Undergraduate GPA | -0.0246 | -0.0052 | 0.0114 | -0.0155 | -0.0098 | -0.0065 | 1.1711 | 0.7352 | -0.3999 |
| | (0.0158) | (0.0082) | (0.0703) | (0.0240) | (0.0242) | (0.0221) | (1.9187) | (1.9201) | (1.5412) |
| Missing TOEFL/IELTS | 0.0338** | 0.0070 | -0.0104 | -0.0194 | -0.0221 | -0.0008 | 1.9295 | 2.0863 | -1.0922 |
| | (0.0147) | (0.0078) | (0.0839) | (0.0194) | (0.0195) | (0.0182) | (1.6007) | (1.6020) | (1.3613) |
| Committee Score | | 0.0788*** | -0.0371 | | 0.0144** | 0.0007 | | -1.2091** | 0.3160 |
| | | (0.0045) | (0.0322) | | (0.0069) | (0.0062) | | (0.5716) | (0.4649) |
| #Math Courses | -0.0068 | -0.0050 | 0.0107 | 0.0096 | 0.0101 | 0.0001 | -1.1485 | -1.1832 | 0.4185 |
| | (0.0055) | (0.0053) | (0.0239) | (0.0116) | (0.0117) | (0.0112) | (0.8692) | (0.8747) | (0.6922) |
| Advanced Math | 0.0453** | 0.0246 | -0.1950** | -0.0296 | -0.0317 | -0.0287 | 1.4653 | 1.6660 | 0.9161 |
| | (0.0216) | (0.0168) | (0.0793) | (0.0399) | (0.0402) | (0.0372) | (3.1061) | (3.1022) | (2.5621) |
| Letters Length/100 | 0.0097*** | 0.0020 | -0.0129 | 0.0010 | -0.0003 | -0.0019 | -0.1250 | -0.0038 | 0.3308 |
| | (0.0037) | (0.0026) | (0.0100) | (0.0051) | (0.0052) | (0.0049) | (0.3978) | (0.4012) | (0.3253) |
| Letters Terms (%) PC1 | 0.0108 | 0.0055 | 0.0671** | 0.0056 | 0.0038 | 0.0159 | 0.7752 | 0.8855 | -0.2576 |
| | (0.0066) | (0.0054) | (0.0283) | (0.0112) | (0.0114) | (0.0115) | (0.9905) | (0.9892) | (0.8348) |
| Letters Terms (%) PC2 | -0.0065 | -0.0093 | 0.0387 | -0.0188 | -0.0183 | -0.0179 | 0.3431 | 0.2706 | 0.0530 |
| | (0.0105) | (0.0077) | (0.0371) | (0.0178) | (0.0179) | (0.0160) | (1.3495) | (1.3535) | (1.0765) |
| CV Length/100 | 0.0026 | 0.0010 | -0.0282 | 0.0046 | 0.0042 | 0.0044 | -0.5040 | -0.4877 | -0.3768 |
| | (0.0044) | (0.0022) | (0.0190) | (0.0086) | (0.0086) | (0.0075) | (0.6374) | (0.6328) | (0.4706) |
| CV Coding | -0.0031 | -0.0044* | -0.0414** | 0.0110 | 0.0108 | 0.0125 | -0.9448 | -0.9129 | -0.9246 |
| | (0.0055) | (0.0026) | (0.0193) | (0.0082) | (0.0081) | (0.0082) | (0.7454) | (0.7403) | (0.6139) |
| CV Publication/Working Paper | 0.0441* | 0.0154 | -0.0237 | 0.0451 | 0.0433 | 0.0165 | -5.5511 | -5.2459 | -2.1210 |
| | (0.0244) | (0.0173) | (0.0820) | (0.0460) | (0.0463) | (0.0427) | (3.5094) | (3.5215) | (3.0760) |
| Essay Length/100 | 0.0125* | 0.0041 | 0.0253 | -0.0001 | -0.0001 | -0.0090 | -0.2293 | -0.2579 | 0.8458 |
| | (0.0067) | (0.0073) | (0.0219) | (0.0115) | (0.0117) | (0.0103) | (0.8988) | (0.9022) | (0.7451) |
| Essay Specific Professor | 0.0275 | 0.0077 | 0.0974 | 0.0329 | 0.0271 | 0.0362 | -3.3108 | -2.8484 | -3.4539 |
| | (0.0208) | (0.0130) | (0.0724) | (0.0369) | (0.0368) | (0.0351) | (2.9044) | (2.8921) | (2.4207) |
| Essay Economics Terms (%) | -0.0012 | 0.0001 | 0.0366** | -0.0173** | -0.0175** | -0.0104 | 1.7613*** | 1.7965*** | 0.9962* |
| | (0.0039) | (0.0016) | (0.0162) | (0.0082) | (0.0081) | (0.0078) | (0.6048) | (0.6027) | (0.5134) |
| Essay Research Terms (%) | -0.0001 | -0.0009 | -0.0266 | 0.0107 | 0.0098 | 0.0019 | -1.7875** | -1.7733** | -0.4855 |
| | (0.0053) | (0.0031) | (0.0206) | (0.0097) | (0.0098) | (0.0085) | (0.7813) | (0.7808) | (0.6382) |
| Essay Positive Terms (%) | -0.0014 | -0.0013 | -0.0072 | 0.0111 | 0.0118* | 0.0123* | -0.4461 | -0.4785 | -0.3335 |
| | (0.0037) | (0.0019) | (0.0189) | (0.0069) | (0.0070) | (0.0068) | (0.5609) | (0.5541) | (0.4475) |
| Essay Negative Terms (%) | -0.0011 | -0.0016 | 0.0003 | -0.0059 | -0.0069 | -0.0060 | 0.3935 | 0.4548 | 0.4364 |
| | (0.0055) | (0.0029) | (0.0152) | (0.0102) | (0.0101) | (0.0094) | (0.7865) | (0.7837) | (0.6420) |
| Missing Committee Score | | -0.0096 | | | -0.0528** | -0.0277 | | 4.4451*** | 1.3124 |
| | | (0.0120) | | | (0.0212) | (0.0191) | | (1.6677) | (1.3892) |
| Missing Application Pdf | -0.0723*** | 0.0036 | -0.1054 | 0.0224 | 0.0410 | 0.0305 | -2.4877 | -4.0722 | -3.1533 |
| | (0.0227) | (0.0149) | (0.0858) | (0.0375) | (0.0379) | (0.0348) | (2.9090) | (2.9614) | (2.4962) |
| Open House | | | 0.2615*** | | | | | | |
| | | | (0.0512) | | | | | | |
| Program Rank | | | | | | -0.0011*** | | | 0.0832*** |
| | | | | | | (0.0003) | | | (0.0185) |
| Missing Program Rank | | | | | | -0.3965*** | | | 41.0729*** |
| | | | | | | (0.0512) | | | (2.6010) |
| Sample | All | All | Accept = 1 | All | All | All | All | All | All |
| Observations | 2329 | 2329 | 257 | 2329 | 2329 | 2329 | 2329 | 2329 | 2329 |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

# Table 11a: Logistic Regression Results (Synthetic)

| Dependent Variable | Accept | Accept | Matriculate | T100 Placement | T100 Placement | T100 Placement | Placement Rank | Placement Rank | Placement Rank |
|---|---|---|---|---|---|---|---|---|---|
| 2016 | -0.2091 | -0.1311 | 0.1291 | 0.0997 | 0.0892 | 0.1020 | -0.8042 | -0.5831 | -1.3927 |
| | (0.2288) | (0.2577) | (0.6478) | (0.1812) | (0.1814) | (0.1970) | (4.5726) | (4.5702) | (4.1436) |
| 2017 | -0.3355 | -0.5461** | 0.6767 | 0.2827 | 0.2903 | 0.1565 | -12.4495*** | -12.2002*** | -6.3639 |
| | (0.2390) | (0.2627) | (0.6543) | (0.1761) | (0.1767) | (0.1925) | (4.4305) | (4.4347) | (3.9262) |
| 2018 | 0.2889 | -0.2993 | 0.2947 | 0.1627 | 0.2192 | 0.1272 | -9.3709** | -9.2465** | -6.0098 |
| | (0.2056) | (0.2212) | (0.5517) | (0.1806) | (0.1827) | (0.1972) | (4.4278) | (4.4749) | (3.9975) |
| 2019 | -0.1933 | -0.3208 | -0.0662 | 0.0116 | 0.0023 | -0.1488 | -5.3768 | -4.7378 | 1.0400 |
| | (0.2227) | (0.2490) | (0.6069) | (0.1927) | (0.1933) | (0.2061) | (4.4927) | (4.5026) | (3.9171) |
| Age | -0.0508** | -0.0414* | 0.0065 | -0.0141 | -0.0150 | -0.0094 | 0.4466 | 0.4383 | 0.2496 |
| | (0.0220) | (0.0243) | (0.0600) | (0.0170) | (0.0170) | (0.0185) | (0.4147) | (0.4141) | (0.3766) |
| Female | -0.1826 | -0.1535 | 0.0691 | 0.1351 | 0.1379 | 0.2109* | -1.5097 | -1.6305 | -4.7144* |
| | (0.1446) | (0.1590) | (0.4006) | (0.1167) | (0.1168) | (0.1259) | (2.9357) | (2.9318) | (2.6789) |
| U.S. | 0.0002 | -0.0549 | 0.0998 | 0.0198 | 0.0241 | -0.0123 | 2.8625 | 2.5697 | 4.2450 |
| | (0.2240) | (0.2468) | (0.6663) | (0.1794) | (0.1801) | (0.1997) | (4.4691) | (4.4669) | (4.1868) |
| International Asian | -0.1230 | -0.0646 | -0.1205 | 0.2014 | 0.1972 | 0.1700 | -0.7475 | -0.9433 | 1.1783 |
| | (0.1688) | (0.1838) | (0.5028) | (0.1394) | (0.1392) | (0.1541) | (3.3754) | (3.3732) | (3.1231) |
| GRE Verbal Percentile | -0.0025 | -0.0001 | 0.0008 | 0.0004 | 0.0001 | -0.0021 | -0.0668 | -0.0623 | 0.0181 |
| | (0.0034) | (0.0036) | (0.0089) | (0.0026) | (0.0026) | (0.0029) | (0.0661) | (0.0663) | (0.0595) |
| GRE Quantitative Percentile | -0.0028 | -0.0005 | -0.0125 | 0.0145** | 0.0143** | -0.0003 | -0.5982*** | -0.5916*** | -0.0184 |
| | (0.0075) | (0.0080) | (0.0195) | (0.0065) | (0.0065) | (0.0069) | (0.1570) | (0.1569) | (0.1376) |
| GRE Analytic Percentile | 0.0041 | 0.0019 | -0.0010 | 0.0009 | 0.0011 | 0.0019 | 0.0181 | 0.0156 | -0.0008 |
| | (0.0031) | (0.0033) | (0.0074) | (0.0024) | (0.0024) | (0.0026) | (0.0587) | (0.0587) | (0.0529) |
| Undergraduate GPA | -0.4230 | -0.3157 | 0.9077 | 0.4075 | 0.3986 | 0.4627 | -2.7870 | -2.7400 | -5.8113 |
| | (0.3862) | (0.4536) | (1.8572) | (0.3984) | (0.3990) | (0.4326) | (9.4800) | (9.4624) | (8.8952) |
| TOEFL/IELTS | 0.2329 | 0.2776 | -0.2073 | -0.0682 | -0.0625 | -0.0142 | 2.4237 | 2.2404 | -0.4376 |
| | (0.2154) | (0.2202) | (0.4100) | (0.1509) | (0.1507) | (0.1670) | (3.7070) | (3.7021) | (3.1595) |
| Elite U.S. University | -0.0473 | -0.1157 | -0.2546 | 0.3145 | 0.3119 | 0.3760* | -6.7562 | -6.6352 | -7.1987 |
| | (0.2618) | (0.2943) | (0.7160) | (0.2054) | (0.2062) | (0.2230) | (5.3790) | (5.3875) | (4.8870) |
| Elite U.S. Liberal Arts College | -0.6642 | -0.6053 | | 0.0069 | -0.0044 | -0.0958 | -2.2643 | -2.2283 | 3.3486 |
| | (0.5448) | (0.5744) | | (0.3656) | (0.3659) | (0.3964) | (8.9098) | (8.8713) | (7.5578) |
| Elite Foreign University | -0.1797 | -0.1798 | 0.5156 | -0.0538 | -0.0692 | -0.0298 | -1.0302 | -0.9315 | -0.2765 |
| | (0.3900) | (0.4341) | (1.2147) | (0.3273) | (0.3294) | (0.3495) | (8.2453) | (8.2793) | (7.1254) |
| Other Foreign University | -0.0406 | -0.1089 | 0.6613 | 0.1055 | 0.1125 | 0.0046 | -6.0523 | -6.2650 | -0.6409 |
| | (0.2078) | (0.2262) | (0.5235) | (0.1688) | (0.1689) | (0.1840) | (4.2123) | (4.2076) | (3.8417) |
| Missing Undergraduate Institution | -0.0012 | 0.0377 | -0.6367 | 0.1037 | 0.0966 | 0.0297 | -5.9887 | -5.9936 | -2.3404 |
| | (0.2217) | (0.2413) | (0.5829) | (0.1844) | (0.1842) | (0.1977) | (4.5947) | (4.5865) | (4.1622) |
| Graduate Degree | -0.0846 | -0.1354 | -0.0100 | 0.1515 | 0.1611 | 0.1906 | -1.4756 | -1.6341 | -2.0725 |
| | (0.1458) | (0.1626) | (0.3834) | (0.1166) | (0.1165) | (0.1252) | (2.8643) | (2.8619) | (2.5933) |
| Work Experience | 0.2547* | 0.2149 | -0.2181 | 0.0750 | 0.0805 | 0.1443 | 0.4084 | 0.3542 | -2.7442 |
| | (0.1463) | (0.1588) | (0.3573) | (0.1161) | (0.1160) | (0.1250) | (2.8682) | (2.8651) | (2.6268) |
| #Professor Recommenders | 0.0194 | -0.0565 | 0.1806 | 0.0038 | 0.0074 | -0.0302 | -0.2230 | -0.1761 | 1.2584 |
| | (0.0950) | (0.1021) | (0.2725) | (0.0759) | (0.0761) | (0.0794) | (1.9087) | (1.9104) | (1.7006) |
| #Prolific Recommenders | -0.0388 | -0.0071 | 0.0168 | 0.0027 | -0.0000 | 0.0325 | 0.3327 | 0.3170 | -1.1079 |
| | (0.0671) | (0.0713) | (0.1685) | (0.0521) | (0.0522) | (0.0575) | (1.3245) | (1.3242) | (1.2261) |
| Missing GRE | 0.3002 | 0.0736 | | 0.3136 | 0.3314 | 0.2674 | -12.0035 | -11.9570 | -11.5860 |
| | (0.3757) | (0.3872) | | (0.2858) | (0.2857) | (0.3297) | (8.1036) | (8.0998) | (8.0641) |
| Missing Undergraduate GPA | 0.1997 | 0.1720 | 0.2910 | -0.1321 | -0.1290 | -0.0816 | 5.2305 | 5.1674 | 4.3351 |
| | (0.1863) | (0.2016) | (0.5443) | (0.1391) | (0.1389) | (0.1534) | (3.6167) | (3.6124) | (3.3094) |
| Missing TOEFL/IELTS | -0.1740 | -0.0792 | -0.3009 | 0.1948 | 0.1815 | 0.2286* | -1.5099 | -1.4121 | -3.1183 |
| | (0.1609) | (0.1741) | (0.4256) | (0.1280) | (0.1281) | (0.1380) | (3.1162) | (3.1192) | (2.7616) |
| Committee Score | | 0.0372 | -0.1004 | | 0.0465 | 0.0330 | | -1.6627 | -0.4906 |
| | | (0.0423) | (0.1060) | | (0.0505) | (0.0558) | | (1.1655) | (1.1645) |
| #Math Courses | 0.0632 | 0.0756 | -0.1683 | -0.1269 | -0.1305* | -0.1764** | 1.1826 | 1.2073 | 3.0509** |
| | (0.0687) | (0.0733) | (0.1601) | (0.0786) | (0.0780) | (0.0863) | (1.7287) | (1.7209) | (1.5010) |
| Advanced Math | 0.0222 | -0.0336 | 0.9873 | -0.1639 | -0.1630 | -0.2842 | 1.1644 | 1.2290 | 6.1671 |
| | (0.2590) | (0.2747) | (0.7209) | (0.3004) | (0.3013) | (0.3256) | (6.4828) | (6.4875) | (5.6351) |
| Letters Length/100 | -0.0254 | -0.0128 | -0.0575 | -0.1049*** | -0.1057*** | -0.1218*** | 1.6052** | 1.6308** | 2.1010*** |
| | (0.0333) | (0.0361) | (0.0885) | (0.0381) | (0.0376) | (0.0428) | (0.7873) | (0.7869) | (0.6934) |
| Letters Terms (%) PC1 | 0.0280 | -0.0073 | 0.0381 | 0.0574 | 0.0643 | 0.0566 | 1.1364 | 1.1816 | 1.0356 |
| | (0.0779) | (0.0821) | (0.2097) | (0.0828) | (0.0834) | (0.0930) | (1.8933) | (1.9000) | (1.6894) |
| Letters Terms (%) PC2 | -0.2328* | -0.2369* | -0.4921 | 0.0802 | 0.0718 | 0.1318 | -2.8589 | -2.7586 | -5.0669** |
| | (0.1226) | (0.1255) | (0.3028) | (0.1205) | (0.1211) | (0.1250) | (2.6915) | (2.7005) | (2.1872) |
| CV Length/100 | 0.0971** | 0.0795 | 0.1022 | -0.0358 | -0.0336 | -0.0095 | 2.4879** | 2.5172** | 0.5354 |
| | (0.0492) | (0.0508) | (0.1340) | (0.0548) | (0.0551) | (0.0641) | (1.2406) | (1.2458) | (1.0685) |
| CV Coding | -0.0198 | -0.0473 | -0.0834 | 0.0688 | 0.0720* | 0.0795* | -1.6529 | -1.6778 | -1.0250 |
| | (0.0411) | (0.0448) | (0.1069) | (0.0434) | (0.0434) | (0.0474) | (1.0435) | (1.0451) | (0.8904) |
| CV Publication/Working Paper | 0.3079 | 0.3005 | 1.3491* | 0.1111 | 0.1183 | 0.1111 | 2.7524 | 2.4246 | 5.5955 |
| | (0.2852) | (0.2962) | (0.7475) | (0.3264) | (0.3272) | (0.3482) | (7.3505) | (7.3749) | (6.2211) |
| Essay Length/100 | 0.0679 | 0.0790 | -0.0558 | -0.0667 | -0.0636 | -0.0694 | 0.5523 | 0.4398 | 0.3231 |
| | (0.0770) | (0.0831) | (0.1489) | (0.0780) | (0.0794) | (0.0871) | (1.7254) | (1.7321) | (1.3973) |
| Essay Specific Professor | -0.1756 | -0.2494 | -0.1115 | -0.0547 | -0.0524 | 0.0009 | 1.2639 | 1.2219 | -3.1507 |
| | (0.2564) | (0.2714) | (0.6302) | (0.2774) | (0.2798) | (0.3022) | (6.3588) | (6.3580) | (5.5747) |
| Essay Economics Terms (%) | -0.0123 | -0.0094 | -0.0406 | -0.0091 | -0.0082 | -0.0222 | 0.2276 | 0.2960 | 0.6769 |
| | (0.0542) | (0.0572) | (0.1334) | (0.0539) | (0.0547) | (0.0624) | (1.3179) | (1.3189) | (1.1799) |
| Essay Research Terms (%) | 0.0303 | 0.0066 | -0.0464 | 0.1042 | 0.1084 | 0.1176 | -1.8679 | -1.9069 | -1.8170 |
| | (0.0656) | (0.0702) | (0.1578) | (0.0703) | (0.0705) | (0.0723) | (1.5218) | (1.5245) | (1.2596) |
| Essay Positive Terms (%) | 0.0180 | 0.0074 | 0.3699*** | 0.0871* | 0.0868* | 0.0947 | -1.5542 | -1.4775 | -1.1629 |
| | (0.0474) | (0.0484) | (0.1213) | (0.0511) | (0.0517) | (0.0609) | (1.1518) | (1.1568) | (1.1006) |
| Essay Negative Terms (%) | 0.0333 | 0.0312 | -0.0731 | -0.0304 | -0.0280 | -0.0168 | 0.5011 | 0.4716 | 0.2911 |
| | (0.0314) | (0.0351) | (0.0769) | (0.0359) | (0.0359) | (0.0382) | (0.8608) | (0.8614) | (0.7100) |
| Missing Committee Score | | -4.3141*** | | | 0.2635** | 0.4129*** | | -0.9440 | -6.1957** |
| | | (0.4312) | | | (0.1267) | (0.1388) | | (3.0258) | (2.8364) |
| Missing Application Pdf | -0.9321*** | -0.0588 | 0.9473 | 0.0574 | -0.0560 | -0.1100 | 4.5136 | 4.8399 | 5.6614 |
| | (0.2534) | (0.2753) | (0.6793) | (0.2628) | (0.2678) | (0.2943) | (5.9238) | (6.0503) | (5.4160) |
| Open House | | | 1.0643 | | | | | | |
| | | | (0.8268) | | | | | | |
| Program Rank | | | | | | -0.0049** | | | 0.1176*** |
| | | | | | | (0.0019) | | | (0.0386) |
| Missing Program Rank | | | | | | -2.5925*** | | | 94.5255*** |
| | | | | | | (0.3273) | | | (5.6183) |
| Sample | All | All | Accept = 1 | All | All | All | All | All | All |
| Observations | 2329 | 2329 | 237 | 2329 | 2329 | 2329 | 2329 | 2329 | 2329 |
| Pseudo-$R^2$ | 0.0552 | 0.2804 | 0.1402 | 0.0221 | 0.0245 | 0.1935 | 0.0040 | 0.0042 | 0.0953 |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

# Table 11b: Logistic Regression AMEs (Synthetic)

| Dependent Variable | Accept | Accept | Matriculate | T100 Placement | T100 Placement | T100 Placement | Placement Rank | Placement Rank | Placement Rank |
|---|---|---|---|---|---|---|---|---|---|
| 2016 | -0.0185 | -0.0100 | 0.0201 | 0.0136 | 0.0122 | 0.0120 | -0.3764 | -0.2729 | -0.6112 |
| | (0.0203) | (0.0197) | (0.1005) | (0.0248) | (0.0248) | (0.0233) | (2.1401) | (2.1390) | (1.8189) |
| 2017 | -0.0297 | -0.0417** | 0.1052 | 0.0387 | 0.0396 | 0.0185 | -5.8265*** | -5.7099*** | -2.7929 |
| | (0.0212) | (0.0199) | (0.1005) | (0.0241) | (0.0241) | (0.0227) | (2.0748) | (2.0770) | (1.7228) |
| 2018 | 0.0256 | -0.0229 | 0.0458 | 0.0223 | 0.0299 | 0.0150 | -4.3857** | -4.3275** | -2.6375 |
| | (0.0182) | (0.0168) | (0.0856) | (0.0247) | (0.0250) | (0.0233) | (2.0732) | (2.0958) | (1.7555) |
| 2019 | -0.0171 | -0.0245 | -0.0103 | 0.0016 | 0.0003 | -0.0176 | -2.5164 | -2.2174 | 0.4564 |
| | (0.0197) | (0.0189) | (0.0944) | (0.0264) | (0.0264) | (0.0243) | (2.1013) | (2.1061) | (1.7192) |
| Age | -0.0045** | -0.0032* | 0.0010 | -0.0019 | -0.0020 | -0.0011 | 0.2090 | 0.2051 | 0.1095 |
| | (0.0020) | (0.0019) | (0.0093) | (0.0023) | (0.0023) | (0.0022) | (0.1941) | (0.1938) | (0.1654) |
| Female | -0.0162 | -0.0117 | 0.0107 | 0.0185 | 0.0188 | 0.0249* | -0.7066 | -0.7631 | -2.0690* |
| | (0.0128) | (0.0121) | (0.0621) | (0.0160) | (0.0159) | (0.0149) | (1.3745) | (1.3728) | (1.1767) |
| U.S. | 0.0000 | -0.0042 | 0.0155 | 0.0027 | 0.0033 | -0.0015 | 1.3397 | 1.2026 | 1.8630 |
| | (0.0198) | (0.0189) | (0.1035) | (0.0246) | (0.0246) | (0.0236) | (2.0914) | (2.0905) | (1.8379) |
| International Asian | -0.0109 | -0.0049 | -0.0187 | 0.0276 | 0.0269 | 0.0201 | -0.3498 | -0.4415 | 0.5171 |
| | (0.0150) | (0.0140) | (0.0782) | (0.0191) | (0.0190) | (0.0182) | (1.5798) | (1.5789) | (1.3706) |
| GRE Verbal Percentile | -0.0002 | -0.0000 | 0.0001 | 0.0001 | 0.0000 | -0.0002 | -0.0313 | -0.0291 | 0.0080 |
| | (0.0003) | (0.0003) | (0.0014) | (0.0004) | (0.0004) | (0.0003) | (0.0309) | (0.0310) | (0.0261) |
| GRE Quantitative Percentile | -0.0002 | -0.0000 | -0.0019 | 0.0020** | 0.0020** | -0.0000 | -0.2800*** | -0.2769*** | -0.0081 |
| | (0.0007) | (0.0006) | (0.0030) | (0.0009) | (0.0009) | (0.0008) | (0.0734) | (0.0734) | (0.0604) |
| GRE Analytic Percentile | 0.0004 | 0.0001 | -0.0002 | 0.0001 | 0.0002 | 0.0002 | 0.0085 | 0.0073 | -0.0004 |
| | (0.0003) | (0.0003) | (0.0011) | (0.0003) | (0.0003) | (0.0003) | (0.0275) | (0.0275) | (0.0232) |
| Undergraduate GPA | -0.0375 | -0.0241 | 0.1411 | 0.0558 | 0.0544 | 0.0546 | -1.3043 | -1.2824 | -2.5504 |
| | (0.0342) | (0.0346) | (0.2875) | (0.0545) | (0.0545) | (0.0510) | (4.4373) | (4.4292) | (3.9043) |
| TOEFL/IELTS | 0.0206 | 0.0212 | -0.0322 | -0.0093 | -0.0085 | -0.0017 | 1.1343 | 1.0486 | -0.1921 |
| | (0.0191) | (0.0168) | (0.0637) | (0.0207) | (0.0206) | (0.0197) | (1.7354) | (1.7331) | (1.3866) |
| Elite U.S. University | -0.0042 | -0.0088 | -0.0396 | 0.0431 | 0.0426 | 0.0444* | -3.1620 | -3.1054 | -3.1593 |
| | (0.0232) | (0.0225) | (0.1109) | (0.0281) | (0.0282) | (0.0263) | (2.5179) | (2.5220) | (2.1436) |
| Elite U.S. Liberal Arts College | -0.0588 | -0.0462 | | 0.0009 | -0.0006 | -0.0113 | -1.0597 | -1.0429 | 1.4696 |
| | (0.0483) | (0.0439) | | (0.0500) | (0.0500) | (0.0468) | (4.1696) | (4.1517) | (3.3177) |
| Elite Foreign University | -0.0159 | -0.0137 | 0.0801 | -0.0074 | -0.0095 | -0.0035 | -0.4822 | -0.4360 | -0.1214 |
| | (0.0346) | (0.0332) | (0.1891) | (0.0448) | (0.0450) | (0.0412) | (3.8590) | (3.8749) | (3.1271) |
| Other Foreign University | -0.0036 | -0.0083 | 0.1028 | 0.0144 | 0.0154 | 0.0005 | -2.8325 | -2.9321 | -0.2813 |
| | (0.0184) | (0.0173) | (0.0813) | (0.0231) | (0.0231) | (0.0217) | (1.9702) | (1.9680) | (1.6858) |
| Missing Undergraduate Institution | -0.0001 | 0.0029 | -0.0989 | 0.0142 | 0.0132 | 0.0035 | -2.8028 | -2.8051 | -1.0271 |
| | (0.0196) | (0.0184) | (0.0902) | (0.0252) | (0.0251) | (0.0233) | (2.1499) | (2.1460) | (1.8264) |
| Graduate Degree | -0.0075 | -0.0103 | -0.0015 | 0.0207 | 0.0220 | 0.0225 | -0.6906 | -0.7648 | -0.9095 |
| | (0.0129) | (0.0124) | (0.0596) | (0.0160) | (0.0159) | (0.0147) | (1.3406) | (1.3395) | (1.1381) |
| Work Experience | 0.0226* | 0.0164 | -0.0339 | 0.0103 | 0.0110 | 0.0170 | 0.1911 | 0.1658 | -1.2044 |
| | (0.0129) | (0.0121) | (0.0551) | (0.0159) | (0.0158) | (0.0147) | (1.3422) | (1.3408) | (1.1533) |
| #Professor Recommenders | 0.0017 | -0.0043 | 0.0281 | 0.0005 | 0.0010 | -0.0036 | -0.1044 | -0.0824 | 0.5523 |
| | (0.0084) | (0.0078) | (0.0423) | (0.0104) | (0.0104) | (0.0094) | (0.8933) | (0.8941) | (0.7465) |
| #Prolific Recommenders | -0.0034 | -0.0005 | 0.0026 | 0.0004 | -0.0000 | 0.0038 | 0.1557 | 0.1484 | -0.4862 |
| | (0.0059) | (0.0054) | (0.0262) | (0.0071) | (0.0071) | (0.0068) | (0.6197) | (0.6196) | (0.5384) |
| Missing GRE | 0.0266 | 0.0056 | | 0.0429 | 0.0453 | 0.0316 | -5.6177 | -5.5961 | -5.0848 |
| | (0.0333) | (0.0296) | | (0.0391) | (0.0390) | (0.0389) | (3.7950) | (3.7934) | (3.5412) |
| Missing Undergraduate GPA | 0.0177 | 0.0131 | 0.0452 | -0.0181 | -0.0176 | -0.0096 | 2.4479 | 2.4184 | 1.9026 |
| | (0.0165) | (0.0154) | (0.0847) | (0.0190) | (0.0190) | (0.0181) | (1.6942) | (1.6921) | (1.4526) |
| Missing TOEFL/IELTS | -0.0154 | -0.0060 | -0.0468 | 0.0267 | 0.0248 | 0.0270* | -0.7067 | -0.6609 | -1.3685 |
| | (0.0143) | (0.0133) | (0.0659) | (0.0175) | (0.0175) | (0.0162) | (1.4590) | (1.4604) | (1.2133) |
| Committee Score | | 0.0028 | -0.0156 | | 0.0064 | 0.0039 | | -0.7782 | -0.2153 |
| | | (0.0032) | (0.0165) | | (0.0069) | (0.0066) | | (0.5455) | (0.5111) |
| #Math Courses | 0.0056 | 0.0058 | -0.0262 | -0.0174 | -0.0178* | -0.0208** | 0.5535 | 0.5650 | 1.3390** |
| | (0.0061) | (0.0056) | (0.0248) | (0.0107) | (0.0106) | (0.0101) | (0.8095) | (0.8059) | (0.6591) |
| Advanced Math | 0.0020 | -0.0026 | 0.1534 | -0.0224 | -0.0223 | -0.0335 | 0.5450 | 0.5752 | 2.7066 |
| | (0.0229) | (0.0210) | (0.1122) | (0.0411) | (0.0411) | (0.0384) | (3.0339) | (3.0362) | (2.4720) |
| Letters Length/100 | -0.0022 | -0.0010 | -0.0089 | -0.0144*** | -0.0144*** | -0.0144*** | 0.7512** | 0.7632** | 0.9221*** |
| | (0.0029) | (0.0028) | (0.0137) | (0.0052) | (0.0051) | (0.0050) | (0.3694) | (0.3693) | (0.3050) |
| Letters Terms (%) PC1 | 0.0025 | -0.0006 | 0.0059 | 0.0079 | 0.0088 | 0.0067 | 0.5318 | 0.5530 | 0.4545 |
| | (0.0069) | (0.0063) | (0.0325) | (0.0113) | (0.0114) | (0.0110) | (0.8858) | (0.8889) | (0.7413) |
| Letters Terms (%) PC2 | -0.0206* | -0.0181* | -0.0765* | 0.0110 | 0.0098 | 0.0156 | -1.3380 | -1.2911 | -2.2237** |
| | (0.0108) | (0.0095) | (0.0462) | (0.0165) | (0.0165) | (0.0147) | (1.2609) | (1.2651) | (0.9607) |
| CV Length/100 | 0.0086** | 0.0061 | 0.0159 | -0.0049 | -0.0046 | -0.0011 | 1.1643** | 1.1781** | 0.2350 |
| | (0.0043) | (0.0039) | (0.0210) | (0.0075) | (0.0075) | (0.0076) | (0.5802) | (0.5825) | (0.4688) |
| CV Coding | -0.0018 | -0.0036 | -0.0130 | 0.0094 | 0.0098* | 0.0094* | -0.7736 | -0.7852 | -0.4499 |
| | (0.0036) | (0.0034) | (0.0167) | (0.0059) | (0.0059) | (0.0056) | (0.4887) | (0.4894) | (0.3909) |
| CV Publication/Working Paper | 0.0273 | 0.0230 | 0.2097* | 0.0152 | 0.0162 | 0.0131 | 1.2881 | 1.1348 | 2.4557 |
| | (0.0252) | (0.0226) | (0.1128) | (0.0447) | (0.0447) | (0.0411) | (3.4400) | (3.4515) | (2.7307) |
| Essay Length/100 | 0.0060 | 0.0060 | -0.0087 | -0.0091 | -0.0087 | -0.0082 | 0.2585 | 0.2058 | 0.1418 |
| | (0.0068) | (0.0063) | (0.0231) | (0.0107) | (0.0108) | (0.0103) | (0.8075) | (0.8107) | (0.6133) |
| Essay Specific Professor | -0.0156 | -0.0190 | -0.0173 | -0.0075 | -0.0072 | 0.0001 | 0.5915 | 0.5719 | -1.3828 |
| | (0.0227) | (0.0207) | (0.0978) | (0.0380) | (0.0382) | (0.0357) | (2.9760) | (2.9757) | (2.4463) |
| Essay Economics Terms (%) | -0.0011 | -0.0007 | -0.0063 | -0.0012 | -0.0011 | -0.0026 | 0.1065 | 0.1385 | 0.2971 |
| | (0.0048) | (0.0044) | (0.0207) | (0.0074) | (0.0075) | (0.0074) | (0.6168) | (0.6173) | (0.5179) |
| Essay Research Terms (%) | 0.0027 | 0.0005 | -0.0072 | 0.0143 | 0.0148 | 0.0139 | -0.8742 | -0.8925 | -0.7975 |
| | (0.0058) | (0.0054) | (0.0245) | (0.0096) | (0.0096) | (0.0085) | (0.7127) | (0.7141) | (0.5530) |
| Essay Positive Terms (%) | 0.0016 | 0.0006 | 0.0575*** | 0.0119* | 0.0118* | 0.0112 | -0.7274 | -0.6915 | -0.5104 |
| | (0.0042) | (0.0037) | (0.0182) | (0.0070) | (0.0071) | (0.0072) | (0.5394) | (0.5418) | (0.4832) |
| Essay Negative Terms (%) | 0.0030 | 0.0024 | -0.0114 | -0.0042 | -0.0038 | -0.0020 | 0.2345 | 0.2207 | 0.1278 |
| | (0.0028) | (0.0027) | (0.0120) | (0.0049) | (0.0049) | (0.0045) | (0.4029) | (0.4032) | (0.3117) |
| Missing Committee Score | | -0.3296*** | | | 0.0360** | 0.0487*** | | -0.4418 | -2.7191** |
| | | (0.0330) | | | (0.0173) | (0.0162) | | (1.4166) | (1.2447) |
| Missing Application Pdf | -0.0826*** | -0.0045 | 0.1472 | 0.0079 | -0.0076 | -0.0130 | 2.1124 | 2.2652 | 2.4846 |
| | (0.0225) | (0.0210) | (0.1064) | (0.0360) | (0.0366) | (0.0347) | (2.7720) | (2.8317) | (2.3765) |
| Open House | | | 0.1654 | | | | | | |
| | | | (0.1278) | | | | | | |
| Program Rank | | | | | | -0.0006** | | | 0.0516*** |
| | | | | | | (0.0002) | | | (0.0169) |
| Missing Program Rank | | | | | | -0.3060*** | | | 41.4849*** |
| | | | | | | (0.0381) | | | (2.4608) |
| Sample | All | All | Accept = 1 | All | All | All | All | All | All |
| Observations | 2329 | 2329 | 237 | 2329 | 2329 | 2329 | 2329 | 2329 | 2329 |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 12: Lasso Logistic Regression Results

| Dependent Variable | Accept | Accept | Matriculate | T100 Placement | T100 Placement | T100 Placement | Placement Rank | Placement Rank | Placement Rank |
|---|---|---|---|---|---|---|---|---|---|
| 2016 | -0.1107 | | | | | | | | |
| 2017 | | -0.4928 | | | | | | | |
| 2018 | | -2.4456 | 0.0186 | | | | | | |
| 2019 | | -1.7752 | | | | | | | |
| Age | | | | 0.0084 | 0.0047 | 0.0184 | | | |
| Female | 0.2674 | 0.5589 | | | | | | | |
| U.S. | -0.1959 | | | | | | | 0.0477 | |
| International Asian | -0.8686 | -0.8651 | 0.1271 | | | | -0.5295 | -0.7927 | |
| GRE Verbal Percentile | 0.0227 | 0.0047 | -0.0015 | 0.0015 | | | -0.0085 | | |
| GRE Quantitative Percentile | 0.1027 | 0.0174 | -0.0273 | 0.0179 | 0.0125 | | -0.2706 | -0.2243 | -0.0070 |
| GRE Analytic Percentile | 0.0056 | 0.0035 | | | | | | | |
| Undergraduate GPA | 0.9812 | 0.7135 | | | | -0.1316 | | | 0.2561 |
| TOEFL/IELTS | 0.3055 | | -0.2943 | 0.0300 | | 0.0934 | -0.6903 | -0.5850 | -2.2692 |
| Elite U.S. University | 0.4024 | | | | | | | | |
| Elite U.S. Liberal Arts College | | 0.1112 | | | | | | | |
| Elite Foreign University | | -0.4025 | | | | | | | |
| Other Foreign University | 0.0790 | 0.0940 | 0.1998 | | | | | | -0.2265 |
| Missing Undergraduate Institution | | | -0.0036 | | | | | | |
| Graduate Degree | 0.1178 | | | | | | | | |
| Work Experience | 0.6006 | 0.4068 | 0.0404 | 0.0961 | 0.0470 | 0.1976 | | | -2.0809 |
| #Professor Recommenders | | | | | | | | | 0.0922 |
| #Prolific Recommenders | 0.1534 | 0.0301 | -0.0870 | 0.0526 | 0.0363 | | -0.0789 | -0.0358 | |
| Missing GRE | -0.6534 | | | -0.0697 | | | | | |
| Missing Undergraduate GPA | -0.2348 | -0.1148 | | | | | | | |
| Missing TOEFL/IELTS | 0.1447 | | | -0.1243 | -0.1295 | | 1.8402 | 2.2548 | |
| Committee Score | | 3.4243 | | | 0.0353 | | | -0.4899 | |
| #Math Courses | -0.0553 | -0.0354 | | | | | | | |
| Advanced Math | 0.4302 | 0.7352 | | | | | | | |
| Letters Length/100 | 0.1196 | 0.0587 | | | | | | | |
| Letters Terms (%) PC1 | 0.0746 | 0.1485 | 0.1258 | | | 0.0102 | | | |
| Letters Terms (%) PC2 | -0.0162 | -0.2901 | | | | | | | |
| CV Length/100 | 0.0219 | 0.0298 | | 0.0014 | | | -0.0286 | -0.1179 | |
| CV Coding | -0.0038 | -0.1754 | | 0.0165 | 0.0076 | | -0.2540 | -0.4154 | -0.0998 |
| CV Publication/Working Paper | 0.5563 | 0.6436 | | 0.0892 | | | -0.0987 | | |
| Essay Length/100 | 0.1202 | 0.0295 | | | | | | | |
| Essay Specific Professor | 0.3403 | 0.3958 | 0.0674 | | | | | | |
| Essay Economics Terms (%) | -0.0016 | | | -0.0405 | -0.0325 | -0.0104 | 0.4473 | 0.6570 | 0.2117 |
| Essay Research Terms (%) | | -0.0278 | | | | | | -0.1506 | |
| Essay Positive Terms (%) | -0.0049 | -0.0382 | | 0.0192 | 0.0140 | 0.0267 | | | |
| Essay Negative Terms (%) | | -0.0417 | | | | | | | |
| Missing Committee Score | | -0.8416 | | | -0.2615 | -0.0714 | | 4.0146 | 0.7971 |
| Missing Application Pdf | -1.0365 | | | | | | | | |
| Open House | | | 1.0982 | | | | | | |
| Program Rank | | | | | | -0.0082 | | | 0.1385 |
| Missing Program Rank | | | | | | -2.2557 | | | 19.4273 |
| Sample | All | All | Accept = 1 | All | All | All | All | All | All |
| Observations | 2329 | 2329 | 257 | 2329 | 2329 | 2329 | 2329 | 2329 | 2329 |
| Pseudo-$R^2$ | 0.2841 | 0.8007 | 0.1281 | 0.0196 | 0.0213 | 0.2265 | 0.0206 | 0.0278 | 0.2693 |
| CV Pseudo-$R^2$ | 0.2441 | 0.7605 | 0.0596 | 0.0082 | 0.0129 | 0.2199 | 0.0121 | 0.0183 | 0.2638 |

# C Additional Structural Results

Table 13: Unrestricted Dynamic Estimates

| | Committee Score | Transition | | Accept | | Success | |
|---|---|---|---|---|---|---|---|
| | Coefficient | Coefficient | AME | Coefficient | AME | Coefficient | AME |
| Intercept | -2.7966 | -23.0601*** | -3.4656*** | -28.5278*** | -1.2765*** | -4.4553** | -0.5635** |
| | (2.1751) | (2.0491) | (0.2371) | (6.9081) | (0.2933) | (2.1831) | (0.2752) |
| 2016 | 0.0915 | -0.2896 | -0.0435 | -0.5294 | -0.0237 | 0.0762 | 0.0096 |
| | (0.1856) | (0.1794) | (0.0268) | (0.5264) | (0.0230) | (0.1883) | (0.0238) |
| 2017 | 0.2354 | -0.1533 | -0.0230 | -1.3366** | -0.0598** | 0.1266 | 0.0160 |
| | (0.2130) | (0.1726) | (0.0260) | (0.5312) | (0.0235) | (0.1929) | (0.0244) |
| 2018 | 0.5999*** | 1.5429*** | 0.2319*** | -3.6854*** | -0.1649*** | 0.0095 | 0.0012 |
| | (0.1777) | (0.1858) | (0.0251) | (0.6281) | (0.0209) | (0.2011) | (0.0254) |
| 2019 | 0.6118*** | 0.2905 | 0.0437 | -3.0765*** | -0.1377*** | 0.0114 | 0.0014 |
| | (0.1907) | (0.1833) | (0.0273) | (0.6899) | (0.0257) | (0.1974) | (0.0250) |
| Age | -0.0621*** | 0.0628*** | 0.0094*** | -0.0133 | -0.0006 | 0.0428* | 0.0054* |
| | (0.0235) | (0.0218) | (0.0032) | (0.0612) | (0.0027) | (0.0235) | (0.0030) |
| Female | -0.0543 | 0.3919*** | 0.0589*** | 0.8909** | 0.0399*** | -0.0422 | -0.0053 |
| | (0.1125) | (0.1109) | (0.0164) | (0.3672) | (0.0154) | (0.1246) | (0.0158) |
| U.S. | -0.5071** | -0.1124 | -0.0169 | 0.0293 | 0.0013 | 0.1597 | 0.0202 |
| | (0.2043) | (0.1894) | (0.0285) | (0.4978) | (0.0223) | (0.2219) | (0.0280) |
| International Asian | -0.7530*** | 0.0087 | 0.0013 | -1.1484*** | -0.0514*** | -0.0921 | -0.0116 |
| | (0.1452) | (0.1396) | (0.0210) | (0.4133) | (0.0172) | (0.1540) | (0.0195) |
| GRE Verbal Percentile | 0.0066* | 0.0326*** | 0.0049*** | 0.0039 | 0.0002 | 0.0006 | 0.0001 |
| | (0.0036) | (0.0034) | (0.0005) | (0.0131) | (0.0006) | (0.0036) | (0.0005) |
| GRE Quantitative Percentile | 0.0447*** | 0.1829*** | 0.0275*** | 0.0402 | 0.0018 | 0.0028 | 0.0004 |
| | (0.0111) | (0.0163) | (0.0018) | (0.0326) | (0.0015) | (0.0080) | (0.0010) |
| GRE Analytic Percentile | 0.0029 | 0.0152*** | 0.0023*** | 0.0078 | 0.0004 | -0.0022 | -0.0003 |
| | (0.0027) | (0.0026) | (0.0004) | (0.0090) | (0.0004) | (0.0029) | (0.0004) |
| TOEFL/IELTS | 0.4407*** | -0.0097 | -0.0015 | -0.1286 | -0.0058 | 0.3375* | 0.0427* |
| | (0.1696) | (0.1632) | (0.0245) | (0.5621) | (0.0250) | (0.1752) | (0.0221) |
| Elite U.S. University | 0.5507*** | 0.9066*** | 0.1363*** | -0.5317 | -0.0238 | -0.0941 | -0.0119 |
| | (0.1994) | (0.2103) | (0.0312) | (0.6438) | (0.0293) | (0.2319) | (0.0293) |
| Elite U.S. Liberal Arts College | -0.0711 | 0.3268 | 0.0491 | 0.3253 | 0.0146 | -0.0866 | -0.0110 |
| | (0.3683) | (0.3402) | (0.0510) | (0.8010) | (0.0356) | (0.3922) | (0.0496) |
| Elite Foreign University | 0.2902 | 0.0518 | 0.0078 | -0.9107 | -0.0408 | 0.0958 | 0.0121 |
| | (0.2991) | (0.2959) | (0.0445) | (0.9288) | (0.0424) | (0.3439) | (0.0435) |
| Other Foreign University | 0.4797** | 0.1161 | 0.0174 | 0.2293 | 0.0103 | 0.1627 | 0.0206 |
| | (0.2022) | (0.1969) | (0.0296) | (0.6888) | (0.0306) | (0.2168) | (0.0274) |
| Missing Undergraduate Institution | 0.5097** | -0.4337** | -0.0652** | 0.0505 | 0.0023 | 0.1768 | 0.0224 |
| | (0.2157) | (0.1996) | (0.0300) | (0.6688) | (0.0299) | (0.2321) | (0.0294) |
| Graduate Degree | -0.0757 | 1.0229*** | 0.1537*** | -0.3419 | -0.0153 | -0.1684 | -0.0213 |
| | (0.1279) | (0.1262) | (0.0177) | (0.3883) | (0.0171) | (0.1367) | (0.0172) |
| Work Experience | 0.3458** | 0.7870*** | 0.1183*** | 0.5914* | 0.0265* | 0.2740* | 0.0347* |
| | (0.1345) | (0.1256) | (0.0179) | (0.3412) | (0.0154) | (0.1424) | (0.0179) |
| #Professor Recommenders | 0.0618 | -0.0211 | -0.0032 | -0.1896 | -0.0085 | -0.0622 | -0.0079 |
| | (0.0769) | (0.0839) | (0.0126) | (0.2574) | (0.0113) | (0.0859) | (0.0109) |
| #Prolific Recommenders | 0.1667*** | 0.1325*** | 0.0199*** | 0.1903 | 0.0085 | 0.0667 | 0.0084 |
| | (0.0548) | (0.0513) | (0.0077) | (0.1757) | (0.0077) | (0.0582) | (0.0073) |
| Missing GRE | -1.6087* | -1.8142*** | -0.2727*** | 1.6533 | 0.0740 | 0.0247 | 0.0031 |
| | (0.9422) | (0.5268) | (0.0772) | (1.8092) | (0.0811) | (0.3772) | (0.0477) |
| Missing TOEFL/IELTS | 0.3199** | 0.0947 | 0.0142 | 0.4511 | 0.0202 | -0.0036 | -0.0005 |
| | (0.1319) | (0.1295) | (0.0194) | (0.4238) | (0.0189) | (0.1435) | (0.0181) |
| Committee Score | | | | 3.7089*** | 0.1660*** | -0.0006 | -0.0001 |
| | | | | (0.3984) | (0.0065) | (0.0486) | (0.0062) |
| #Math Courses | -0.0790 | | | -0.2494 | -0.0112 | 0.0007 | 0.0001 |
| | (0.0519) | | | (0.2386) | (0.0108) | (0.0879) | (0.0111) |
| Advanced Math | 0.2979 | | | 1.1003 | 0.0492 | -0.2220 | -0.0281 |
| | (0.1889) | | | (0.7916) | (0.0347) | (0.2926) | (0.0370) |
| Letters Length/100 | 0.0858*** | | | 0.0770 | 0.0034 | -0.0135 | -0.0017 |
| | (0.0245) | | | (0.1080) | (0.0048) | (0.0385) | (0.0049) |
| Letters Terms (%) PC1 | 0.1027* | | | 0.2451 | 0.0110 | 0.1278 | 0.0162 |
| | (0.0551) | | | (0.2489) | (0.0112) | (0.0911) | (0.0115) |
| Letters Terms (%) PC2 | -0.0208 | | | -0.3767 | -0.0169 | -0.1364 | -0.0172 |
| | (0.0823) | | | (0.3450) | (0.0153) | (0.1257) | (0.0159) |
| CV Length/100 | 0.0513 | | | 0.0391 | 0.0017 | 0.0347 | 0.0044 |
| | (0.0329) | | | (0.1060) | (0.0047) | (0.0592) | (0.0075) |
| CV Coding | 0.0381 | | | -0.2228* | -0.0100* | 0.1024 | 0.0130 |
| | (0.0424) | | | (0.1301) | (0.0060) | (0.0651) | (0.0082) |
| CV Publication/Working Paper | 0.5482*** | | | 0.7611 | 0.0341 | 0.1440 | 0.0182 |
| | (0.1912) | | | (0.8128) | (0.0364) | (0.3359) | (0.0425) |
| Essay Length/100 | 0.0927 | | | 0.2047 | 0.0092 | -0.0718 | -0.0091 |
| | (0.0597) | | | (0.3483) | (0.0157) | (0.0818) | (0.0103) |
| Essay Specific Professor | 0.1411 | | | 0.4125 | 0.0185 | 0.2894 | 0.0366 |
| | (0.1744) | | | (0.6138) | (0.0275) | (0.2770) | (0.0350) |
| Essay Economics Terms (%) | 0.0287 | | | 0.0154 | 0.0007 | -0.0854 | -0.0108 |
| | (0.0312) | | | (0.0814) | (0.0036) | (0.0614) | (0.0077) |
| Essay Research Terms (%) | 0.0396 | | | -0.0751 | -0.0034 | 0.0146 | 0.0018 |
| | (0.0426) | | | (0.1492) | (0.0067) | (0.0672) | (0.0085) |
| Essay Positive Terms (%) | 0.0171 | | | -0.0689 | -0.0031 | 0.0977* | 0.0124* |
| | (0.0345) | | | (0.0968) | (0.0043) | (0.0539) | (0.0068) |
| Essay Negative Terms (%) | 0.0441 | | | -0.0863 | -0.0039 | -0.0441 | -0.0056 |
| | (0.0443) | | | (0.1368) | (0.0061) | (0.0745) | (0.0094) |
| Missing Committee Score | | | | | | -0.2179 | -0.0276 |
| | | | | | | (0.1515) | (0.0191) |
| Missing Application Pdf | -0.5376*** | | | 0.2614 | 0.0117 | 0.2498 | 0.0316 |
| | (0.1841) | | | (0.7005) | (0.0311) | (0.2756) | (0.0348) |
| Program Rank | | | | | | -0.0086*** | -0.0011*** |
| | | | | | | (0.0021) | (0.0003) |
| Missing Program Rank | | | | | | -3.1438*** | -0.3976*** |
| | | | | | | (0.4076) | (0.0514) |
| $\sqrt{\text{MSE}}$ | 1.6467*** | | | | | | |
| | (0.0370) | | | | | | |
| Obs \| Param \| LL \| AIC | 2329 | 582 | -14135.6 | 29435.1 | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

# Table 14: Restricted Dynamic Estimates

| | Committee Score | Cutoff 1 | $L(\text{Cutoff})_1$ | Cutoff 2 | $P(\text{Cutoff})_2$ | Success | Sigma |
|---|---|---|---|---|---|---|---|
| Intercept | -5.6768** | -0.9860*** | 0.3731 | -0.5503*** | 36.58% | -0.7153* | |
| | (2.3401) | (0.1000) | (0.3067, 0.4539) | (0.1561) | (29.81%, 43.92%) | (0.3751) | |
| 2016 | 0.0001 | 0.0171 | 0.3795 | 0.0269 | 37.21% | 0.0155 | |
| | (0.1827) | (0.1265) | (0.3025, 0.4761) | (0.2038) | (29.20%, 45.99%) | (0.2039) | |
| 2017 | 0.4867** | 0.1006 | 0.4126 | 0.1561 | 40.27% | 0.1404 | |
| | (0.2144) | (0.1085) | (0.3563, 0.4778) | (0.1786) | (34.52%, 46.30%) | (0.1795) | |
| 2018 | 0.9329*** | 0.0277 | 0.3836 | 0.0473 | 37.68% | -0.0004 | |
| | (0.1967) | (0.1208) | (0.3271, 0.4497) | (0.1943) | (31.79%, 43.97%) | (0.2015) | |
| 2019 | 0.6228*** | 0.0411 | 0.3887 | 0.0667 | 38.14% | 0.0232 | |
| | (0.1963) | (0.1142) | (0.3268, 0.4624) | (0.1852) | (31.89%, 44.81%) | (0.1863) | |
| Age | -0.0454* | | | | | -0.0004 | |
| | (0.0246) | | | | | (0.0009) | |
| Female | -0.1026 | | | | | 0.0114 | |
| | (0.1055) | | | | | (0.0091) | |
| U.S. | -0.3132 | | | | | 0.0035 | |
| | (0.2098) | | | | | (0.0081) | |
| International Asian | -0.3624** | | | | | -0.0051 | |
| | (0.1577) | | | | | (0.0070) | |
| GRE Verbal Percentile | 0.0110*** | | | | | 0.0002 | |
| | (0.0036) | | | | | (0.0003) | |
| GRE Quantitative Percentile | 0.0580*** | | | | | 0.0007 | |
| | (0.0107) | | | | | (0.0008) | |
| GRE Analytic Percentile | 0.0026 | | | | | 0.0001 | |
| | (0.0028) | | | | | (0.0001) | |
| TOEFL/IELTS | 0.3161* | | | | | 0.0007 | |
| | (0.1703) | | | | | (0.0073) | |
| Elite U.S. University | 0.5975*** | | | | | -0.0108 | |
| | (0.1775) | | | | | (0.0108) | |
| Elite U.S. Liberal Arts College | -0.4500 | | | | | 0.0047 | |
| | (0.3224) | | | | | (0.0151) | |
| Elite Foreign University | 0.2329 | | | | | -0.0086 | |
| | (0.2727) | | | | | (0.0136) | |
| Other Foreign University | 0.4356** | | | | | 0.0048 | |
| | (0.1932) | | | | | (0.0112) | |
| Missing Undergraduate Institution | 0.3922* | | | | | -0.0001 | |
| | (0.2038) | | | | | (0.0098) | |
| Graduate Degree | -0.0531 | | | | | -0.0022 | |
| | (0.1498) | | | | | (0.0063) | |
| Work Experience | 0.0819 | | | | | 0.0088 | |
| | (0.1415) | | | | | (0.0079) | |
| #Professor Recommenders | 0.1025 | | | | | -0.0005 | |
| | (0.0784) | | | | | (0.0037) | |
| #Prolific Recommenders | 0.0536 | | | | | -0.0002 | |
| | (0.0598) | | | | | (0.0026) | |
| Missing GRE | -1.2274 | | | | | 0.0216 | |
| | (0.9140) | | | | | (0.0228) | |
| Missing TOEFL/IELTS | 0.2149* | | | | | 0.0036 | |
| | (0.1252) | | | | | (0.0061) | |
| Committee Score | | | | | | 0.0514 | |
| | | | | | | (0.0355) | |
| #Math Courses | -0.1443 | | | | | -0.0029 | |
| | (0.0920) | | | | | (0.0047) | |
| Advanced Math | 0.4000 | | | | | -0.0022 | |
| | (0.2955) | | | | | (0.0108) | |
| Letters Length/100 | 0.0906** | | | | | 0.0005 | |
| | (0.0439) | | | | | (0.0020) | |
| Letters Terms (%) PC1 | 0.2131** | | | | | 0.0053 | |
| | (0.0942) | | | | | (0.0056) | |
| Letters Terms (%) PC2 | -0.1480 | | | | | -0.0045 | |
| | (0.1963) | | | | | (0.0067) | |
| CV Length/100 | 0.0283 | | | | | 0.0013 | |
| | (0.0724) | | | | | (0.0021) | |
| CV Coding | 0.0306 | | | | | -0.0044 | |
| | (0.0981) | | | | | (0.0040) | |
| CV Publication/Working Paper | 1.1465*** | | | | | 0.0006 | |
| | (0.4362) | | | | | (0.0134) | |
| Essay Length/100 | 0.2719*** | | | | | 0.0080 | |
| | (0.1044) | | | | | (0.0090) | |
| Essay Specific Professor | 0.1374 | | | | | -0.0060 | |
| | (0.2212) | | | | | (0.0136) | |
| Essay Economics Terms (%) | 0.0664 | | | | | 0.0004 | |
| | (0.0550) | | | | | (0.0015) | |
| Essay Research Terms (%) | 0.1364* | | | | | -0.0002 | |
| | (0.0821) | | | | | (0.0030) | |
| Essay Positive Terms (%) | 0.0422 | | | | | -0.0004 | |
| | (0.0718) | | | | | (0.0029) | |
| Essay Negative Terms (%) | 0.1003 | | | | | 0.0008 | |
| | (0.0766) | | | | | (0.0033) | |
| Missing Committee Score | | | | | | -0.2001 | |
| | | | | | | (0.1292) | |
| Missing Application Pdf | -0.7069** | | | | | -0.0267 | |
| | (0.2763) | | | | | (0.0180) | |
| Program Rank | | | | | | -0.0080*** | |
| | | | | | | (0.0020) | |
| Missing Program Rank | | | | | | -3.0773*** | |
| | | | | | | (0.4051) | |
| $\sqrt{\text{MSE}}$ | 1.7300*** | | | | | | |
| | (0.0475) | | | | | | |
| Sigma | | | | | | | 0.0020 |
| | | | | | | | (0.0013) |
| Obs \| Param \| LL \| AIC | 2329 | 529 | -14131.0 | 29319.9 | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

# Table 15: Unrestricted Static Estimates

| | Transition | | Accept | | Success 1 | | Success 2 | |
|---|---|---|---|---|---|---|---|---|
| | Coefficient | AME | Coefficient | AME | Coefficient | AME | Coefficient | AME |
| Intercept | -23.0601*** | -3.4656*** | -28.5278*** | -1.2765*** | -4.6847*** | -0.5998*** | -4.4553** | -0.5635** |
| | (2.0491) | (0.2371) | (6.9081) | (0.2933) | (1.6180) | (0.2054) | (2.1831) | (0.2752) |
| 2016 | -0.2896 | -0.0435 | -0.5294 | -0.0237 | 0.0683 | 0.0087 | 0.0762 | 0.0096 |
| | (0.1794) | (0.0268) | (0.5264) | (0.0230) | (0.1852) | (0.0237) | (0.1883) | (0.0238) |
| 2017 | -0.1533 | -0.0230 | -1.3366** | -0.0598** | 0.1166 | 0.0149 | 0.1266 | 0.0160 |
| | (0.1726) | (0.0260) | (0.5312) | (0.0235) | (0.1877) | (0.0240) | (0.1929) | (0.0244) |
| 2018 | 1.5429*** | 0.2319*** | -3.6854*** | -0.1649*** | 0.0112 | 0.0014 | 0.0095 | 0.0012 |
| | (0.1858) | (0.0251) | (0.6281) | (0.0209) | (0.1920) | (0.0246) | (0.2011) | (0.0254) |
| 2019 | 0.2905 | 0.0437 | -3.0765*** | -0.1377*** | 0.0177 | 0.0023 | 0.0114 | 0.0014 |
| | (0.1833) | (0.0273) | (0.6899) | (0.0257) | (0.1913) | (0.0245) | (0.1974) | (0.0250) |
| Age | 0.0628*** | 0.0094*** | -0.0133 | -0.0006 | 0.0439* | 0.0056* | 0.0428* | 0.0054* |
| | (0.0218) | (0.0032) | (0.0612) | (0.0027) | (0.0229) | (0.0029) | (0.0235) | (0.0030) |
| Female | 0.3919*** | 0.0589*** | 0.8909** | 0.0399*** | -0.0342 | -0.0044 | -0.0422 | -0.0053 |
| | (0.1109) | (0.0164) | (0.3672) | (0.0154) | (0.1232) | (0.0158) | (0.1246) | (0.0158) |
| U.S. | -0.1124 | -0.0169 | 0.0293 | 0.0013 | 0.1634 | 0.0209 | 0.1597 | 0.0202 |
| | (0.1894) | (0.0285) | (0.4978) | (0.0223) | (0.2198) | (0.0281) | (0.2219) | (0.0280) |
| International Asian | 0.0087 | 0.0013 | -1.1484*** | -0.0514*** | -0.0894 | -0.0114 | -0.0921 | -0.0116 |
| | (0.1396) | (0.0210) | (0.4133) | (0.0172) | (0.1489) | (0.0191) | (0.1540) | (0.0195) |
| GRE Verbal Percentile | 0.0326*** | 0.0049*** | 0.0039 | 0.0002 | 0.0015 | 0.0002 | 0.0006 | 0.0001 |
| | (0.0034) | (0.0005) | (0.0131) | (0.0006) | (0.0034) | (0.0004) | (0.0036) | (0.0005) |
| GRE Quantitative Percentile | 0.1829*** | 0.0275*** | 0.0402 | 0.0018 | 0.0065 | 0.0008 | 0.0028 | 0.0004 |
| | (0.0163) | (0.0018) | (0.0326) | (0.0015) | (0.0076) | (0.0010) | (0.0080) | (0.0010) |
| GRE Analytic Percentile | 0.0152*** | 0.0023*** | 0.0078 | 0.0004 | -0.0011 | -0.0001 | -0.0022 | -0.0003 |
| | (0.0026) | (0.0004) | (0.0090) | (0.0004) | (0.0028) | (0.0004) | (0.0029) | (0.0004) |
| TOEFL/IELTS | -0.0097 | -0.0015 | -0.1286 | -0.0058 | 0.3267* | 0.0418* | 0.3375* | 0.0427* |
| | (0.1632) | (0.0245) | (0.5621) | (0.0250) | (0.1721) | (0.0220) | (0.1752) | (0.0221) |
| Elite U.S. University | 0.9066*** | 0.1363*** | -0.5317 | -0.0238 | -0.0506 | -0.0065 | -0.0941 | -0.0119 |
| | (0.2103) | (0.0312) | (0.6438) | (0.0293) | (0.2283) | (0.0292) | (0.2319) | (0.0293) |
| Elite U.S. Liberal Arts College | 0.3268 | 0.0491 | 0.3253 | 0.0146 | -0.0812 | -0.0104 | -0.0866 | -0.0110 |
| | (0.3402) | (0.0510) | (0.8010) | (0.0356) | (0.3863) | (0.0495) | (0.3922) | (0.0496) |
| Elite Foreign University | 0.0518 | 0.0078 | -0.9107 | -0.0408 | 0.0910 | 0.0117 | 0.0958 | 0.0121 |
| | (0.2959) | (0.0445) | (0.9288) | (0.0424) | (0.3413) | (0.0437) | (0.3439) | (0.0435) |
| Other Foreign University | 0.1161 | 0.0174 | 0.2293 | 0.0103 | 0.2023 | 0.0259 | 0.1627 | 0.0206 |
| | (0.1969) | (0.0296) | (0.6888) | (0.0306) | (0.2140) | (0.0274) | (0.2168) | (0.0274) |
| Missing Undergraduate Institution | -0.4337** | -0.0652** | 0.0505 | 0.0023 | 0.1941 | 0.0249 | 0.1768 | 0.0224 |
| | (0.1996) | (0.0300) | (0.6688) | (0.0299) | (0.2292) | (0.0293) | (0.2321) | (0.0294) |
| Graduate Degree | 1.0229*** | 0.1537*** | -0.3419 | -0.0153 | -0.1377 | -0.0176 | -0.1684 | -0.0213 |
| | (0.1262) | (0.0177) | (0.3883) | (0.0171) | (0.1312) | (0.0168) | (0.1367) | (0.0172) |
| Work Experience | 0.7870*** | 0.1183*** | 0.5914* | 0.0265* | 0.3144** | 0.0403** | 0.2740* | 0.0347* |
| | (0.1256) | (0.0179) | (0.3412) | (0.0154) | (0.1361) | (0.0173) | (0.1424) | (0.0179) |
| #Professor Recommenders | -0.0211 | -0.0032 | -0.1896 | -0.0085 | -0.0570 | -0.0073 | -0.0622 | -0.0079 |
| | (0.0839) | (0.0126) | (0.2574) | (0.0113) | (0.0846) | (0.0108) | (0.0859) | (0.0109) |
| #Prolific Recommenders | 0.1325*** | 0.0199*** | 0.1903 | 0.0085 | 0.0559 | 0.0072 | 0.0667 | 0.0084 |
| | (0.0513) | (0.0077) | (0.1757) | (0.0077) | (0.0574) | (0.0073) | (0.0582) | (0.0073) |
| Missing GRE | -1.8142*** | -0.2727*** | 1.6533 | 0.0740 | -0.0087 | -0.0011 | 0.0247 | 0.0031 |
| | (0.5268) | (0.0772) | (1.8092) | (0.0811) | (0.3716) | (0.0476) | (0.3772) | (0.0477) |
| Missing TOEFL/IELTS | 0.0947 | 0.0142 | 0.4511 | 0.0202 | -0.0080 | -0.0010 | -0.0036 | -0.0005 |
| | (0.1295) | (0.0194) | (0.4238) | (0.0189) | (0.1419) | (0.0182) | (0.1435) | (0.0181) |
| Committee Score | | | 3.7089*** | 0.1660*** | | | -0.0006 | -0.0001 |
| | | | (0.3984) | (0.0065) | | | (0.0486) | (0.0062) |
| #Math Courses | | | -0.2494 | -0.0112 | | | 0.0007 | 0.0001 |
| | | | (0.2386) | (0.0108) | | | (0.0879) | (0.0111) |
| Advanced Math | | | 1.1003 | 0.0492 | | | -0.2220 | -0.0281 |
| | | | (0.7916) | (0.0347) | | | (0.2926) | (0.0370) |
| Letters Length/100 | | | 0.0770 | 0.0034 | | | -0.0135 | -0.0017 |
| | | | (0.1080) | (0.0048) | | | (0.0385) | (0.0049) |
| Letters Terms (%) PC1 | | | 0.2451 | 0.0110 | | | 0.1278 | 0.0162 |
| | | | (0.2489) | (0.0112) | | | (0.0911) | (0.0115) |
| Letters Terms (%) PC2 | | | -0.3767 | -0.0169 | | | -0.1364 | -0.0172 |
| | | | (0.3450) | (0.0153) | | | (0.1257) | (0.0159) |
| CV Length/100 | | | 0.0391 | 0.0017 | | | 0.0347 | 0.0044 |
| | | | (0.1060) | (0.0047) | | | (0.0592) | (0.0075) |
| CV Coding | | | -0.2228* | -0.0100* | | | 0.1024 | 0.0130 |
| | | | (0.1301) | (0.0060) | | | (0.0651) | (0.0082) |
| CV Publication/Working Paper | | | 0.7611 | 0.0341 | | | 0.1440 | 0.0182 |
| | | | (0.8128) | (0.0364) | | | (0.3359) | (0.0425) |
| Essay Length/100 | | | 0.2047 | 0.0092 | | | -0.0718 | -0.0091 |
| | | | (0.3483) | (0.0157) | | | (0.0818) | (0.0103) |
| Essay Specific Professor | | | 0.4125 | 0.0185 | | | 0.2894 | 0.0366 |
| | | | (0.6138) | (0.0275) | | | (0.2770) | (0.0350) |
| Essay Economics Terms (%) | | | 0.0154 | 0.0007 | | | -0.0854 | -0.0108 |
| | | | (0.0814) | (0.0036) | | | (0.0614) | (0.0077) |
| Essay Research Terms (%) | | | -0.0751 | -0.0034 | | | 0.0146 | 0.0018 |
| | | | (0.1492) | (0.0067) | | | (0.0672) | (0.0085) |
| Essay Positive Terms (%) | | | -0.0689 | -0.0031 | | | 0.0977* | 0.0124* |
| | | | (0.0968) | (0.0043) | | | (0.0539) | (0.0068) |
| Essay Negative Terms (%) | | | -0.0863 | -0.0039 | | | -0.0441 | -0.0056 |
| | | | (0.1368) | (0.0061) | | | (0.0745) | (0.0094) |
| Missing Committee Score | | | | | | | -0.2179 | -0.0276 |
| | | | | | | | (0.1515) | (0.0191) |
| Missing Application Pdf | | | 0.2614 | 0.0117 | | | 0.2498 | 0.0316 |
| | | | (0.7005) | (0.0311) | | | (0.2756) | (0.0348) |
| Program Rank | | | | | -0.0084*** | -0.0011*** | -0.0086*** | -0.0011*** |
| | | | | | (0.0020) | (0.0003) | (0.0021) | (0.0003) |
| Missing Program Rank | | | | | -3.1385*** | -0.4018*** | -3.1438*** | -0.3976*** |
| | | | | | (0.4051) | (0.0517) | (0.4076) | (0.0514) |
| Obs \| Param \| LL \| AIC | 2329 | 133 | -2964.5 | 6195.0 | | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

# Table 16: Restricted Static Estimates

| | Cutoff 1 | $P(\text{Cutoff})_1$ | Cutoff 2 | $P(\text{Cutoff})_2$ | Success 1 | Success 2 | Sigma |
|---|---|---|---|---|---|---|---|
| Intercept | -0.8781*** | 29.36% | -0.4389*** | 39.20% | -1.2521*** | -0.9262** | |
| | (0.1628) | (23.20%, 36.38%) | (0.1432) | (32.75%, 46.05%) | (0.4466) | (0.4540) | |
| 2016 | -0.0091 | 29.17% | 0.0554 | 40.53% | -0.0178 | 0.0421 | |
| | (0.2466) | (22.21%, 37.27%) | (0.1767) | (33.55%, 47.92%) | (0.2477) | (0.1771) | |
| 2017 | 0.3482* | 37.06% | 0.1035 | 41.69% | 0.3447* | 0.0698 | |
| | (0.1963) | (32.10%, 42.29%) | (0.1727) | (34.83%, 48.90%) | (0.1964) | (0.1725) | |
| 2018 | 0.3936** | 38.12% | 0.0040 | 39.30% | 0.4390** | -0.0912 | |
| | (0.1856) | (33.98%, 42.44%) | (0.1908) | (31.80%, 47.33%) | (0.1859) | (0.1872) | |
| 2019 | 0.2163 | 34.04% | 0.0175 | 39.62% | 0.2258 | -0.0617 | |
| | (0.2017) | (28.67%, 39.85%) | (0.1791) | (32.54%, 47.16%) | (0.2018) | (0.1778) | |
| Age | | | | | 0.0018 | 0.0000 | |
| | | | | | (0.0012) | (0.0016) | |
| Female | | | | | 0.0118 | 0.0233 | |
| | | | | | (0.0073) | (0.0146) | |
| U.S. | | | | | -0.0034 | 0.0022 | |
| | | | | | (0.0058) | (0.0131) | |
| International Asian | | | | | -0.0006 | -0.0302 | |
| | | | | | (0.0041) | (0.0187) | |
| GRE Verbal Percentile | | | | | 0.0010* | 0.0001 | |
| | | | | | (0.0005) | (0.0003) | |
| GRE Quantitative Percentile | | | | | 0.0055* | 0.0011 | |
| | | | | | (0.0032) | (0.0011) | |
| GRE Analytic Percentile | | | | | 0.0005* | 0.0002 | |
| | | | | | (0.0003) | (0.0002) | |
| TOEFL/IELTS | | | | | -0.0009 | -0.0015 | |
| | | | | | (0.0047) | (0.0144) | |
| Elite U.S. University | | | | | 0.0252 | -0.0131 | |
| | | | | | (0.0155) | (0.0178) | |
| Elite U.S. Liberal Arts College | | | | | 0.0087 | 0.0095 | |
| | | | | | (0.0113) | (0.0216) | |
| Elite Foreign University | | | | | 0.0013 | -0.0229 | |
| | | | | | (0.0086) | (0.0272) | |
| Other Foreign University | | | | | 0.0038 | 0.0075 | |
| | | | | | (0.0062) | (0.0180) | |
| Missing Undergraduate Institution | | | | | -0.0119 | 0.0030 | |
| | | | | | (0.0084) | (0.0173) | |
| Graduate Degree | | | | | 0.0306* | -0.0103 | |
| | | | | | (0.0175) | (0.0117) | |
| Work Experience | | | | | 0.0235* | 0.0168 | |
| | | | | | (0.0141) | (0.0130) | |
| #Professor Recommenders | | | | | -0.0004 | -0.0057 | |
| | | | | | (0.0025) | (0.0074) | |
| #Prolific Recommenders | | | | | 0.0039 | 0.0052 | |
| | | | | | (0.0026) | (0.0053) | |
| Missing GRE | | | | | -0.0566 | 0.0458 | |
| | | | | | (0.0361) | (0.0532) | |
| Missing TOEFL/IELTS | | | | | 0.0029 | 0.0123 | |
| | | | | | (0.0041) | (0.0127) | |
| Committee Score | | | | | | 0.0961* | |
| | | | | | | (0.0491) | |
| #Math Courses | | | | | | -0.0064 | |
| | | | | | | (0.0069) | |
| Advanced Math | | | | | | 0.0287 | |
| | | | | | | (0.0244) | |
| Letters Length/100 | | | | | | 0.0019 | |
| | | | | | | (0.0029) | |
| Letters Terms (%) PC1 | | | | | | 0.0063 | |
| | | | | | | (0.0072) | |
| Letters Terms (%) PC2 | | | | | | -0.0103 | |
| | | | | | | (0.0104) | |
| CV Length/100 | | | | | | 0.0010 | |
| | | | | | | (0.0027) | |
| CV Coding | | | | | | -0.0054 | |
| | | | | | | (0.0043) | |
| CV Publication/Working Paper | | | | | | 0.0207 | |
| | | | | | | (0.0245) | |
| Essay Length/100 | | | | | | 0.0050 | |
| | | | | | | (0.0089) | |
| Essay Specific Professor | | | | | | 0.0106 | |
| | | | | | | (0.0168) | |
| Essay Economics Terms (%) | | | | | | 0.0002 | |
| | | | | | | (0.0021) | |
| Essay Research Terms (%) | | | | | | -0.0019 | |
| | | | | | | (0.0039) | |
| Essay Positive Terms (%) | | | | | | -0.0014 | |
| | | | | | | (0.0025) | |
| Essay Negative Terms (%) | | | | | | -0.0023 | |
| | | | | | | (0.0038) | |
| Missing Committee Score | | | | | | -0.2328** | |
| | | | | | | (0.1162) | |
| Missing Application Pdf | | | | | | 0.0079 | |
| | | | | | | (0.0185) | |
| Program Rank | | | | | -0.0085*** | -0.0078*** | |
| | | | | | (0.0021) | (0.0020) | |
| Missing Program Rank | | | | | -3.0030*** | -3.1171*** | |
| | | | | | (0.4080) | (0.4041) | |
| Sigma | | | | | | | 0.0064* |
| | | | | | | | (0.0035) |
| Obs \| Param \| LL \| AIC | 2329 | 80 | -2992.0 | 6144.1 | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

# Table 17: Unrestricted Myopic Estimates

| | Committee Score | Transition | | Accept | | Success | |
|---|---|---|---|---|---|---|---|
| | Coefficient | Coefficient | AME | Coefficient | AME | Coefficient | AME |
| Intercept | 4.0405*** | -23.0601*** | -3.4656*** | -28.5278*** | -1.2765*** | -4.4553** | -0.5635** |
| | (0.9990) | (2.0491) | (0.2371) | (6.9081) | (0.2933) | (2.1831) | (0.2752) |
| 2016 | 0.1701 | -0.2896 | -0.0435 | -0.5294 | -0.0237 | 0.0762 | 0.0096 |
| | (0.1966) | (0.1794) | (0.0268) | (0.5264) | (0.0230) | (0.1883) | (0.0238) |
| 2017 | 0.4355** | -0.1533 | -0.0230 | -1.3366** | -0.0598** | 0.1266 | 0.0160 |
| | (0.2214) | (0.1726) | (0.0260) | (0.5312) | (0.0235) | (0.1929) | (0.0244) |
| 2018 | 0.8410*** | 1.5429*** | 0.2319*** | -3.6854*** | -0.1649*** | 0.0095 | 0.0012 |
| | (0.1873) | (0.1858) | (0.0251) | (0.6281) | (0.0209) | (0.2011) | (0.0254) |
| 2019 | 0.7560*** | 0.2905 | 0.0437 | -3.0765*** | -0.1377*** | 0.0114 | 0.0014 |
| | (0.1987) | (0.1833) | (0.0273) | (0.6899) | (0.0257) | (0.1974) | (0.0250) |
| Age | | 0.0628*** | 0.0094*** | -0.0133 | -0.0006 | 0.0428* | 0.0054* |
| | | (0.0218) | (0.0032) | (0.0612) | (0.0027) | (0.0235) | (0.0030) |
| Female | | 0.3919*** | 0.0589*** | 0.8909** | 0.0399*** | -0.0422 | -0.0053 |
| | | (0.1109) | (0.0164) | (0.3672) | (0.0154) | (0.1246) | (0.0158) |
| U.S. | | -0.1124 | -0.0169 | 0.0293 | 0.0013 | 0.1597 | 0.0202 |
| | | (0.1894) | (0.0285) | (0.4978) | (0.0223) | (0.2219) | (0.0280) |
| International Asian | | 0.0087 | 0.0013 | -1.1484*** | -0.0514*** | -0.0921 | -0.0116 |
| | | (0.1396) | (0.0210) | (0.4133) | (0.0172) | (0.1540) | (0.0195) |
| GRE Verbal Percentile | | 0.0326*** | 0.0049*** | 0.0039 | 0.0002 | 0.0006 | 0.0001 |
| | | (0.0034) | (0.0005) | (0.0131) | (0.0006) | (0.0036) | (0.0005) |
| GRE Quantitative Percentile | | 0.1829*** | 0.0275*** | 0.0402 | 0.0018 | 0.0028 | 0.0004 |
| | | (0.0163) | (0.0018) | (0.0326) | (0.0015) | (0.0080) | (0.0010) |
| GRE Analytic Percentile | | 0.0152*** | 0.0023*** | 0.0078 | 0.0004 | -0.0022 | -0.0003 |
| | | (0.0026) | (0.0004) | (0.0090) | (0.0004) | (0.0029) | (0.0004) |
| TOEFL/IELTS | | -0.0097 | -0.0015 | -0.1286 | -0.0058 | 0.3375* | 0.0427* |
| | | (0.1632) | (0.0245) | (0.5621) | (0.0250) | (0.1752) | (0.0221) |
| Elite U.S. University | | 0.9066*** | 0.1363*** | -0.5317 | -0.0238 | -0.0941 | -0.0119 |
| | | (0.2103) | (0.0312) | (0.6438) | (0.0293) | (0.2319) | (0.0293) |
| Elite U.S. Liberal Arts College | | 0.3268 | 0.0491 | 0.3253 | 0.0146 | -0.0866 | -0.0110 |
| | | (0.3402) | (0.0510) | (0.8010) | (0.0356) | (0.3922) | (0.0496) |
| Elite Foreign University | | 0.0518 | 0.0078 | -0.9107 | -0.0408 | 0.0958 | 0.0121 |
| | | (0.2959) | (0.0445) | (0.9288) | (0.0424) | (0.3439) | (0.0435) |
| Other Foreign University | | 0.1161 | 0.0174 | 0.2293 | 0.0103 | 0.1627 | 0.0206 |
| | | (0.1969) | (0.0296) | (0.6888) | (0.0306) | (0.2168) | (0.0274) |
| Missing Undergraduate Institution | | -0.4337** | -0.0652** | 0.0505 | 0.0023 | 0.1768 | 0.0224 |
| | | (0.1996) | (0.0300) | (0.6688) | (0.0299) | (0.2321) | (0.0294) |
| Graduate Degree | | 1.0229*** | 0.1537*** | -0.3419 | -0.0153 | -0.1684 | -0.0213 |
| | | (0.1262) | (0.0177) | (0.3883) | (0.0171) | (0.1367) | (0.0172) |
| Work Experience | | 0.7870*** | 0.1183*** | 0.5914* | 0.0265* | 0.2740* | 0.0347* |
| | | (0.1256) | (0.0179) | (0.3412) | (0.0154) | (0.1424) | (0.0179) |
| #Professor Recommenders | | -0.0211 | -0.0032 | -0.1896 | -0.0085 | -0.0622 | -0.0079 |
| | | (0.0839) | (0.0126) | (0.2574) | (0.0113) | (0.0859) | (0.0109) |
| #Prolific Recommenders | | 0.1325*** | 0.0199*** | 0.1903 | 0.0085 | 0.0667 | 0.0084 |
| | | (0.0513) | (0.0077) | (0.1757) | (0.0077) | (0.0582) | (0.0073) |
| Missing GRE | | -1.8142*** | -0.2727*** | 1.6533 | 0.0740 | 0.0247 | 0.0031 |
| | | (0.5268) | (0.0772) | (1.8092) | (0.0811) | (0.3772) | (0.0477) |
| Missing TOEFL/IELTS | | 0.0947 | 0.0142 | 0.4511 | 0.0202 | -0.0036 | -0.0005 |
| | | (0.1295) | (0.0194) | (0.4238) | (0.0189) | (0.1435) | (0.0181) |
| Committee Score | | | | 3.7089*** | 0.1660*** | -0.0006 | -0.0001 |
| | | | | (0.3984) | (0.0065) | (0.0486) | (0.0062) |
| #Math Courses | -0.0804 | | | -0.2494 | -0.0112 | 0.0007 | 0.0001 |
| | (0.0520) | | | (0.2386) | (0.0108) | (0.0879) | (0.0111) |
| Advanced Math | 0.3560* | | | 1.1003 | 0.0492 | -0.2220 | -0.0281 |
| | (0.1909) | | | (0.7916) | (0.0347) | (0.2926) | (0.0370) |
| Letters Length/100 | 0.1225*** | | | 0.0770 | 0.0034 | -0.0135 | -0.0017 |
| | (0.0242) | | | (0.1080) | (0.0048) | (0.0385) | (0.0049) |
| Letters Terms (%) PC1 | 0.0974* | | | 0.2451 | 0.0110 | 0.1278 | 0.0162 |
| | (0.0536) | | | (0.2489) | (0.0112) | (0.0911) | (0.0115) |
| Letters Terms (%) PC2 | -0.0925 | | | -0.3767 | -0.0169 | -0.1364 | -0.0172 |
| | (0.0856) | | | (0.3450) | (0.0153) | (0.1257) | (0.0159) |
| CV Length/100 | 0.0588* | | | 0.0391 | 0.0017 | 0.0347 | 0.0044 |
| | (0.0323) | | | (0.1060) | (0.0047) | (0.0592) | (0.0075) |
| CV Coding | 0.0551 | | | -0.2228* | -0.0100* | 0.1024 | 0.0130 |
| | (0.0417) | | | (0.1301) | (0.0060) | (0.0651) | (0.0082) |
| CV Publication/Working Paper | 0.5324*** | | | 0.7611 | 0.0341 | 0.1440 | 0.0182 |
| | (0.1975) | | | (0.8128) | (0.0364) | (0.3359) | (0.0425) |
| Essay Length/100 | 0.0763 | | | 0.2047 | 0.0092 | -0.0718 | -0.0091 |
| | (0.0567) | | | (0.3483) | (0.0157) | (0.0818) | (0.0103) |
| Essay Specific Professor | 0.1442 | | | 0.4125 | 0.0185 | 0.2894 | 0.0366 |
| | (0.1767) | | | (0.6138) | (0.0275) | (0.2770) | (0.0350) |
| Essay Economics Terms (%) | 0.0042 | | | 0.0154 | 0.0007 | -0.0854 | -0.0108 |
| | (0.0314) | | | (0.0814) | (0.0036) | (0.0614) | (0.0077) |
| Essay Research Terms (%) | 0.0304 | | | -0.0751 | -0.0034 | 0.0146 | 0.0018 |
| | (0.0446) | | | (0.1492) | (0.0067) | (0.0672) | (0.0085) |
| Essay Positive Terms (%) | 0.0175 | | | -0.0689 | -0.0031 | 0.0977* | 0.0124* |
| | (0.0373) | | | (0.0968) | (0.0043) | (0.0539) | (0.0068) |
| Essay Negative Terms (%) | 0.0429 | | | -0.0863 | -0.0039 | -0.0441 | -0.0056 |
| | (0.0460) | | | (0.1368) | (0.0061) | (0.0745) | (0.0094) |
| Missing Committee Score | | | | | | -0.2179 | -0.0276 |
| | | | | | | (0.1515) | (0.0191) |
| Missing Application Pdf | -0.4006** | | | 0.2614 | 0.0117 | 0.2498 | 0.0316 |
| | (0.1850) | | | (0.7005) | (0.0311) | (0.2756) | (0.0348) |
| Program Rank | | | | | | -0.0086*** | -0.0011*** |
| | | | | | | (0.0021) | (0.0003) |
| Missing Program Rank | | | | | | -3.1438*** | -0.3976*** |
| | | | | | | (0.4076) | (0.0514) |
| $\sqrt{\text{MSE}}$ | 1.7525*** | | | | | | |
| | (0.0403) | | | | | | |
| Obs \| Param \| LL \| AIC | 2329 | 297 | -14470.7 | 29535.4 | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

# Table 18: Restricted Myopic Estimates

| | Committee Score | Cutoff 1 | $L(\text{Cutoff})_1$ | Cutoff 2 | $P(\text{Cutoff})_2$ | Success | Sigma |
|---|---|---|---|---|---|---|---|
| Intercept | 3.4796*** | -0.9771*** | 0.3764 | -0.5522*** | 36.54% | -1.2859** | |
| | (1.2648) | (0.0960) | (0.3118, 0.4544) | (0.1553) | (29.80%, 43.84%) | (0.5090) | |
| 2016 | 0.0978 | -0.0005 | 0.3762 | -0.0048 | 36.42% | -0.0164 | |
| | (0.1990) | (0.1320) | (0.2992, 0.4731) | (0.2135) | (28.21%, 45.51%) | (0.2130) | |
| 2017 | 0.7100*** | 0.1077 | 0.4192 | 0.1614 | 40.35% | 0.1384 | |
| | (0.2332) | (0.1059) | (0.3701, 0.4749) | (0.1760) | (35.00%, 45.95%) | (0.1772) | |
| 2018 | 1.0737*** | 0.0146 | 0.3820 | 0.0286 | 37.20% | -0.0305 | |
| | (0.1730) | (0.1219) | (0.3237, 0.4506) | (0.1968) | (31.07%, 43.76%) | (0.1993) | |
| 2019 | 0.8095*** | 0.0005 | 0.3766 | 0.0004 | 36.54% | -0.0654 | |
| | (0.1948) | (0.1254) | (0.3086, 0.4596) | (0.2030) | (29.43%, 44.30%) | (0.2045) | |
| Age | | | | | | 0.0010 | |
| | | | | | | (0.0007) | |
| Female | | | | | | 0.0132** | |
| | | | | | | (0.0063) | |
| U.S. | | | | | | 0.0010 | |
| | | | | | | (0.0048) | |
| International Asian | | | | | | -0.0075 | |
| | | | | | | (0.0051) | |
| GRE Verbal Percentile | | | | | | 0.0008** | |
| | | | | | | (0.0003) | |
| GRE Quantitative Percentile | | | | | | 0.0039** | |
| | | | | | | (0.0017) | |
| GRE Analytic Percentile | | | | | | 0.0004** | |
| | | | | | | (0.0002) | |
| TOEFL/IELTS | | | | | | 0.0009 | |
| | | | | | | (0.0042) | |
| Elite U.S. University | | | | | | 0.0105 | |
| | | | | | | (0.0071) | |
| Elite U.S. Liberal Arts College | | | | | | 0.0064 | |
| | | | | | | (0.0086) | |
| Elite Foreign University | | | | | | -0.0012 | |
| | | | | | | (0.0077) | |
| Other Foreign University | | | | | | 0.0066 | |
| | | | | | | (0.0059) | |
| Missing Undergraduate Institution | | | | | | -0.0034 | |
| | | | | | | (0.0053) | |
| Graduate Degree | | | | | | 0.0194** | |
| | | | | | | (0.0090) | |
| Work Experience | | | | | | 0.0189** | |
| | | | | | | (0.0089) | |
| #Professor Recommenders | | | | | | -0.0014 | |
| | | | | | | (0.0022) | |
| #Prolific Recommenders | | | | | | 0.0016 | |
| | | | | | | (0.0016) | |
| Missing GRE | | | | | | -0.0449 | |
| | | | | | | (0.0282) | |
| Missing TOEFL/IELTS | | | | | | 0.0012 | |
| | | | | | | (0.0035) | |
| Committee Score | | | | | | 0.0686** | |
| | | | | | | (0.0302) | |
| #Math Courses | -0.0627 | | | | | -0.0011 | |
| | (0.0689) | | | | | (0.0071) | |
| Advanced Math | 0.3298 | | | | | 0.0023 | |
| | (0.2533) | | | | | (0.0161) | |
| Letters Length/100 | 0.1467*** | | | | | 0.0023 | |
| | (0.0317) | | | | | (0.0029) | |
| Letters Terms (%) PC1 | 0.1155 | | | | | 0.0016 | |
| | (0.0772) | | | | | (0.0087) | |
| Letters Terms (%) PC2 | -0.1020 | | | | | -0.0069 | |
| | (0.1152) | | | | | (0.0089) | |
| CV Length/100 | 0.0628 | | | | | 0.0022 | |
| | (0.0407) | | | | | (0.0026) | |
| CV Coding | 0.0466 | | | | | -0.0051 | |
| | (0.0580) | | | | | (0.0036) | |
| CV Publication/Working Paper | 0.7501*** | | | | | -0.0009 | |
| | (0.2653) | | | | | (0.0155) | |
| Essay Length/100 | 0.1355 | | | | | 0.0083 | |
| | (0.0858) | | | | | (0.0131) | |
| Essay Specific Professor | 0.1387 | | | | | 0.0023 | |
| | (0.2579) | | | | | (0.0205) | |
| Essay Economics Terms (%) | -0.0024 | | | | | -0.0018 | |
| | (0.0396) | | | | | (0.0021) | |
| Essay Research Terms (%) | 0.0663 | | | | | 0.0023 | |
| | (0.0596) | | | | | (0.0044) | |
| Essay Positive Terms (%) | 0.0403 | | | | | 0.0011 | |
| | (0.0476) | | | | | (0.0033) | |
| Essay Negative Terms (%) | 0.0459 | | | | | -0.0009 | |
| | (0.0575) | | | | | (0.0048) | |
| Missing Committee Score | | | | | | -0.1177 | |
| | | | | | | (0.1482) | |
| Missing Application Pdf | -0.9529*** | | | | | -0.0359** | |
| | (0.2274) | | | | | (0.0157) | |
| Program Rank | | | | | | -0.0078*** | |
| | | | | | | (0.0020) | |
| Missing Program Rank | | | | | | -3.0747*** | |
| | | | | | | (0.4058) | |
| $\sqrt{\text{MSE}}$ | 1.8070*** | | | | | | |
| | (0.0436) | | | | | | |
| Sigma | | | | | | | 0.0027** |
| | | | | | | | (0.0012) |
| Obs \| Param \| LL \| AIC | 2329 | 244 | -14536.1 | 29560.2 | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

# Table 19: Unrestricted Matriculation-Dynamic Estimates

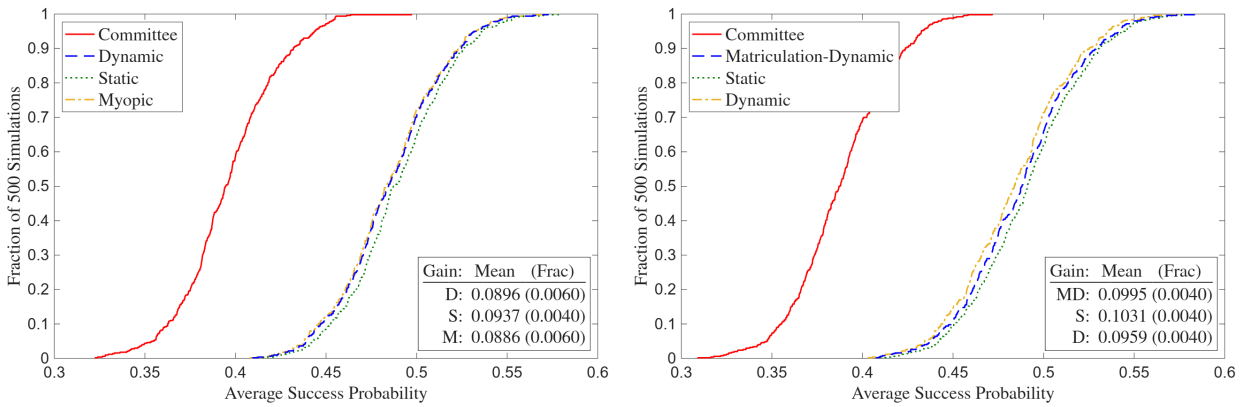
| | Committee Score | Transition | | Accept | | Success | | Matriculate | |
|---|---|---|---|---|---|---|---|---|---|
| | Coefficient | Coefficient | AME | Coefficient | AME | Coefficient | AME | Coefficient | AME |
| Intercept | -2.7966 | -23.0601*** | -3.4656*** | -28.5278*** | -1.2765*** | -4.4553** | -0.5635** | 21.0916*** | 3.3374*** |
| | (2.1751) | (2.0491) | (0.2371) | (6.9081) | (0.2933) | (2.1831) | (0.2752) | (6.9868) | (1.0163) |
| 2016 | 0.0915 | -0.2896 | -0.0435 | -0.5294 | -0.0237 | 0.0762 | 0.0096 | 0.3206 | 0.0507 |
| | (0.1856) | (0.1794) | (0.0268) | (0.5264) | (0.0230) | (0.1883) | (0.0238) | (0.6530) | (0.1030) |
| 2017 | 0.2354 | -0.1533 | -0.0230 | -1.3366** | -0.0598** | 0.1266 | 0.0160 | 0.9971 | 0.1578 |
| | (0.2130) | (0.1726) | (0.0260) | (0.5312) | (0.0235) | (0.1929) | (0.0244) | (0.6410) | (0.1008) |
| 2018 | 0.5999*** | 1.5429*** | 0.2319*** | -3.6854*** | -0.1649*** | 0.0095 | 0.0012 | 1.6496*** | 0.2610*** |
| | (0.1777) | (0.1858) | (0.0251) | (0.6281) | (0.0209) | (0.2011) | (0.0254) | (0.5920) | (0.0886) |
| 2019 | 0.6118*** | 0.2905 | 0.0437 | -3.0765*** | -0.1377*** | 0.0114 | 0.0014 | 1.0916* | 0.1727* |
| | (0.1907) | (0.1833) | (0.0273) | (0.6899) | (0.0257) | (0.1974) | (0.0250) | (0.6190) | (0.0971) |
| Age | -0.0621*** | 0.0628*** | 0.0094*** | -0.0133 | -0.0006 | 0.0428* | 0.0054* | -0.1396* | -0.0221* |
| | (0.0235) | (0.0218) | (0.0032) | (0.0612) | (0.0027) | (0.0235) | (0.0030) | (0.0738) | (0.0113) |
| Female | -0.0543 | 0.3919*** | 0.0589*** | 0.8909** | 0.0399*** | -0.0422 | -0.0053 | -0.4562 | -0.0722 |
| | (0.1125) | (0.1109) | (0.0164) | (0.3672) | (0.0154) | (0.1246) | (0.0158) | (0.3550) | (0.0556) |
| U.S. | -0.5071** | -0.1124 | -0.0169 | 0.0293 | 0.0013 | 0.1597 | 0.0202 | 0.9160 | 0.1449 |
| | (0.2043) | (0.1894) | (0.0285) | (0.4978) | (0.0223) | (0.2219) | (0.0280) | (0.5700) | (0.0896) |
| International Asian | -0.7530*** | 0.0087 | 0.0013 | -1.1484*** | -0.0514*** | -0.0921 | -0.0116 | 1.1667** | 0.1846*** |
| | (0.1452) | (0.1396) | (0.0210) | (0.4133) | (0.0172) | (0.1540) | (0.0195) | (0.4531) | (0.0689) |
| GRE Verbal Percentile | 0.0066* | 0.0326*** | 0.0049*** | 0.0039 | 0.0002 | 0.0006 | 0.0001 | -0.0043 | -0.0007 |
| | (0.0036) | (0.0034) | (0.0005) | (0.0131) | (0.0006) | (0.0036) | (0.0005) | (0.0115) | (0.0018) |
| GRE Quantitative Percentile | 0.0447*** | 0.1829*** | 0.0275*** | 0.0402 | 0.0018 | 0.0028 | 0.0004 | -0.1358*** | -0.0215*** |
| | (0.0111) | (0.0163) | (0.0018) | (0.0326) | (0.0015) | (0.0080) | (0.0010) | (0.0444) | (0.0064) |
| GRE Analytic Percentile | 0.0029 | 0.0152*** | 0.0023*** | 0.0078 | 0.0004 | -0.0022 | -0.0003 | -0.0185** | -0.0029** |
| | (0.0027) | (0.0026) | (0.0004) | (0.0090) | (0.0004) | (0.0029) | (0.0004) | (0.0087) | (0.0013) |
| TOEFL/IELTS | 0.4407*** | -0.0097 | -0.0015 | -0.1286 | -0.0058 | 0.3375* | 0.0427* | -0.5140 | -0.0813 |
| | (0.1696) | (0.1632) | (0.0245) | (0.5621) | (0.0250) | (0.1752) | (0.0221) | (0.5175) | (0.0807) |
| Elite U.S. University | 0.5507*** | 0.9066*** | 0.1363*** | -0.5317 | -0.0238 | -0.0941 | -0.0119 | -0.4342 | -0.0687 |
| | (0.1994) | (0.2103) | (0.0312) | (0.6438) | (0.0293) | (0.2319) | (0.0293) | (0.5475) | (0.0858) |
| Elite U.S. Liberal Arts College | -0.0711 | 0.3268 | 0.0491 | 0.3253 | 0.0146 | -0.0866 | -0.0110 | 1.0020 | 0.1586 |
| | (0.3683) | (0.3402) | (0.0510) | (0.8010) | (0.0356) | (0.3922) | (0.0496) | (0.7518) | (0.1185) |
| Elite Foreign University | 0.2902 | 0.0518 | 0.0078 | -0.9107 | -0.0408 | 0.0958 | 0.0121 | -0.2759 | -0.0437 |
| | (0.2991) | (0.2959) | (0.0445) | (0.9288) | (0.0424) | (0.3439) | (0.0435) | (0.9777) | (0.1548) |
| Other Foreign University | 0.4797** | 0.1161 | 0.0174 | 0.2293 | 0.0103 | 0.1627 | 0.0206 | 0.4536 | 0.0718 |
| | (0.2022) | (0.1969) | (0.0296) | (0.6888) | (0.0306) | (0.2168) | (0.0274) | (0.6285) | (0.0992) |
| Missing Undergraduate Institution | 0.5097** | -0.4337** | -0.0652** | 0.0505 | 0.0023 | 0.1768 | 0.0224 | -0.8208 | -0.1299 |
| | (0.2157) | (0.1996) | (0.0300) | (0.6688) | (0.0299) | (0.2321) | (0.0294) | (0.6805) | (0.1058) |
| Graduate Degree | -0.0757 | 1.0229*** | 0.1537*** | -0.3419 | -0.0153 | -0.1684 | -0.0213 | -0.0653 | -0.0103 |
| | (0.1279) | (0.1262) | (0.0177) | (0.3883) | (0.0171) | (0.1367) | (0.0172) | (0.4719) | (0.0746) |
| Work Experience | 0.3458** | 0.7870*** | 0.1183*** | 0.5914* | 0.0265* | 0.2740* | 0.0347* | 0.9166** | 0.1450** |
| | (0.1345) | (0.1256) | (0.0179) | (0.3412) | (0.0154) | (0.1424) | (0.0179) | (0.4411) | (0.0667) |
| #Professor Recommenders | 0.0618 | -0.0211 | -0.0032 | -0.1896 | -0.0085 | -0.0622 | -0.0079 | 0.3588 | 0.0568 |
| | (0.0769) | (0.0839) | (0.0126) | (0.2574) | (0.0113) | (0.0859) | (0.0109) | (0.2789) | (0.0433) |
| #Prolific Recommenders | 0.1667*** | 0.1325*** | 0.0199*** | 0.1903 | 0.0085 | 0.0667 | 0.0084 | -0.2845 | -0.0450 |
| | (0.0548) | (0.0513) | (0.0077) | (0.1757) | (0.0077) | (0.0582) | (0.0073) | (0.1875) | (0.0288) |
| Missing GRE | -1.6087* | -1.8142*** | -0.2727*** | 1.6533 | 0.0740 | 0.0247 | 0.0031 | | |
| | (0.9422) | (0.5268) | (0.0772) | (1.8092) | (0.0811) | (0.3772) | (0.0477) | | |
| Missing TOEFL/IELTS | 0.3199** | 0.0947 | 0.0142 | 0.4511 | 0.0202 | -0.0036 | -0.0005 | -0.0091 | -0.0014 |
| | (0.1319) | (0.1295) | (0.0194) | (0.4238) | (0.0189) | (0.1435) | (0.0181) | (0.5591) | (0.0885) |
| Committee Score | | | | 3.7089*** | 0.1660*** | -0.0006 | -0.0001 | -0.1637 | -0.0259 |
| | | | | (0.3984) | (0.0065) | (0.0486) | (0.0062) | (0.2099) | (0.0336) |
| #Math Courses | -0.0790 | | | -0.2494 | -0.0112 | 0.0007 | 0.0001 | -0.0182 | -0.0029 |
| | (0.0519) | | | (0.2386) | (0.0108) | (0.0879) | (0.0111) | (0.1671) | (0.0265) |
| Advanced Math | 0.2979 | | | 1.1003 | 0.0492 | -0.2220 | -0.0281 | -1.1143** | -0.1763** |
| | (0.1889) | | | (0.7916) | (0.0347) | (0.2926) | (0.0370) | (0.5578) | (0.0877) |
| Letters Length/100 | 0.0858*** | | | 0.0770 | 0.0034 | -0.0135 | -0.0017 | -0.0744 | -0.0118 |
| | (0.0245) | | | (0.1080) | (0.0048) | (0.0385) | (0.0049) | (0.0711) | (0.0113) |
| Letters Terms (%) PC1 | 0.1027* | | | 0.2451 | 0.0110 | 0.1278 | 0.0162 | 0.3997** | 0.0632** |
| | (0.0551) | | | (0.2489) | (0.0112) | (0.0911) | (0.0115) | (0.1903) | (0.0287) |
| Letters Terms (%) PC2 | -0.0208 | | | -0.3767 | -0.0169 | -0.1364 | -0.0172 | 0.2897 | 0.0458 |
| | (0.0823) | | | (0.3450) | (0.0153) | (0.1257) | (0.0159) | (0.2460) | (0.0388) |
| CV Length/100 | 0.0513 | | | 0.0391 | 0.0017 | 0.0347 | 0.0044 | -0.1109 | -0.0175 |
| | (0.0329) | | | (0.1060) | (0.0047) | (0.0592) | (0.0075) | (0.1105) | (0.0175) |
| CV Coding | 0.0381 | | | -0.2228* | -0.0100* | 0.1024 | 0.0130 | -0.2901** | -0.0459** |
| | (0.0424) | | | (0.1301) | (0.0060) | (0.0651) | (0.0082) | (0.1176) | (0.0181) |
| CV Publication/Working Paper | 0.5482*** | | | 0.7611 | 0.0341 | 0.1440 | 0.0182 | -0.3451 | -0.0546 |
| | (0.1912) | | | (0.8128) | (0.0364) | (0.3359) | (0.0425) | (0.5102) | (0.0812) |
| Essay Length/100 | 0.0927 | | | 0.2047 | 0.0092 | -0.0718 | -0.0091 | 0.0882 | 0.0140 |
| | (0.0597) | | | (0.3483) | (0.0157) | (0.0818) | (0.0103) | (0.1386) | (0.0220) |
| Essay Specific Professor | 0.1411 | | | 0.4125 | 0.0185 | 0.2894 | 0.0366 | 0.6812 | 0.1078 |
| | (0.1744) | | | (0.6138) | (0.0275) | (0.2770) | (0.0350) | (0.4976) | (0.0773) |
| Essay Economics Terms (%) | 0.0287 | | | 0.0154 | 0.0007 | -0.0854 | -0.0108 | 0.2244** | 0.0355** |
| | (0.0312) | | | (0.0814) | (0.0036) | (0.0614) | (0.0077) | (0.1075) | (0.0169) |
| Essay Research Terms (%) | 0.0396 | | | -0.0751 | -0.0034 | 0.0146 | 0.0018 | -0.1904 | -0.0301 |
| | (0.0426) | | | (0.1492) | (0.0067) | (0.0672) | (0.0085) | (0.1440) | (0.0226) |
| Essay Positive Terms (%) | 0.0171 | | | -0.0689 | -0.0031 | 0.0977* | 0.0124* | -0.0170 | -0.0027 |
| | (0.0345) | | | (0.0968) | (0.0043) | (0.0539) | (0.0068) | (0.1151) | (0.0182) |
| Essay Negative Terms (%) | 0.0441 | | | -0.0863 | -0.0039 | -0.0441 | -0.0056 | -0.0176 | -0.0028 |
| | (0.0443) | | | (0.1368) | (0.0061) | (0.0745) | (0.0094) | (0.1043) | (0.0165) |
| Missing Committee Score | | | | | | -0.2179 | -0.0276 | | |
| | | | | | | (0.1515) | (0.0191) | | |
| Missing Application Pdf | -0.5376*** | | | 0.2614 | 0.0117 | 0.2498 | 0.0316 | -0.7440 | -0.1177 |
| | (0.1841) | | | (0.7005) | (0.0311) | (0.2756) | (0.0348) | (0.5803) | (0.0919) |
| Program Rank | | | | | | -0.0086*** | -0.0011*** | | |
| | | | | | | (0.0021) | (0.0003) | | |
| Missing Program Rank | | | | | | -3.1438*** | -0.3976*** | | |
| | | | | | | (0.4076) | (0.0514) | | |
| $\sqrt{\text{MSE}}$ | 1.6467*** | | | | | | | | |
| | (0.0370) | | | | | | | | |
| Obs \| Param \| LL \| AIC | 2329 | 621 | -14259.1 | 29760.1 | | | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

# Table 20: Restricted Matriculation-Dynamic Estimates

| | Committee Score | Cutoff 1 | $L(\text{Cutoff})_1$ | Cutoff 2 | $P(\text{Cutoff})_2$ | Success | Matriculate | Sigma |
|---|---|---|---|---|---|---|---|---|
| Intercept | -6.5631*** | -4.9411*** | 0.0071 | -0.5602*** | 36.35% | -0.8176** | 1.5446 | |
| | (2.5333) | (0.5082) | (0.0026, 0.0194) | (0.1594) | (29.47%, 43.84%) | (0.3643) | (5.3407) | |
| 2016 | 0.2912 | -0.0124 | 0.0071 | 0.0811 | 38.25% | 0.0776 | -1.2678* | |
| | (0.2079) | (0.1800) | (0.0028, 0.0180) | (0.1961) | (31.05%, 46.00%) | (0.1966) | (0.7295) | |
| 2017 | 0.6424*** | 0.3850** | 0.0105 | 0.1816 | 40.65% | 0.1644 | -0.3048 | |
| | (0.2488) | (0.1788) | (0.0041, 0.0267) | (0.1820) | (35.15%, 46.39%) | (0.1821) | (0.8727) | |
| 2018 | 0.8514*** | 0.2894 | 0.0095 | 0.0102 | 36.59% | -0.0518 | 1.3060* | |
| | (0.1924) | (0.1890) | (0.0038, 0.0240) | (0.2053) | (30.01%, 43.71%) | (0.2096) | (0.7318) | |
| 2019 | 0.5883*** | 0.2802 | 0.0095 | 0.0456 | 37.41% | -0.0066 | 0.7687 | |
| | (0.2154) | (0.2153) | (0.0035, 0.0255) | (0.1960) | (31.04%, 44.26%) | (0.1975) | (0.8962) | |
| Age | -0.0569** | | | | | -0.0001 | -0.0312 | |
| | (0.0252) | | | | | (0.0011) | (0.0547) | |
| Female | -0.0776 | | | | | 0.0116 | 0.0927 | |
| | (0.1152) | | | | | (0.0083) | (0.2112) | |
| U.S. | -0.3707* | | | | | -0.0033 | 0.4696 | |
| | (0.2114) | | | | | (0.0084) | (0.4504) | |
| International Asian | -0.5051*** | | | | | -0.0103 | 0.5968* | |
| | (0.1631) | | | | | (0.0083) | (0.3430) | |
| GRE Verbal Percentile | 0.0082** | | | | | 0.0000 | 0.0098 | |
| | (0.0037) | | | | | (0.0002) | (0.0075) | |
| GRE Quantitative Percentile | 0.0704*** | | | | | 0.0010 | -0.0610* | |
| | (0.0105) | | | | | (0.0009) | (0.0338) | |
| GRE Analytic Percentile | 0.0016 | | | | | 0.0001 | 0.0021 | |
| | (0.0029) | | | | | (0.0002) | (0.0066) | |
| TOEFL/IELTS | 0.2848 | | | | | -0.0023 | -0.0629 | |
| | (0.1746) | | | | | (0.0101) | (0.2753) | |
| Elite U.S. University | 0.6391*** | | | | | -0.0098 | -0.0637 | |
| | (0.1892) | | | | | (0.0142) | (0.3925) | |
| Elite U.S. Liberal Arts College | -0.3528 | | | | | 0.0127 | -0.2757 | |
| | (0.3489) | | | | | (0.0174) | (0.6783) | |
| Elite Foreign University | 0.1906 | | | | | -0.0187 | 0.4536 | |
| | (0.2875) | | | | | (0.0195) | (0.5403) | |
| Other Foreign University | 0.5511*** | | | | | 0.0038 | 0.2515 | |
| | (0.2049) | | | | | (0.0148) | (0.3764) | |
| Missing Undergraduate Institution | 0.5458*** | | | | | 0.0033 | -0.3893 | |
| | (0.2109) | | | | | (0.0140) | (0.4067) | |
| Graduate Degree | -0.1826 | | | | | -0.0069 | 0.4784 | |
| | (0.1589) | | | | | (0.0082) | (0.3581) | |
| Work Experience | 0.0765 | | | | | 0.0080 | 0.5662 | |
| | (0.1486) | | | | | (0.0075) | (0.3731) | |
| #Professor Recommenders | 0.0824 | | | | | -0.0021 | 0.2152 | |
| | (0.0822) | | | | | (0.0054) | (0.1592) | |
| #Prolific Recommenders | 0.0919 | | | | | -0.0002 | -0.1769* | |
| | (0.0590) | | | | | (0.0032) | (0.1041) | |
| Missing GRE | -1.3931 | | | | | 0.0140 | | |
| | (0.8753) | | | | | (0.0192) | | |
| Missing TOEFL/IELTS | 0.2632** | | | | | 0.0023 | 0.0851 | |
| | (0.1311) | | | | | (0.0066) | (0.2494) | |
| Committee Score | | | | | | 0.0649** | 0.5117*** | |
| | | | | | | (0.0299) | (0.1497) | |
| #Math Courses | -0.1660 | | | | | -0.0048 | -0.1664 | |
| | (0.1024) | | | | | (0.0057) | (0.1619) | |
| Advanced Math | 0.7224* | | | | | 0.0257 | 0.0920 | |
| | (0.4161) | | | | | (0.0208) | (0.5297) | |
| Letters Length/100 | 0.1095** | | | | | 0.0013 | -0.0122 | |
| | (0.0471) | | | | | (0.0022) | (0.0646) | |
| Letters Terms (%) PC1 | 0.2635*** | | | | | 0.0089 | 0.4342*** | |
| | (0.0988) | | | | | (0.0071) | (0.1642) | |
| Letters Terms (%) PC2 | -0.3026* | | | | | -0.0108 | -0.2680 | |
| | (0.1764) | | | | | (0.0091) | (0.2167) | |
| CV Length/100 | 0.0263 | | | | | 0.0004 | -0.0601 | |
| | (0.0752) | | | | | (0.0023) | (0.0819) | |
| CV Coding | 0.0417 | | | | | -0.0049 | -0.0663 | |
| | (0.1086) | | | | | (0.0044) | (0.1017) | |
| CV Publication/Working Paper | 1.3411*** | | | | | 0.0190 | -0.2251 | |
| | (0.3435) | | | | | (0.0192) | (0.5320) | |
| Essay Length/100 | 0.2771* | | | | | 0.0065 | 0.1436 | |
| | (0.1475) | | | | | (0.0103) | (0.1533) | |
| Essay Specific Professor | 0.4040 | | | | | 0.0133 | 0.6561 | |
| | (0.3655) | | | | | (0.0168) | (0.4106) | |
| Essay Economics Terms (%) | 0.0881 | | | | | 0.0010 | 0.0869 | |
| | (0.0633) | | | | | (0.0022) | (0.0809) | |
| Essay Research Terms (%) | 0.1320 | | | | | 0.0005 | -0.1728 | |
| | (0.0843) | | | | | (0.0030) | (0.1284) | |
| Essay Positive Terms (%) | 0.0538 | | | | | -0.0006 | 0.0066 | |
| | (0.0761) | | | | | (0.0025) | (0.0966) | |
| Essay Negative Terms (%) | 0.0781 | | | | | -0.0021 | -0.0494 | |
| | (0.0830) | | | | | (0.0035) | (0.1032) | |
| Missing Committee Score | | | | | | -0.1920 | | |
| | | | | | | (0.1290) | | |
| Missing Application Pdf | -0.4098 | | | | | 0.0124 | -1.5977*** | |
| | (0.2846) | | | | | (0.0125) | (0.2218) | |
| Program Rank | | | | | | -0.0079*** | | |
| | | | | | | (0.0020) | | |
| Missing Program Rank | | | | | | -3.0808*** | | |
| | | | | | | (0.4053) | | |
| $\sqrt{\text{MSE}}$ | 1.7632*** | | | | | | | |
| | (0.0469) | | | | | | | |
| Sigma | | | | | | | | 0.0020** |
| | | | | | | | | (0.0010) |
| Obs \| Param \| LL \| AIC | 2329 | 568 | -14260.9 | 29657.8 | | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.




(a) Acceptance Set

(b) Matriculation Set

Figure 8: Uncorrected Deviation Gains

Table 21: Deviation Gains Attribution Table

(a) Acceptance Set

| | Gain (%pt) | Sample Mean | Committee Mean | 1-Variable Optimal Mean | Joint Optimal Mean | 1-Variable Difference (SD) | Joint Difference (SD) |
|---|---|---|---|---|---|---|---|
| Age | 2.4692 | 25.9910 | 26.2840 | 29.9572 | 29.5331 | 1.1674 | 1.0326 |
| Work Experience | 2.2037 | 0.5247 | 0.7276 | 0.9844 | 0.9650 | 0.5141 | 0.4752 |
| Essay Positive Terms (%) | 1.8431 | 11.7499 | 11.6521 | 12.5376 | 12.1183 | 0.7858 | 0.4137 |
| Missing TOEFL/IELTS | 1.2872 | 0.6076 | 0.6654 | 0.0739 | 0.4086 | -1.2110 | -0.5258 |
| CV Coding | 1.1426 | 2.8001 | 2.7820 | 3.3399 | 2.9415 | 0.6244 | 0.1785 |
| Essay Economics Terms (%) | 1.0382 | 9.7228 | 9.7423 | 9.0284 | 9.5388 | -0.6411 | -0.1828 |
| Elite U.S. University | 1.0086 | 0.1267 | 0.2335 | 0.0078 | 0.0389 | -0.6784 | -0.5848 |
| Letters Terms (%) PC1 | 0.8868 | -0.0106 | 0.1374 | 0.4969 | 0.1387 | 0.5202 | 0.0018 |
| TOEFL/IELTS | 0.7541 | 7.4376 | 7.5675 | 7.7759 | 7.6827 | 0.5957 | 0.3294 |

(b) Matriculation Set

| | Gain (%pt) | Sample Mean | Committee Mean | 1-Variable Optimal Mean | Joint Optimal Mean | 1-Variable Difference (SD) | Joint Difference (SD) |
|---|---|---|---|---|---|---|---|
| Age | 2.0176 | 25.9910 | 26.3284 | 30.9159 | 29.7737 | 1.4580 | 1.0950 |
| Essay Positive Terms (%) | 1.9874 | 11.7499 | 11.6107 | 12.9136 | 12.2763 | 1.1534 | 0.5905 |
| Work Experience | 1.8659 | 0.5247 | 0.7938 | 0.9949 | 0.9799 | 0.4077 | 0.3725 |
| TOEFL/IELTS | 1.2973 | 7.4376 | 7.4939 | 7.8501 | 7.6128 | 1.0238 | 0.3397 |
| Essay Economics Terms (%) | 1.0726 | 9.7228 | 9.9994 | 9.0856 | 9.5738 | -0.8271 | -0.3821 |
| CV Coding | 1.0219 | 2.8001 | 2.6661 | 3.3395 | 2.8782 | 0.7618 | 0.2374 |

# D Two-Round Portfolio Choice

## D.1 Without Matriculation

In Section 4.1, following Bai et al. (2022), I show that the committee's one-round portfolio choice problem reduces to a cutoff rule problem. Then in Sections 4.1.1 and 4.1.2, I write out dynamic and static two-round cutoff rule models with Round 1 screening and Round 2 final selections. In this appendix, I solve a two-round portfolio choice problem by backward induction, showing that each round's solution takes the cutoff rule form in the paper. I start with Round 2:

$$\max_{(d_{2,1},\ldots,d_{2,N_T})\in\{A,R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S|x_n,z_n)\mathbb{1}(d_{2,n}=A), \text{ where: } \sum_{n=1}^{N_T}\mathbb{1}(d_{2,n}=A)\leq N_A. \tag{43}$$

Because success probabilities are never negative, the unconstrained objective increases every time $d_{2,n}=A$, which means that the constraint binds. The Lagrangian with multiplier $\lambda_2$ is:

$$\begin{aligned}\mathcal{L}(d_{2,1},\ldots,d_{2,N_T};\lambda_2) &= \sum_{n=1}^{N_T} P(y_n=S|x_n,z_n)\mathbb{1}(d_{2,n}=A)\\ &\quad+\lambda_2\left\{N_A-\left[N_T-\sum_{n=1}^{N_T}\mathbb{1}(d_{2,n}=R)\right]\right\}\\ &= \sum_{n=1}^{N_T}\left[P(y_n=S|x_n,z_n)\mathbb{1}(d_{2,n}=A)+\lambda_2\mathbb{1}(d_{2,n}=R)\right]\\ &\quad+\lambda_2(N_A-N_T).\end{aligned} \tag{44}$$

By the same logic in Section 4.1, the Round 2 solution is as follows with set-identified cutoff probability $\lambda_2^* = P(y_2^*=S)\in\left[P(y_{N_A+1}=S|x_{N_A+1},z_{N_A+1}),P(y_{N_A}=S|x_{N_A},z_{N_A})\right]$:

$$\forall n\in\{1,\ldots,N_T\},\ d_{2,n}(x_n,z_n|\lambda_2^*) = \begin{cases} A & \text{if } P(y_n=S|x_n,z_n)>\lambda_2^*\\ \{A,R\} & \text{if } P(y_n=S|x_n,z_n)=\lambda_2^*\\ R & \text{if } P(y_n=S|x_n,z_n)<\lambda_2^*.\end{cases} \tag{45}$$

When the committee solves Round 1, it must anticipate $\lambda_2^*$, which is a function of the $(\mathbf{x},\mathbf{z})$ of the entire transitioned pool. I make two assumptions so that this anticipation is tractable. First, conditional on $\mathbf{x}$, I assume that $\mathbf{z}$ is independent across applicants. Second, I assume that $N$ is large. Under these assumptions, the same Glivenko-Cantelli argument as in the proof of Theorem 1, with $\mathbf{z}$ in place of the preference shocks, makes $\lambda_2^*$ deterministic for large $N$.

Denote $\hat{G}_{N_T}(c)=\frac{1}{N_T}\sum_{n=1}^{N}\mathbb{1}\{P(y_n=S|x_n,z_n)\leq c\}\mathbb{1}\{d_{1,n}=T\}$ as the empirical distribution of the success probabilities of the $N_T$ transitioned applicants. The cutoff $\lambda_2^*$ is the $(1-N_A/N_T)$ quantile of this distribution, since the committee accepts the $N_A$ such applicants with the highest success probabilities. Because only $\mathbf{x}$ is observed in Round 1, transition depends on $x_n$ alone; the

Round 1 cutoff rule derived below transitions applicant $n$ when $x_n$ lies in transition region $\mathcal{X}_T$. For large $N$, the cutoff $\lambda_1^*$ converges to a constant by the same argument, so $\mathcal{X}_T$ is deterministic.

Since the pairs $(x_n, z_n)$ are iid and membership in $\mathcal{X}_T$ depends only on $x_n$, the transitioned success probabilities are themselves iid, with conditional distribution $G(c) = P(P(y = S|x,z) \leq c | x \in \mathcal{X}_T)$. So by the Glivenko-Cantelli theorem, $\hat{G}_{N_T}$ converges almost surely to $G$ for large $N_T$. Therefore, as $N_A/N_T \to q_2$, $\hat{G}_{N_T}$'s $(1 - N_A/N_T)$ quantile converges to $G$'s deterministic $(1-q_2)$ quantile. In turn, the set-identified interval for $\lambda_2^*$ collapses to a point, denoted $\overline{\lambda_2^*}$. As in the proof of Theorem 1, independence is essential; a common component across applicants would slide $\hat{G}_{N_T}$ in unison and leave $\lambda_2^*$ random in the limit. This convergence is a fixed point because the transition region $\mathcal{X}_T$ is pinned down by the Round 1 cutoff $\lambda_1^*$, which ranks applicants by their anticipated Round 2 payoff, which itself embeds $\overline{\lambda_2^*}$. The pair $(\lambda_1^*, \overline{\lambda_2^*})$ is jointly determined. Because $\overline{\lambda_2^*}$ is deterministic, no individual applicant's signal moves the cutoff, the committee's anticipated Round 2 payoff separates across applicants, and Round 1 reduces to a cutoff rule.

Let $EV_{2,n}(x_n) = \mathbb{E}[\max\{P(y_n = S|x_n, z_n), \overline{\lambda_2^*}\}|x_n] = \int_{\widetilde{z}_n} \max\{P(y_n = S|x_n, \widetilde{z}_n), \overline{\lambda_2^*}\} dF_{z|x}(\widetilde{z}_n|x_n)$. Then the Round 1 portfolio choice problem is:

$$\max_{(d_{1,1},\ldots,d_{1,N}) \in \{T,R\}^N} \sum_{n=1}^{N} EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T), \text{ where: } \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = T) \leq N_T. \tag{46}$$

Because success probabilities are never negative, the unconstrained objective increases every time $d_{1,n} = T$, which means that the constraint binds. The Lagrangian with multiplier $\lambda_1$ is:

$$\begin{aligned} \mathcal{L}(d_{1,1}, \ldots, d_{1,N}; \lambda_1) &= \sum_{n=1}^{N} EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \left\{ N_T - \left[ N - \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = R) \right] \right\} \\ &= \sum_{n=1}^{N} \{ EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R) \} + \lambda_1 (N_T - N). \end{aligned} \tag{47}$$

By the same logic in Section 4.1, the Round 1 solution is as follows with set-identified cutoff level $\lambda_1^* = L(y_1^* = S) \in \left[ EV_{2,N_T+1}(x_{N_T+1}), EV_{2,N_T}(x_{N_T}) \right]$:

$$\forall n \in \{1, \ldots, N\}, \, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T, R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^*. \end{cases} \tag{48}$$

There is one discrepancy between $\lambda_1^* = L(y_1^* = S)$ in this appendix vs. Section 4.1.1. Here, $\lambda_1^*$ is bounded above by 1 because for all $n$, it is the case that $\max\{P(y_n = S|x_n, \widetilde{z}_n), \overline{\lambda_2^*}\} \leq 1$ pointwise in $\widetilde{z}_n$. Therefore, $EV_{2,n}(x_n) \leq \int_{\widetilde{z}_n} dF_{z|x}(\widetilde{z}_n|x_n) = 1$. But in Section 4.1.1, the type 1 extreme value

preference shocks increase this upper bound to $1+\sigma\log 2$, where I now denote $\overline{\lambda_2^*}$ by $P(y_2^*=S)$:

$$\begin{aligned}\mathbb{E}[v_2(x_n,z_n,\varepsilon_{2,n})|x_n] &= \int_{\widetilde{z}_n} \sigma\log\left(e^{P(y_n=S|x_n,\widetilde{z_n})/\sigma}+e^{P(y_2^*=S)/\sigma}\right)dF(\widetilde{z_n}|x_n)\\ &\leq \int_{\widetilde{z}_n} \sigma\log\left(2e^{\max\{P(y_n=S|x_n,\widetilde{z_n}),P(y_2^*=S)\}/\sigma}\right)dF(\widetilde{z_n}|x_n)\\ &= \int_{\widetilde{z}_n} \max\{P(y_n=S|x_n,\widetilde{z_n}),P(y_2^*=S)\}dF(\widetilde{z_n}|x_n)+\sigma\log 2\\ &\leq 1+\sigma\log 2.\end{aligned} \tag{49}$$

In the static version of the two-round portfolio choice model, the Round 1 problem is:

$$\max_{(d_{1,1},\ldots,d_{1,N})\in\{T,R\}^N}\sum_{n=1}^N P(y_n=S|x_n)\mathbb{1}(d_{1,n}=T),\text{ where: }\sum_{n=1}^N\mathbb{1}(d_{1,n}=T)\leq N_T. \tag{50}$$

Because success probabilities are never negative, the unconstrained objective increases every time $d_{1,n}=T$, which means that the constraint binds. The Lagrangian with multiplier $\lambda_1$ is:

$$\begin{aligned}\mathcal{L}(d_{1,1},\ldots,d_{1,N};\lambda_1) &= \sum_{n=1}^N P(y_n=S|x_n)\mathbb{1}(d_{1,n}=T)+\lambda_1\left\{N_T-\left[N-\sum_{n=1}^N\mathbb{1}(d_{1,n}=R)\right]\right\}\\ &= \sum_{n=1}^N\left[P(y_n=S|x_n)\mathbb{1}(d_{1,n}=T)+\lambda_1\mathbb{1}(d_{1,n}=R)\right]+\lambda_1(N_T-N).\end{aligned} \tag{51}$$

By the same logic in Section 4.1, the solution is as follows with set-identified cutoff probability $\lambda_1^*=P(y_1^*=S)\in\left[P(y_{N_T+1}=S|x_{N_T+1}),P(y_{N_T}=S|x_{N_T})\right]$:

$$\forall n\in\{1,\ldots,N\},\ d_{1,n}(x_n|\lambda_1^*)=\begin{cases}T & \text{if } P(y_n=S|x_n)>\lambda_1^*\\ \{T,R\} & \text{if } P(y_n=S|x_n)=\lambda_1^*\\ R & \text{if } P(y_n=S|x_n)<\lambda_1^*.\end{cases} \tag{52}$$

## D.2 With Matriculation

In Section 4.2, I show that because the committee's capacity constraint is on the number of matriculations rather than acceptances, a one-round portfolio choice problem with matriculation reduces to a cutoff rule problem on success probabilities. However, a two-round portfolio choice problem with matriculation is less straightforward because matriculation ends up mattering in Round 1. In this appendix, I present such a model with Round 1 screening and Round 2 final selections. For readability, I keep finite-sum notation rather than the integrals in the proof of Theorem 3 in Appendix E. But I invoke that theorem's large-$N$ framework throughout because it removes the finite-$N$ edge cases and, as in Appendix D.1, collapses $\lambda_2^*$ to the deterministic $\overline{\lambda_2^*}$.

Working backward, I start with Round 2, where for brevity, I denote $w = (x, z)$:

$$\begin{aligned} &\max_{(d_{2,1},\ldots,d_{2,N_T})\in\{A,R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S|w_n)P(m_n = M|w_n)\mathbb{1}(d_{2,n} = A), \\ &\text{where: } \sum_{n=1}^{N_T} P(m_n = M|w_n)\mathbb{1}(d_{2,n} = A) \leq N_M. \end{aligned} \tag{53}$$

Because success and matriculation probabilities are never negative, the unconstrained objective increases every time $d_{2,n} = A$, which means that the constraint binds. The Lagrangian is:

$$\begin{aligned} \mathcal{L}(d_{2,1},\ldots,d_{2,N_T};\lambda_2) &= \sum_{n=1}^{N_T} P(y_n = S|w_n)P(m_n = M|w_n)\mathbb{1}(d_{2,n} = A) \\ &\quad + \lambda_2\left\{N_M - \left[\overline{N_M} - \sum_{n=1}^{N_T} P(m_n = M|w_n)\mathbb{1}(d_{2,n} = R)\right]\right\} \\ &= \sum_{n=1}^{N_T} \left[P(y_n = S|w_n)\mathbb{1}(d_{2,n} = A) + \lambda_2\mathbb{1}(d_{2,n} = R)\right]P(m_n = M|w_n) \\ &\quad + \lambda_2(N_M - \overline{N_M}), \end{aligned} \tag{54}$$

where $\overline{N_M} = \sum_{n=1}^{N_T} P(m_n = M|w_n)$ is the expected number of matriculants if everyone is accepted.

Edge cases aside, the Round 2 solution, with cutoff probability $\lambda_2^* = P(y_2^* = S)$, is:

$$\forall n \in \{1,\ldots,N_T\},\ d_{2,n}(w_n|\lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S|w_n) > \lambda_2^* \\ \{A,R\} & \text{if } P(y_n = S|w_n) = \lambda_2^* \\ R & \text{if } P(y_n = S|w_n) < \lambda_2^*. \end{cases} \tag{55}$$

Let $EV_{2,n}(x_n) = \int_{\widetilde{z}_n} \max\{P(y_n = S|x_n,\widetilde{z}_n) - \overline{\lambda_2^*}, 0\}P(m_n = M|x_n,\widetilde{z}_n)dF_{z|x}(\widetilde{z}_n|x_n)$. This netted form, and not $\int_{\widetilde{z}_n} \max\{P(y_n = S|x_n,\widetilde{z}_n), \overline{\lambda_2^*}\}P(m_n = M|x_n,\widetilde{z}_n)dF_{z|x}(\widetilde{z}_n|x_n)$, is the correct transition payoff. To see why, evaluate the Round 2 Lagrangian at the cutoff rule solution with deterministic cutoff $\overline{\lambda_2^*}$, where $P(y_n = S|w_n)\mathbb{1}(d_{2,n} = A) + \lambda_2\mathbb{1}(d_{2,n} = R)$ becomes $\max\{P(y_n = S|w_n), \overline{\lambda_2^*}\}$:

$$\mathcal{L}(d_2^*;\overline{\lambda_2^*}) = \sum_{n=1}^{N_T} \max\{P(y_n = S|w_n), \overline{\lambda_2^*}\}P(m_n = M|w_n) + \overline{\lambda_2^*}(N_M - \overline{N_M}). \tag{56}$$

By $\max\{P(y_n = S|w_n), \overline{\lambda_2^*}\} = \max\{P(y_n = S|w_n) - \overline{\lambda_2^*}, 0\} + \overline{\lambda_2^*}$, and $\overline{N_M} = \sum_{n=1}^{N_T} P(m_n = M|w_n)$:

$$\begin{aligned} \mathcal{L}(d_2^*;\overline{\lambda_2^*}) &= \sum_{n=1}^{N_T} \max\{P(y_n = S|w_n) - \overline{\lambda_2^*}, 0\}P(m_n = M|w_n) + \overline{\lambda_2^*}\,\overline{N_M} + \overline{\lambda_2^*}(N_M - \overline{N_M}) \\ &= \sum_{n=1}^{N_T} \max\{P(y_n = S|w_n) - \overline{\lambda_2^*}, 0\}P(m_n = M|w_n) + \overline{\lambda_2^*}N_M. \end{aligned} \tag{57}$$

Although $\overline{N_M}$ varies with the Round 1 transition decisions, the $\overline{\lambda_2^*}\overline{N_M}$ terms cancel, leaving the constant $\overline{\lambda_2^*}N_M$. The Round 1 transition payoff is therefore the per-applicant netted term $\max\{P(y_n = S|w_n) - \overline{\lambda_2^*}, 0\}P(m_n = M|w_n)$, whose expectation over $\widetilde{z}_n$ defines $EV_{2,n}$.

The netting distinguishes the matriculation case from the case without it. Without matriculation, the netted and non-netted forms differ by $\int_{\widetilde{z}_n} \overline{\lambda_2^*} dF_{z|x}(\widetilde{z}_n|x_n) = \overline{\lambda_2^*}$, a constant across applicants that leaves the Round 1 ordering unchanged. With matriculation, they differ by $\int_{\widetilde{z}_n} \overline{\lambda_2^*} P(m_n = M|x_n, \widetilde{z}_n) dF_{z|x}(\widetilde{z}_n|x_n)$, which varies with $x_n$. The non-netted form would therefore order applicants differently, and incorrectly. The Round 1 portfolio choice problem is:

$$\max_{(d_{1,1},\ldots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} EV_{2,n}(x_n)\mathbb{1}(d_{1,n} = T), \text{ where: } \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = T) \leq N_T. \tag{58}$$

Because success and matriculation probabilities are never negative, the unconstrained objective increases every time $d_{1,n} = T$, which means that the constraint binds. The Lagrangian is:

$$\begin{aligned} \mathcal{L}(d_{1,1},\ldots,d_{1,N};\lambda_1) &= \sum_{n=1}^{N} EV_{2,n}(x_n)\mathbb{1}(d_{1,n} = T) + \lambda_1 \left\{ N_T - \left[ N - \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = R) \right] \right\} \\ &= \sum_{n=1}^{N} [EV_{2,n}(x_n)\mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1(N_T - N). \end{aligned} \tag{59}$$

Edge cases aside, the Round 1 solution with cutoff level $\lambda_1^* = L(y_1^* = S, m_1^* = M)$ is:

$$\forall n \in \{1,\ldots,N\},\ d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T,R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^*. \end{cases} \tag{60}$$

The Round 1 transition payoff with Type 1 extreme value shocks is therefore:

$$\mathbb{E}[v_2(x_n, z_n, \varepsilon_{2,n})|x_n] = \int_{\widetilde{z}_n} \sigma \log\left( e^{[P(y_n=S|x_n,\widetilde{z}_n) - P(y_2^*=S)]/\sigma} + 1 \right) P(m_n = M|x_n, \widetilde{z}_n) dF(\widetilde{z}_n|x_n). \tag{61}$$

The committee prefers to transition likely matriculants so that it can be more selective in Round 2. As a simple example, suppose that $N = 4$, $N_T = 2$, and $N_M = 1$, and suppose that recommendation letters matter more for success than GRE scores. Suppose also that Applicant 1 has a high GRE and strong recommendation letter, Applicant 2 has a high GRE and weak letter, Applicant 3 has a low GRE and strong letter, and Applicant 4 has a low GRE and weak letter. If the committee ignores matriculation probabilities in Round 1, it will transition Applicants 1 and 2 and then accept Applicant 1. But suppose that because Applicant 1 is so strong, they choose a higher-ranked university. Then the committee has to settle for Applicant 2. Whereas if the committee had transitioned Applicants 3 and 4, it could have accepted and landed Applicant 3.

In the static version of the model with matriculation, the Round 1 problem, with $\alpha \in [0,1]$, is:

$$\begin{aligned} \max_{(d_{1,1},\ldots,d_{1,N})\in\{T,R\}^N} & \sum_{n=1}^{N} P(y_n = S|x_n)^{\alpha} P(m_n = M|x_n)^{1-\alpha} \mathbb{1}(d_{1,n} = T), \\ \text{where: } & \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = T) \leq N_T. \end{aligned} \tag{62}$$

Because success and matriculation probabilities are never negative, the unconstrained objective increases every time $d_{1,n} = T$, which means that the constraint binds. The Lagrangian is:

$$\begin{aligned} \mathcal{L}(d_{1,1},\ldots,d_{1,N};\lambda_1) &= \sum_{n=1}^{N} P(y_n = S|x_n)^{\alpha} P(m_n = M|x_n)^{1-\alpha} \mathbb{1}(d_{1,n} = T) \\ &\quad + \lambda_1 \left\{ N_T - \left[ N - \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = R) \right] \right\} \\ &= \sum_{n=1}^{N} \left[ P(y_n = S|x_n)^{\alpha} P(m_n = M|x_n)^{1-\alpha} \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R) \right] \\ &\quad + \lambda_1 (N_T - N). \end{aligned} \tag{63}$$

Edge cases aside, the solution is as follows with cutoff probability $\lambda_1^* = P(y_1^* = S, m_1^* = M)$:

$$\forall n \in \{1,\ldots,N\},\ d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } P(y_n = S|x_n)^{\alpha} P(m_n = M|x_n)^{1-\alpha} > \lambda_1^* \\ \{T,R\} & \text{if } P(y_n = S|x_n)^{\alpha} P(m_n = M|x_n)^{1-\alpha} = \lambda_1^* \\ R & \text{if } P(y_n = S|x_n)^{\alpha} P(m_n = M|x_n)^{1-\alpha} < \lambda_1^*. \end{cases} \tag{64}$$

But per the dynamic $EV_{2,n}$ integrand without shocks, $\max\{P(y_n = S|w_n) - \overline{\lambda_2^*}, 0\} P(m_n = M|w_n)$, matriculation probabilities matter less than success probabilities. For applicants with $P(y_n = S|w_n) \leq \overline{\lambda_2^*}$, matriculation has a weight of zero. Even for the other applicants, the weight is only large when $P(y_n = S|w_n) \gg \overline{\lambda_2^*}$. Thus, the static model without matriculation approximates the matriculation-dynamic model well enough empirically that $\alpha$ can be set to 1. Doing so also aligns with committee members indicating that they do not consider matriculation in Round 1.

# E Proofs of Theorems

## E.1 Structural Analysis

**Theorem 1.** *For large N, the committee's portfolio choice problem with preference shocks has a cutoff rule solution with logistic conditional choice probabilities.*

**Proof.** The portfolio choice problem with type 1 extreme value preference shocks is:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} \left[(P(y_n = S|x_n,z_n) + \varepsilon_n(A))\mathbb{1}(d_n = A) + \varepsilon_n(R)\mathbb{1}(d_n = R)\right], \\ \text{where } \sum_{n=1}^{N} \mathbb{1}(d_n = A) \leq N_A. \tag{65}$$

Because success probabilities are never negative, the unconstrained objective increases in expectation every time $d_n = A$, and so the constraint binds. The Lagrangian with multiplier $\lambda$ is:

$$\begin{aligned}\mathcal{L}(d_1,\ldots,d_N;\lambda) &= \sum_{n=1}^{N} \left[(P(y_n = S|x_n,z_n) + \varepsilon_n(A))\mathbb{1}(d_n = A) + \varepsilon_n(R)\mathbb{1}(d_n = R)\right] \\ &\quad + \lambda\left\{N_A - \left[N - \sum_{n=1}^{N} \mathbb{1}(d_n = R)\right]\right\} \\ &= \sum_{n=1}^{N} \left[(P(y_n = S|x_n,z_n) + \varepsilon_n(A))\mathbb{1}(d_n = A) + (\lambda + \varepsilon_n(R))\mathbb{1}(d_n = R)\right] \\ &\quad + \lambda(N_A - N).\end{aligned} \tag{66}$$

Suppose without loss of generality that the applicants are sorted in decreasing order of their shock-perturbed success probabilities such that $P(y_1 = S|x_1,z_1) + \varepsilon_1(A) - \varepsilon_1(R) \geq \cdots \geq P(y_N = S|x_N,z_N) + \varepsilon_N(A) - \varepsilon_N(R)$. To ensure that the committee accepts exactly $N_A$ applicants, it must be the case that $\lambda^* = \lambda^*(\boldsymbol{\varepsilon}(A),\boldsymbol{\varepsilon}(R)) \in \left[P(y_{N_A+1} = S|x_{N_A+1},z_{N_A+1}) + \varepsilon_{N_A+1}(A) - \varepsilon_{N_A+1}(R), P(y_{N_A} = S|x_{N_A},z_{N_A}) + \varepsilon_{N_A}(A) - \varepsilon_{N_A}(R)\right]$. That is, in this setup, $\lambda^*$ is set identified. Then the solution is to accept all applicants whose shock-perturbed success probability exceeds $\lambda^*$ and reject the rest:

$$\forall n \in \{1,\ldots,N\},\ d_n(x_n,z_n,\varepsilon_n|\lambda^*) = \begin{cases} A & \text{if } P(y_n = S|x_n,z_n) + \varepsilon_n(A) > \lambda^* + \varepsilon_n(R) \\ \{A,R\} & \text{if } P(y_n = S|x_n,z_n) + \varepsilon_n(A) = \lambda^* + \varepsilon_n(R) \\ R & \text{if } P(y_n = S|x_n,z_n) + \varepsilon_n(A) < \lambda^* + \varepsilon_n(R). \end{cases} \tag{67}$$

This cutoff rule holds conditional on the realized shocks. Given the shocks, the committee computes $\lambda^*$ and applies the cutoff, so there is no difficulty for the committee. But there is a complication regarding estimation, namely that unlike the committee, the econometrician does not observe the shocks. Therefore, the conditional choice probabilities used in estimation require

taking expectations across each applicant's shocks, which would be fine except $\lambda^*$ is a function of each applicant's shocks simultaneously. Suppose for now that $\lambda^*$ is instead deterministic. Then:

$$\begin{aligned} P(d_n = A|x_n, z_n) &= P(P(y_n = S|x_n, z_n) + \varepsilon_n(A) \geq \lambda^* + \varepsilon_n(R)) \\ &= P(\varepsilon_n(R) - \varepsilon_n(A) \leq P(y_n = S|x_n, z_n) - \lambda^*) \\ &= \frac{\exp(P(y_n = S|x_n, z_n)/\sigma)}{\exp(P(y_n = S|x_n, z_n)/\sigma) + \exp(\lambda^*/\sigma)}, \end{aligned} \tag{68}$$

and the likelihood becomes the product of individual binary logistics. The third equality holds because the difference of two independent T1EV random variables has a logistic distribution.

That said, $\lambda^*$ is not deterministic for finite $N$, so the logistic conditional choice probabilities do not hold exactly. But for large $N$, $\lambda^*$ converges to a constant. Denote $\hat{F}_N(c) = \frac{1}{N}\sum_{n=1}^{N} \mathbb{1}\{P(y_n = S|x_n, z_n) + \varepsilon_n(A) - \varepsilon_n(R) \leq c\}$ as the empirical distribution of the shock-perturbed success probabilities. The cutoff $\lambda^*$ is the $(1 - N_A/N)$ quantile of this distribution. Because the shocks are iid, the perturbed indices in $\hat{F}_N$ are iid. So by the Glivenko-Cantelli theorem, $\hat{F}_N$ converges almost surely to the population distribution $F$ for large $N$. Therefore, as $N_A/N \to q$, $\hat{F}_N$'s $(1 - N_A/N)$ quantile converges to $F$'s deterministic $(1-q)$ quantile. The iid assumption is essential, and not only because each applicant's shock individually moves the cutoff by $O(1/N)$. A common perturbation component across applicants would have the same vanishing per-applicant influence, but it would slide the entire empirical distribution in unison, leaving $\lambda^*$ random in the limit. Independence makes the simultaneous fluctuations of the $N$ shocks cancel, so $\hat{F}_N$ and $\lambda^*$ converge. With $\lambda^*$ deterministic, no individual applicant's shock affects $\lambda^*$, the expectation defining each conditional choice probability factors across applicants, and the above logistic form holds. □

**Theorem 2.** *For simplicity, suppose there is one round. The maximum likelihood estimate $\hat{P}_{y^*} = \hat{P}(y^* = S|t)$ is such that the number of acceptances $N_A$ equals the expected $N_A$. Also, for any $\hat{P}_{y^*}$ and $N_A$, $\mathcal{L}^R$ is highest when the committee accepts the $N_A$ highest $P_{y_n} = P(y_n = S|x_n, z_n, t_n, \overline{k}_n)$.*

**Proof.** Supposing one round, the restricted likelihood is:

$$\begin{aligned} \mathcal{L}^R = \prod_{n=1}^{N} &\left[\frac{e^{P_{y_n}/\sigma}}{e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma}}\right]^{\mathbb{1}(d_n=A)} \left[\frac{e^{P_{y^*}/\sigma}}{e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma}}\right]^{\mathbb{1}(d_n=R)} \\ &\left[\frac{e^{P_{y_n}}}{1 + e^{P_{y_n}}}\right]^{\mathbb{1}(y_n=S)} \left[\frac{1}{1 + e^{P_{y_n}}}\right]^{\mathbb{1}(y_n=F)} \end{aligned} \tag{69}$$

When an applicant is accepted, the resulting term in $\mathcal{L}^R$ decreases in $P_{y^*}$, providing downward pressure on $\hat{P}_{y^*}$. Also, when one is rejected, the term increases in $P_{y^*}$, providing upward pressure.

Since $\mathbb{1}(d_n = A) + \mathbb{1}(d_n = R) = \mathbb{1}(y_n = S) + \mathbb{1}(y_n = F) = 1$, the restricted log-likelihood is:

$$\begin{aligned}\ell^R &= \sum_{n=1}^{N} \left[ \mathbb{1}(d_n = A) \log\left( \frac{e^{P_{y_n}/\sigma}}{e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma}} \right) + \mathbb{1}(d_n = R) \log\left( \frac{e^{P_{y^*}/\sigma}}{e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma}} \right) \right. \\ &\quad \left. + \mathbb{1}(y_n = S) \log\left( \frac{e^{P_{y_n}}}{1 + e^{P_{y_n}}} \right) + \mathbb{1}(y_n = F) \log\left( \frac{1}{1 + e^{P_{y_n}}} \right) \right] \\ &= \sum_{n=1}^{N} \left[ \frac{1}{\sigma} \left( \mathbb{1}(d_n = A) P_{y_n} + \mathbb{1}(d_n = R) P_{y^*} \right) - \log\left( e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma} \right) \right. \\ &\quad \left. + \mathbb{1}(y_n = S) P_{y_n} - \log\left( 1 + e^{P_{y_n}} \right) \right].\end{aligned} \tag{70}$$

Since $\ell^R$ is strictly concave in $P_{y^*}$, I find the unique maximizer $\hat{P}_{y^*}$ by the first-order condition:

$$\frac{1}{\sigma} \left[ N - N_A - \sum_{n=1}^{N} \frac{e^{\hat{P}_{y^*}/\sigma}}{e^{P_{y_n}/\sigma} + e^{\hat{P}_{y^*}/\sigma}} \right] = 0 \quad \Rightarrow \quad \sum_{n=1}^{N} \frac{e^{P_{y_n}/\sigma}}{e^{P_{y_n}/\sigma} + e^{\hat{P}_{y^*}/\sigma}} = N_A. \tag{71}$$

The left side of the latter equation is each applicant's acceptance probability summed over the number of applicants; that is, expected $N_A$. Therefore, $\hat{P}_{y^*}$ is such that the expected $N_A$ equals $N_A$.

Given that $\hat{P}_{y^*}$ is identified, I will show that $\ell^R$, and therefore $\mathcal{L}^R$, is highest when the committee accepts the $N_A$ applicants with the highest $P_{y_n}$. The only part of $\ell^R$ that depends on $d_n$ is $\sum_{n=1}^{N} [\mathbb{1}(d_n = A) P_{y_n} + \mathbb{1}(d_n = R) P_{y^*}] = N P_{y^*} + \sum_{n=1}^{N} \mathbb{1}(d_n = A)(P_{y_n} - P_{y^*})$; this equality holds since $\mathbb{1}(d_n = A) + \mathbb{1}(d_n = R) = 1$. As such, the only part depending on $d_n$ is $\sum_{n=1}^{N} \mathbb{1}(d_n = A)(P_{y_n} - P_{y^*})$.

This expression is highest when $d_n = A$ for the $N_A$ highest values of $P_{y_n}$, so the likelihood is highest when the committee accepts the $N_A$ highest $P_{y_n}$. Intuitively, the committee's selectivity identifies $\hat{P}_{y^*}$, and it is not necessary for identification for the committee to accept applicants with higher $P_{y_n}$. But if the committee engages in positive sorting, the model's fit will improve. □

**Theorem 3.** *For large N, the committee's portfolio choice problem with matriculation probabilities has a cutoff rule solution on success probabilities.*

**Proof.** For brevity, let $w = (x, z)$. Then for large $N$, the portfolio choice problem is:

$$\begin{aligned}&\max_{\delta \in \{\delta | \delta : \mathbb{R}^K \to [0,1]\}} \int_{w \in \mathbb{R}^K} P(y = S|w) P(m = M|w) \delta(w) f(w) dw, \\ &\text{where: } \int_{w \in \mathbb{R}^K} P(m = M|w) \delta(w) f(w) dw \leq N_M,\end{aligned} \tag{72}$$

where $\delta$ is the committee's decision rule, and $f$ is the density of $w$. Because success and matriculation probabilities, as well as densities, are never negative, the unconstrained objective increases

in $\delta$, which means that the constraint binds. The Lagrangian with multiplier $\lambda$ is:

$$\begin{aligned}\mathcal{L}(\delta;\lambda) &= \int_{w\in\mathbb{R}^K} P(y=S|w)P(m=M|w)\delta(w)f(w)dw \\ &\quad + \lambda\left\{N_M - \left[\int_{w\in\mathbb{R}^K} P(m=M|w)f(w)dw - \int_{w\in\mathbb{R}^K} P(m=M|w)(1-\delta(w))f(w)dw\right]\right\} \\ &= \int_{w\in\mathbb{R}^K} \big[P(y=S|w)\delta(w) + \lambda(1-\delta(w))\big]P(m=M|w)f(w)dw \\ &\quad + \lambda\left[N_M - \int_{w\in\mathbb{R}^K} P(m=M|w)f(w)dw\right]. \end{aligned} \tag{73}$$

In the discrete case, there is not necessarily a cutoff rule solution. But in the continuous case, the committee can accept the necessary fraction of applicants such that $\int_{w\in\mathbb{R}^K} P(m=M|w)\delta(w|\lambda^*)f(w)dw = N_M$, which results in a cutoff rule solution on success probabilities alone:

$$\forall w\in\mathbb{R}^K,\ \delta(w|\lambda^*) = \begin{cases} 1 & \text{if } P(y=S|w) > \lambda^* \\ [0,1] & \text{if } P(y=S|w) = \lambda^* \\ 0 & \text{if } P(y=S|w) < \lambda^*. \end{cases} \tag{74}$$

The committee accepts all applicants with success probabilities $P(y=S|w)$ strictly greater than the cutoff probability $\lambda^*$, rejects all applicants with $P(y=S|w)$ strictly less than $\lambda^*$, and may accept or reject any applicants with $P(y=S|w)$ equal to $\lambda^*$, if such a mass of applicants exists. All that remains is to identify $\lambda^*$. Like Bai et al. (2022), I define the following two functions:

$$\begin{aligned} M^H(\lambda) &= \int_{\{w\in\mathbb{R}^K | P(y=S|w)\geq\lambda\}} P(m=M|w)f(w)dw \\ M^L(\lambda) &= \int_{\{w\in\mathbb{R}^K | P(y=S|w)>\lambda\}} P(m=M|w)f(w)dw. \end{aligned} \tag{75}$$

The upper bound function $M^H$ is the fraction of applicants, scaled by their matriculation probabilities, whose success probabilities are greater than or equal to some $\lambda$, and the lower bound function $M^L$ is the fraction whose success probabilities are strictly greater than $\lambda$. Both functions are between 0 and 1, both are decreasing in $\lambda$, and for any $\lambda$, it must be that $M^H(\lambda) \geq M^L(\lambda)$.

Then it is the case that $\lambda^* = \inf_\lambda\{\lambda | M^L(\lambda) \leq N_M\}$. Intuitively, $\lambda^*$ is the smallest cutoff probability where the fraction of matriculation-probability-scaled applicants with $P(y=S|w) > \lambda$ does not exceed the maximum number of matriculants $N_M$. If there is a positive mass of matriculation-probability-scaled applicants with $P(y=S|w) = \lambda^*$ (that is, if $M^H(\lambda^*) - M^L(\lambda^*) > 0$), then the committee will randomly accept the exact fraction $\alpha^* \in [0,1]$ of that mass of applicants such that $M^L(\lambda^*) + \alpha^*[M^H(\lambda^*) - M^L(\lambda^*)] = N_M$. That is, the committee will accept every applicant with $P(y=S|w) > \lambda^*$ and accept just enough applicants with $P(y=S|w) = \lambda^*$ to fill the cohort. □

## E.2 Optimality and Gains

**Theorem 4.** *For two sets that are not chosen using $\hat{\theta}_S^U$, as $B \to \infty$, the fraction of perturbations in which the model's set does no better than the committee's, $g(\tilde{\theta}_{S,b}^U) \leq 0$, converges to the p-value of the test of $H_0\colon g_0 = 0$ vs. $H_1\colon g_0 > 0$ that uses the observed gain $\hat{g}$ as the test statistic.*

**Proof.** Let $g$ be the deviation gain defined in Section 5, with both the model's set and the committee's set held fixed. Since the two sets are determined before performing the test, the proof works the same way for matriculation sets as for acceptance sets. Suppose the maximum likelihood estimate $\hat{\theta}_S^U \overset{a}{\sim} \mathcal{N}(\theta_S^0, \Sigma_0)$, and each perturbation draw $\tilde{\theta}_{S,b}^U \sim \mathcal{N}(\hat{\theta}_S^U, \hat{\Sigma})$, where $\hat{\Sigma}$ is consistent for $\Sigma_0$. Because $g$ is a smooth function of $\theta_S$, it is true by the delta method that $\hat{g} = g(\hat{\theta}_S^U) \overset{a}{\sim} \mathcal{N}(g_0, \sigma_g^2)$ and $\tilde{g} = g(\tilde{\theta}_{S,b}^U) \mid \hat{g} \overset{a}{\sim} \mathcal{N}(\hat{g}, \sigma_g^2)$, where $g_0 = g(\theta_S^0)$ and $\sigma_g^2 = \nabla g(\theta_S^0)' \Sigma_0 \nabla g(\theta_S^0)$.

The test statistic is the observed gain $\hat{g}$, and its p-value is the probability, if the null is true, of an estimated gain at least as large as the observed gain. Denote $\hat{g}' = \sigma_g Z$ as a hypothetical estimated gain generated under $g_0 = 0$, where $Z \sim \mathcal{N}(0,1)$. Then, using $P(Z \geq c) = \Phi(-c)$:

$$P(\hat{g}' \geq \hat{g} \mid g_0 = 0) = P(Z \geq \hat{g}/\sigma_g) = \Phi(-\hat{g}/\sigma_g) = 1 - \Phi(\hat{g}/\sigma_g). \tag{76}$$

As the number of perturbations $B$ becomes large, it is true by the law of large numbers that the fraction of perturbations with $\tilde{g} \leq 0$ converges to the probability that a single perturbation is weakly negative. As such, for large $B$, this fraction, which I denote as $\hat{p}$, is the p-value:

$$\hat{p} = P(\tilde{g} \leq 0 \mid \hat{g}) = \Phi\left(\frac{0 - \hat{g}}{\sigma_g}\right) = \Phi(-\hat{g}/\sigma_g) = 1 - \Phi(\hat{g}/\sigma_g). \tag{77}$$

Finally, although the null hypothesis is $g_0 = 0$, the test is also valid, and conservative, if $g_0 < 0$, which can occur because a fixed model set need not be better than the committee's set at the true parameters. That is, $P(\hat{p} \leq \alpha) \leq \alpha$ whenever $g_0 \leq 0$. Because inverses of strictly increasing functions preserve inequalities, $\Phi$ is strictly increasing, and $\hat{p} = \Phi(-\hat{g}/\sigma_g)$; it is the case that $\hat{p} \leq \alpha$ holds exactly when $\hat{g} \geq -\sigma_g \Phi^{-1}(\alpha)$. Substituting $\hat{g} = g_0 + \sigma_g Z$:

$$P(\hat{p} \leq \alpha) = P(Z \geq -\Phi^{-1}(\alpha) - g_0/\sigma_g) = \Phi(\Phi^{-1}(\alpha) + g_0/\sigma_g). \tag{78}$$

For $g_0 = 0$, it is the case that $P(\hat{p} \leq \alpha) = \Phi(\Phi^{-1}(\alpha)) = \alpha$. For $g_0 < 0$, because $\Phi$ is strictly increasing, it is the case that $P(\hat{p} \leq \alpha) = \Phi(\Phi^{-1}(\alpha) + g_0/\sigma_g) < \alpha$. Therefore, rejecting $H_0$ when $\hat{p} \leq \alpha$ produces Type 1 errors, i.e. false rejections, with probability at most $\alpha$. □

**Theorem 5.** Under a linear probability model, it is the case that:

$$\begin{aligned} \mathbb{E}\left[\Delta(\hat{\theta}_S^U; \theta_S^0)\right] &= \mathbb{E}\left[\overline{P}^{\max}(\beta_0 + \eta)\right] - \overline{P}^{\max}(\beta_0) \qquad \text{and} \\ \mathbb{E}\left[\overline{\Delta}_B\right] &= \mathbb{E}\left[\overline{P}^{\max}(\beta_0 + \sqrt{2}\eta)\right] - \mathbb{E}\left[\overline{P}^{\max}(\beta_0 + \eta)\right], \end{aligned} \tag{79}$$

where the first expectation is over the sampling error and the second over the sampling error and perturbation draws. The winner's curse and its estimate are successive increments of $\mathbb{E}[\overline{P}^{\max}(\beta_0 +$

$\sqrt{\tau}\eta)]$, from $\tau = 0$ to 1 and from 1 to 2. The correction is exact when that function is linear in $\tau$ on $[0,2]$, too small when it is concave in $\tau$ there, and too large when it is convex in $\tau$ there.

**Proof.** For the first identity, at the estimates, the model's set attains $\overline{P}^{\max}(\hat{\beta}^U) = \overline{P}^{\max}(\beta_0+\eta)$. The optimal set $\mathcal{A}^R(\theta_S^0)$, which is fixed, attains $\overline{w}'_{\mathcal{A}^R(\theta_S^0)}(\beta_0+\eta)$, with expectation $\overline{P}^{\max}(\beta_0)$. For the second identity, the set chosen at the perturbation and evaluated there attains $\overline{P}^{\max}(\tilde{\beta}^U) = \overline{P}^{\max}(\beta_0+\eta+(\tilde{\beta}^U-\hat{\beta}^U))$. $\eta$ and $\tilde{\beta}^U-\hat{\beta}^U$ are independent, each normal with mean zero and the same covariance, so their sum is normal with mean zero and twice that covariance, which is also the distribution of $\sqrt{2}\eta$. Therefore, $\mathbb{E}[\overline{P}^{\max}(\tilde{\beta}^U)] = \mathbb{E}[\overline{P}^{\max}(\beta_0+\sqrt{2}\eta)]$. The set chosen at $\hat{\theta}_S^U$ is a function of $\eta$ alone, so at the perturbation, it attains $\overline{w}'_{\mathcal{A}^R(\hat{\theta}_S^U)}\tilde{\beta}^U$. This quantity is $\overline{P}^{\max}(\beta_0+\eta)$ plus $\overline{w}'_{\mathcal{A}^R(\hat{\theta}_S^U)}(\tilde{\beta}^U-\hat{\beta}^U)$, a term with expectation zero given $\eta$. Intuitively, the winner's curse is how much the best-looking average success probability improves when one $\eta$ of noise is added to the truth, and the estimated correction is how much more it improves when a second, independent $\eta$ of noise is added on top. The correction is exact when the second increment equals the first, too small when it is smaller (i.e. concave in $\tau$), and too large when it is larger (i.e. convex in $\tau$). □

**Corollary 1.** If $\beta_0 = 0$, so that every applicant has the same true success probability, then:

$$\mathbb{E}\big[\hat{g}\big] = \mathbb{E}\big[\overline{P}^{\max}(\eta)\big] \qquad \text{and} \qquad \mathbb{E}\big[\Delta(\tilde{\theta}_S^U;\hat{\theta}_S^U)\big] = (\sqrt{2}-1)\,\mathbb{E}\big[\overline{P}^{\max}(\eta)\big]. \tag{80}$$

Here, the estimated correction has expectation $(\sqrt{2}-1) \approx 0.414$ times the winner's curse. This is the worst-case scenario, meaning that in every data-generating process,

$$\mathbb{E}[\overline{\Delta}_B] \geq (\sqrt{2}-1)\,\mathbb{E}[\Delta(\hat{\theta}_S^U;\theta_S^0)].$$

**Proof.** With $\beta_0 = 0$, it is the case that $\overline{P}^{\max}(\beta_0+\sqrt{\tau}\eta) = \sqrt{\tau}\overline{P}^{\max}(\eta)$ because $\overline{P}^{\max}$ is homogeneous of degree one. Therefore, the two increments of Theorem 5 are $\mathbb{E}[\overline{P}^{\max}(\eta)]$ and $(\sqrt{2}-1)\mathbb{E}[\overline{P}^{\max}(\eta)]$. In addition, the committee's set is fixed, so the observed gain $\hat{g} = \overline{P}^{\max}(\eta) - \overline{w}'_{\mathcal{A}^U}\eta$ has expectation $\mathbb{E}[\overline{P}^{\max}(\eta)]$, which equals the winner's curse when $\beta_0 = 0$. As such, the estimated correction in this scenario has expectation $(\sqrt{2}-1) \approx 0.414$ times the winner's curse.

For the claim that this is the worst-case scenario, fix $\eta$ and view $\overline{P}^{\max}(\beta_0+\sqrt{\tau}\eta) = \max_{\mathcal{A}}[\overline{w}'_{\mathcal{A}}\beta_0 + \sqrt{\tau}\overline{w}'_{\mathcal{A}}\eta]$ as a function of $\sqrt{\tau}$. Each term inside the maximum is linear in $\sqrt{\tau}$, a maximum of linear functions is convex, and averaging over $\eta$ preserves convexity. Therefore, $\mathbb{E}[\overline{P}^{\max}(\beta_0+\sqrt{\tau}\eta)]$ is convex in $\sqrt{\tau}$. A convex function lies below a straight line through any two of its points. Also, the values of $\mathbb{E}[\overline{P}^{\max}(\beta_0+\sqrt{\tau}\eta)]$ at $\sqrt{\tau} = \{0,1,\sqrt{2}\}$ are $\overline{P}^{\max}(\beta_0)$, $\mathbb{E}[\overline{P}^{\max}(\beta_0+\eta)]$, and $\mathbb{E}[\overline{P}^{\max}(\beta_0+\sqrt{2}\eta)]$. Finally, $1 = (1-1/\sqrt{2})\cdot 0 + (1/\sqrt{2})\cdot\sqrt{2}$ lies between 0 and $\sqrt{2}$ with weights $1-1/\sqrt{2}$ and $1/\sqrt{2}$, which yields the inequality:

$$\mathbb{E}\big[\overline{P}^{\max}(\beta_0+\eta)\big] \leq (1-1/\sqrt{2})\,\overline{P}^{\max}(\beta_0) + (1/\sqrt{2})\,\mathbb{E}\big[\overline{P}^{\max}(\beta_0+\sqrt{2}\eta)\big]. \tag{81}$$

Multiplying by $\sqrt{2}$ and rearranging:

$$\mathbb{E}\big[\overline{P}^{\max}(\beta_0+\sqrt{2}\eta)\big] - \mathbb{E}\big[\overline{P}^{\max}(\beta_0+\eta)\big] \geq (\sqrt{2}-1)\Big[\mathbb{E}\big[\overline{P}^{\max}(\beta_0+\eta)\big] - \overline{P}^{\max}(\beta_0)\Big], \tag{82}$$

the two increments of Theorem 5. At $\beta_0 = 0$, this holds with equality, so it is the worst case. □